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Name.....

Roll No. Year 20.....20.....

Exam Seat No.....

COMPUTER GROUP | SEMESTER - VI | DIPLOMA IN ENGINEERING AND TECHNOLOGY

A LABORATORY MANUAL FOR

MOBILE APPLICATION DEVELOPMENT <,(22617)"



MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION, MUMBAI
(Autonomous) (ISO 9001 : 2015) (ISO / IEC 27001 : 2013)

Practical No. 2: Install and configure java development kit (JDK), Android studio and android SDK.

I. Practical Significance

Android is based on Linux with a set of native core C/C++ libraries. Android applications are written in Java. However, they run on Android's own Java Virtual Machine, called Dalvik Virtual Machine (DVM) (instead of JDK's JVM) which is optimized to operate on the small and mobile devices. SDK provides a selection of tools required to build Android apps or to ensure the process goes as smoothly as possible. Whether creating an app with Java, Kotlin or C#, SDK should run on an Android device and access unique features of the OS.

II. Relevant Program Outcomes (POs)

PO 1. Basic knowledge

PO 2. Discipline knowledge

PO 7. Ethics

III. Competency and Practical Skills

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to know the android operating system installation and configuration steps.
2. Able to install different versions of android operating system using the knowledge of installation and configuration steps.

IV. Relevant Course Outcome(s)

Configure Android environment and development tools.

V. Practical Outcome (PrOs)

Install and configure java development kit (JDK), android studio and android SDK.

VI. Relevant Affective Domain Related Outcome(s)

1. Work collaboratively in team
2. Follow ethical Practices.

VII. Minimum Theoretical Background

Step 1 - Setup Java Development Kit (JDK) You can download the latest version of Java JDK from Oracle's Java site: Java SE Downloads. You will find instructions for installing JDK in downloaded files, follow the given instructions to install and configure the setup. Finally, set PATH and JAVA_HOME environment variables to refer to the directory that contains java and javac, typically java_install_dir/bin and java_install_dir respectively. If you are running Windows and have installed the JDK in C:\jdk1.6.0_15, you would have to put the following line in your C:\autoexec.batfile.

set

PATH=C:\jdk1.6.0_15\bin;%PATH%

set JAVA_HOME=C:\jdk1.6.0_15

Step 2 - Setup Android SDK You can download the latest version of Android SDK

from Android's official website: <http://developer.android.com/sdk/index.html>. If you are installing SDK on Windows machine, then you will find `ainstaller_rXX-windows.exe`, so just download and run this exe which will launch Android SDK Tool Setup wizard to guide you throughout the installation, so just follow the instructions carefully. Finally, you will have Android SDK Tools installed on your machine. If you are installing SDK either on Mac OS or Linux, check the instructions provided along with the downloaded `android-sdk_rXX-macosx.zip` file for Mac OS and `android-sdk_rXX-linux.tgz` file for Linux. This tutorial will consider that you are going to setup your environment on Windows machine having Windows 7 operating system.

Step 3 - Setup Android Development Tools (ADT) Plugin This step will help you in setting Android Development Tool plugin for Eclipse. Let's start with launching Eclipse and then, choose Help > Software Updates > Install New Software. This will display the following dialogue box.

Step 4 - Create Android Virtual Device to test your Android applications you will need a virtual Android device. So before we start writing our code, let us create an Android virtual device. Launch Android AVD Manager using Eclipse menu options Window > AVD Manager> which will launch Android AVD Manager. Use New button to create a new Android Virtual Device and enter the following information, before clicking Create AVD button.

VIII. Resources required (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks

IX. Practical related Questions

Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

1. List all the steps to install android operating system
2. List various IDEs that can be used to execute android operating system.

(Space for answers)

STEP 1: Download Google Android OS.

STEP 2: Write ISO Image File to CD/DVD or USB Drive.

STEP 3: Boot Your Computer Using CD/DVD or USB Drive.

STEP 4: Install or Test Run Google Android in Your Computer

Ans 2:

Android studio Komodo

Eclipse Cordovo

IntelliJ Idea

Netbeans

Visual Studio

X. Exercise

Note: Faculty must ensure that every group of students use different input value.

(Use blank space for answers or attach more pages if needed)

1. Differentiate between JVM and DVM.
2. What is IDE? Why Java development toolkit is essential to install an android operating system?

(Space for answers)

Ans 1:

JVM is the virtual machine that runs java code on different platforms. It acts as an abstract layer between the program and the platform on which the java code is running. The portability of Java code is possible only because of the JVM. The **javac compiler** converts the source code file(**.java file**) into an intermediate **java bytecode** format which is machine/platform independent.

DVM is a virtual machine to execute Android applications. The Java bytecode(**.class file**) generated by javac compiler is converted into **Dalvik bytecode** to make the application source files executable on the DVM. Since Android devices have a definite processing capacity, memory, and battery life, the DVM design principle aims to optimize itself so Ans it can load fastly and run smoothly even on low memory/powered devices

Ans 2:

An IDE, or Integrated Development Environment, enables programmers to consolidate the different aspects of writing a computer program. IDEs increase programmer productivity by combining common activities of writing software into a single application: editing source code, building executables, and debugging

XI. References / Suggestions for further Reading

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf
4. <https://developer.android.com>

XII. Assessment Scheme

Performance indicators		Weightage
Process related (10 Marks)		30%
1.	Logic Formation	10%
2.	Debugging ability	15%
3.	Follow ethical practices	5%
Product related (15 Marks)		70%
4.	Interactive GUI	20%
5.	Answer to Practical related questions	20%
6.	Expected Output	20%
7.	Timely Submission	10%
Total (25 Marks)		100%

List of student Team Members

1.....

2.....

3.....

4.....

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

Practical No. 3: Configure android development tools (ADT) plug-in and create android virtual device.

I. Practical Significance

Configuration set up steps should be known beforehand as learnt in the practical no.2 Android Development Tools (ADT) is a plugin for the Android studio that is designed to give you a powerful, integrated environment in which to build Android applications. ADT extends the capabilities of Android studio to let you quickly set up new Android projects, create an application UI, add components based on the Android Framework API, debug your applications using the Android SDK tools, and even export signed (or unsigned) APKs in order to distribute your application. Developing in Android studio with ADT is highly recommended and is the fastest way to get started. With the

guided project setup, it provides, as well as tools integration, custom XML editors, and debug output pane, ADT gives you an incredible boost in developing Android applications.

II. Relevant Program Outcomes (POs)

PO 1. Basic knowledge

PO 2. Discipline knowledge

PO 7. Ethics

III. Competency and Practical Skills

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to install android operating system.
2. Able to differentiate between various versions of android operating system.

IV. Relevant Course Outcome(s)

Configure Android environment and development tools.

V. Practical Outcome (PrOs)

Install and configure java development kit (JDK), android studio and android SDK.

VI. Relevant Affective Domain Related Outcome(s)

1. Work collaboratively in team
2. Follow ethical Practices.

VII. Minimum Theoretical Background

- ☐ Setting up Android Development Tools (ADT)
- ☐ Get Android Development Tools (ADT) from ADT Bundle



The ADT Bundle includes everything you need to begin developing apps:

- ☐ Android studio + ADT plugin
- ☐ Android SDK Tools Android Platform-tools
- ☐ The latest Android platform
- ☐ The latest Android system image for the emulator

VIII. Resources required (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks

IX. Practical related Questions

Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

1. List basic requirements for configuring android operating system?
2. Why bytecode cannot run in Android?

(Space for answers)

Ans 1:

- ☐ Microsoft Windows 7/8/10 (32-bit or 64-bit)
- ☐ 3 GB RAM **minimum**, 8 GB RAM recommended (plus 1 GB for the **Android** Emulator)
- ☐ 2 GB of available disk space **minimum**, 4 GB recommended (500 MB for IDE plus 1.5 GB for **Android** SDK and emulator **system** image)
- ☐ 1280 x 800 **minimum** screen resolution.

Ans 2:

Android uses Dalvik Virtual Machine (DVM) which requires a special bytecode. We need to convert Java class files into Dalvik Executable files using an Android tool called "dx". In normal circumstances, developers will not be using this tool directly and build tools will care for the generation of DVM compatible files.

X. Exercise

Note: Faculty must ensure that every group of students use different input value.

(Use blank space for answers or attach more pages if needed)

1. What is a Build Type in Gradle?
2. Explain the build process in Android.

(Space for answers)

Ans 1:

A build type determines how an app is packaged. By default, the Android plug-in for Gradle supports two different types of builds: debug and release. Build types define certain properties that Gradle uses when building and packaging your app, and are typically configured for different stages of your development lifecycle. For example, the debug build type enables debug options and signs the APK with the debug key, while the release build type may shrink, obfuscate, and sign your APK with a release key for distribution.

Ans 2:

The build process for a typical Android app module, as shown in figure 1, follows these general steps:

1. The compilers convert your source code into DEX (Dalvik Executable) files, which include the bytecode that runs on Android devices, and everything else into compiled resources.
2. The APK Packager combines the DEX files and compiled resources into a single APK. Before your app can be installed and deployed onto an Android device, however, the APK must be signed.
3. The APK Packager signs your APK using either the debug or release keystore:
 - a. If you are building a debug version of your app, that is, an app you intend only for testing and profiling, the packager signs your app with the debug keystore. Android Studio automatically configures new projects with a debug keystore.
 - b. If you are building a release version of your app that you intend to release externally, the packager signs your app with the release keystore. To create a release keystore, read about signing your app in android studio.
4. Before generating your final APK, the packager uses the zipline tool to optimize your app to use less memory when running on a device.

XI. References / Suggestions for further Reading

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf
4. <https://developer.android.com>

XII. Assessment Scheme

Performance indicators		Weightage
Process related (10 Marks)		30%
1.	Logic Formation	10%
2.	Debugging ability	15%
3.	Follow ethical practices	5%
Product related (15 Marks)		70%
4.	Interactive GUI	20%
5.	Answer to Practical related questions	20%
6.	Expected Output	20%
7.	Timely Submission	10%
Total (25 Marks)		100%

List of student Team Members

- 1.....
2.....
3.....
4.....

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

Practical No. 4: Develop a program to display Hello World on screen.

I. Practical Significance

In android studio students must be aware of the directory structure and the control flow of the program. Program should be either executed on the android mobile phones or on the suitable emulators. To execute a simple program, like to display Hello World on screen syntax of writing a program in android is pre-requisite as the programming language used is JAVA only. The main activity code is a Java file MainActivity.java. This is the actual application file which ultimately gets converted to a Dalvik executable and runs your application.

II. Relevant Program Outcomes (POs)

- PO 1. Basic knowledge
- PO 2. Discipline knowledge
- PO 3. Experiments and practice
- PO 4. Engineering tools

III. Competency and Practical Skills

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to use basic Java syntax to write a program.
2. Able to analyze the directory structure of android studio.

IV. Relevant Course Outcome(s)

Develop rich user Interfaces by using layouts and controls

V. Practical Outcome (PrOs)

Develop a program to display Hello World on screen.

VI. Relevant Affective Domain Related Outcome(s)

1. Work collaboratively in team
2. Follow ethical Practices.

VII. Minimum Theoretical Background

Following is the default code generated by the application wizard for Hello World! application:

```
package com.example.helloworld; import android.os.Bundle;
import android.app.Activity; import android.view.Menu; import
android.view.MenuItem;
import android.support.v4.app.NavUtils; public class MainActivity extends
Activity
{
@Override
```

```

public void onCreate(Bundle savedInstanceState)

{
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
}
@Override
public boolean onCreateOptionsMenu(Menu menu)
{
    getMenuInflater().inflate(R.menu.activity_main,menu);
    return true;
}
}

```

The Manifest File: Whatever component you develop as a part of your application, you must declare all its components in a manifest file called `AndroidManifest.xml` which resides at the root of the application project directory. This file works as an interface between Android OS and your application, so if you do not declare your component in this file, then it will not be considered by the OS. The `<activity>` tag is used to specify an activity and `android:name` attribute specifies the fully qualified classname of the Activity subclass and the `android:label` attributes specifies a string to use as the label for the activity. You can specify multiple activities using `<activity>` tags. The action for the intent filter is named `android.intent.action.MAIN` to indicate that this activity serves as the entry point for the application. The category for the intent- filter is named `android.intent.category.LAUNCHER` to indicate that the application can be launched from the device's launcher icon. The `@string` refers to the `strings.xml` file explained below. Hence, `@string/app_name` refers to the `app_name` string defined in the `strings.xml` file, which is "HelloWorld". Similar way, other strings get populated in the application. Following is the list of tags which you will use in your manifest file to specify different Android application components:

- 1) `<activity>` elements for activities
- 2) `<service>` elements for services
- 3) `<receiver>` elements for broadcast receivers
- 4) `<provider>` elements for content providers

VIII. Resources required (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

IX. Practical related Questions

Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

1. List the files used to write Hello World program.
2. What is an activity in Android programming?

(Space for answers)

Ans 1:

1. The Main Activity File

The main activity code is a Java file MainActivity.java. This is the actual application file which ultimately gets converted to a Dalvik executable and runs your application.

2. The Manifest File

Whatever component you develop as a part of your application, you must declare all its components in a manifest.xml which resides at the root of the application project directory. This file works as an interface between Android OS and your application, so if you do not declare your component in this file, then it will not be considered by the OS.

3. The Strings File

The strings.xml file is located in the res/values folder and it contains all the text that your application uses. For example, the names of buttons, labels, default text, and similar types of strings go into this file. This file is responsible for their textual content.

4. The Layout File

The activity_main.xml is a layout file available in res/layout directory, that is referenced by your application when building its interface. You will modify this file very frequently to change the layout of your application.

Ans 2:

An activity is the single screen in android. It is like window or frame of Java. By the help of activity, you can place all your UI components or widgets in a single screen. The 7 lifecycle method of Activity describes how activity will behave at different states.

X. Exercise

Note: Faculty must ensure that every group of students use different input value.

(Use blank space for answers or attach more pages if needed)

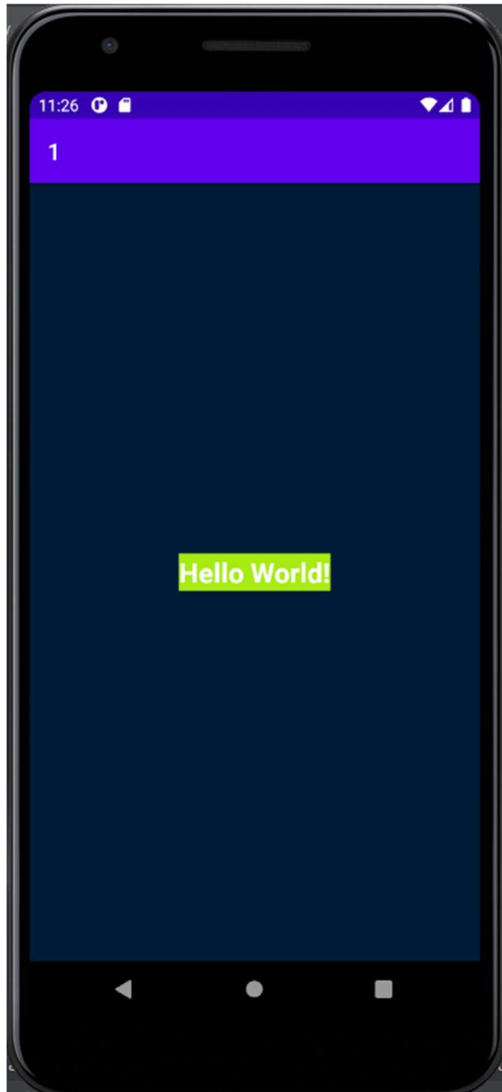
1. Write a program to display HelloWorld.
2. Write a program to display student name and marks.

(Space for answers)

Ans 1:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
android:background="#FFC107"
tools:context=".MainActivity">
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:background="#2196F3"
    android:text="Hello World!"
    android:textColor="#000000"
    android:textSize="24sp"
    android:textStyle="bold"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintLeft_toLeftOf="parent"
    app:layout_constraintRight_toRightOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```



Ans 2:

```
<?xml version="1.0" encoding="utf-8"?>

<TableLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#f6f6f6"
    tools:context=".MainActivity">

    <TableRow android:background="#98ded9" android:padding="5px">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Students_Information"
            android:textColor="#161d6f"
            android:textSize="30sp"
            android:textStyle="bold" />
    </TableRow>

    <TableRow android:background="#c7ffd8" android:padding="5px">

        <TextView
            android:id="@+id/text1"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Name: Heet ; Marks: 96"
            android:textColor="#161d6f"
            android:textSize="30sp"
            android:textStyle="bold" />
    </TableRow>

    <TableRow android:background="#98ded9" android:padding="5px">

        <TextView
            android:id="@+id/text2"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Name: Meet ; Marks: 97"
            android:textColor="#161d6f"
```

```
        android:textSize="30sp"
        android:textStyle="bold" />
</TableRow>

<TableRow android:background="#c7ffd8" android:padding="5px">

    <TextView
        android:id="@+id/text3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Name: Pratik ; Marks: 99"
        android:textColor="#161d6f"
        android:textSize="30sp"
        android:textStyle="bold" />
</TableRow>

<TableRow android:background="#98ded9" android:padding="5px">

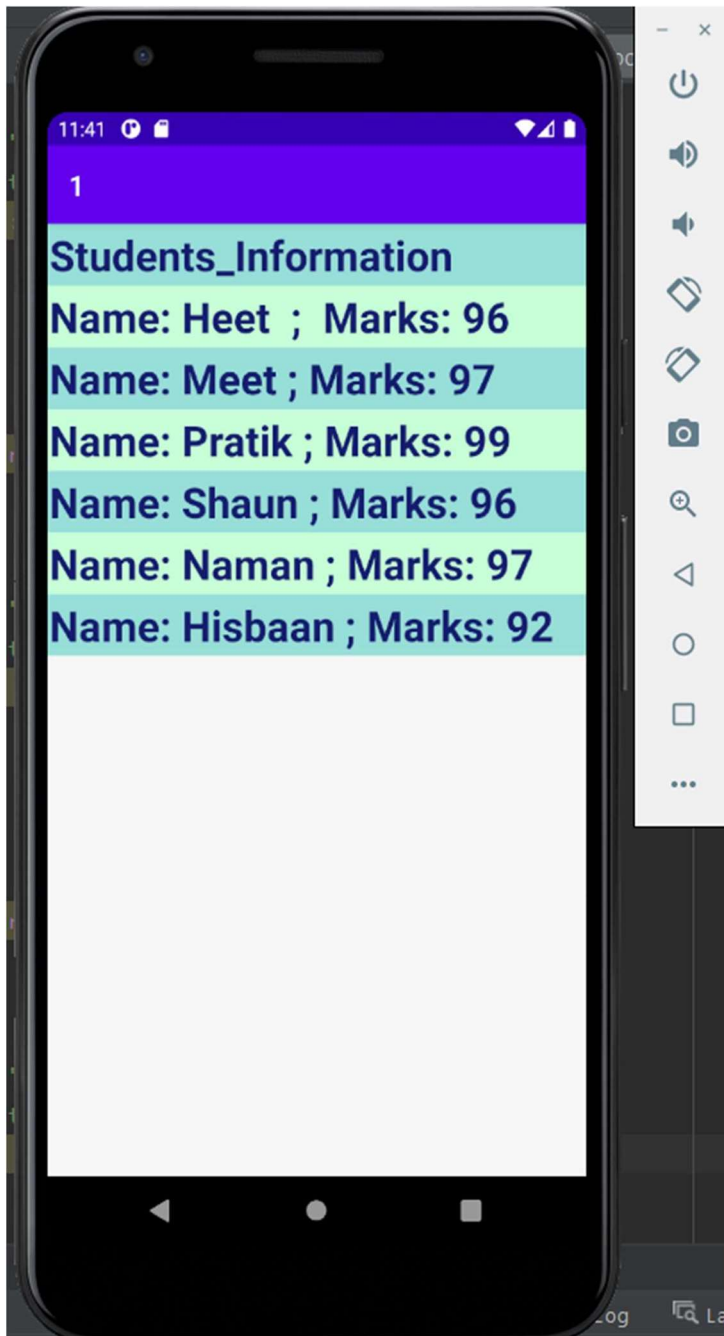
    <TextView
        android:id="@+id/text4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Name: Shaun ; Marks: 96"
        android:textColor="#161d6f"
        android:textSize="30sp"
        android:textStyle="bold" />
</TableRow>

<TableRow android:background="#c7ffd8" android:padding="5px">

    <TextView
        android:id="@+id/text5"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Name: Naman ; Marks: 97"
        android:textColor="#161d6f"
        android:textSize="30sp"
        android:textStyle="bold" />
</TableRow>

<TableRow android:background="#98ded9" android:padding="5px">
```

```
<TextView
    android:id="@+id/text6"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Name: Hisbaan ; Marks: 92"
    android:textColor="#161d6f"
    android:textSize="30sp"
    android:textStyle="bold" />
</TableRow>
</TableLayout>
```



XI. References / Suggestions for further Reading

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf
4. <https://developer.android.com>

XII. Assessment Scheme

Performance indicators		Weightage
Process related (10 Marks)		30%
1.	Logic Formation	10%
2.	Debugging ability	15%
3.	Follow ethical practices	5%
Product related (15 Marks)		70%
4.	Interactive GUI	20%
5.	Answer to Practical related questions	20%
6.	Expected Output	20%
7.	Timely Submission	10%
Total (25 Marks)		100%

List of student Team Members

- 1.....
- 2.....
- 3.....
- 4.....

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

Practical No. 5: Develop a program to implement linear layout and absolute layout.

I. Practical Significance

To develop and place the android components accurately on the display screen, android provides various layout managers. Layout managers can be used on the simple android program too. Various layout managers can be selected as per the program requirements.

II. Relevant Program Outcomes (POs)

PO 1. Basic knowledge

PO 2. Discipline knowledge

PO 3. Experiments and practice

PO 4. Engineering tools

III. Competency and Practical Skills

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to use layout managers to indent the android components on the display screen.
2. Able to analyze different layout managers and select the appropriate layout managers as per the program requirements.

IV. Relevant Course Outcome(s)

Develop rich user Interfaces by using layouts and controls

V. Practical Outcome (PrOs)

Develop a program to implement linear layout and absolute layout.

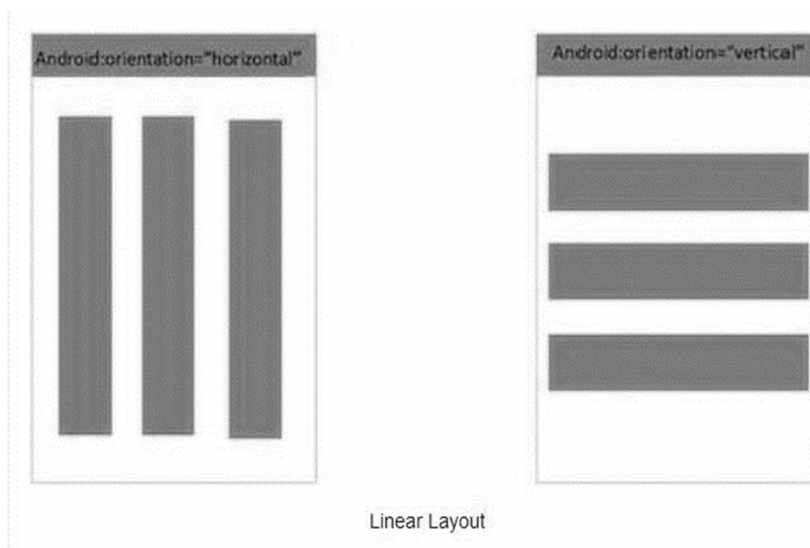
VI. Relevant Affective Domain Related Outcome(s)

1. Work collaboratively in team
2. Follow ethical Practices.

VII. Minimum Theoretical Background

Layouts which are subclasses of View Group class and a typical layout defines the visual structure for an Android user interface and can be created either at run time using View/View Group objects or you can declare your layout using simple XML file main_layout.xml which is located in the res/layout folder of your project. layouts defined in XML file. A layout may contain any type of widgets such as buttons, labels, textboxes etc. Layout Attributes Each layout has a set of attributes which define the visual properties of that layout. There are few common attributes among all the layouts and there are other attributes which are specific to that layout. Types of layouts are Linear and Absolute layouts.

Linear Layout: -



VIII. Resources required (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

IX. Practical related Questions

Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

1. Name any three-layout manager?
2. What is Card View?

Ans. 1.

1. Linear Layout

LinearLayout is a view group that aligns all children in a single direction, vertically or horizontally.

2. Relative Layout

RelativeLayout is a view group that displays child views in relative positions.

3. Table Layout

TableLayout is a view that groups views into rows and columns.

Ans. 2.

CardView is a new widget in Android that can be used to display any sort of data by providing a rounded corner layout along with a specific elevation. CardView is the view that can display views on top of each other. The main usage of CardView is that it helps to give a rich feel and look to the UI design.

X. Exercise

Note: Faculty must ensure that every group of students use different input value.

(Use blank space for answers or attach more pages if needed)

1. Write a program to place Name, Age and mobile number linearly (Vertical) on the display screen using Linear layout.
2. Write a program to place Name, Age and mobile number centrally on the display screen using Absolute layout.

(Space for answers)

Ans 1.

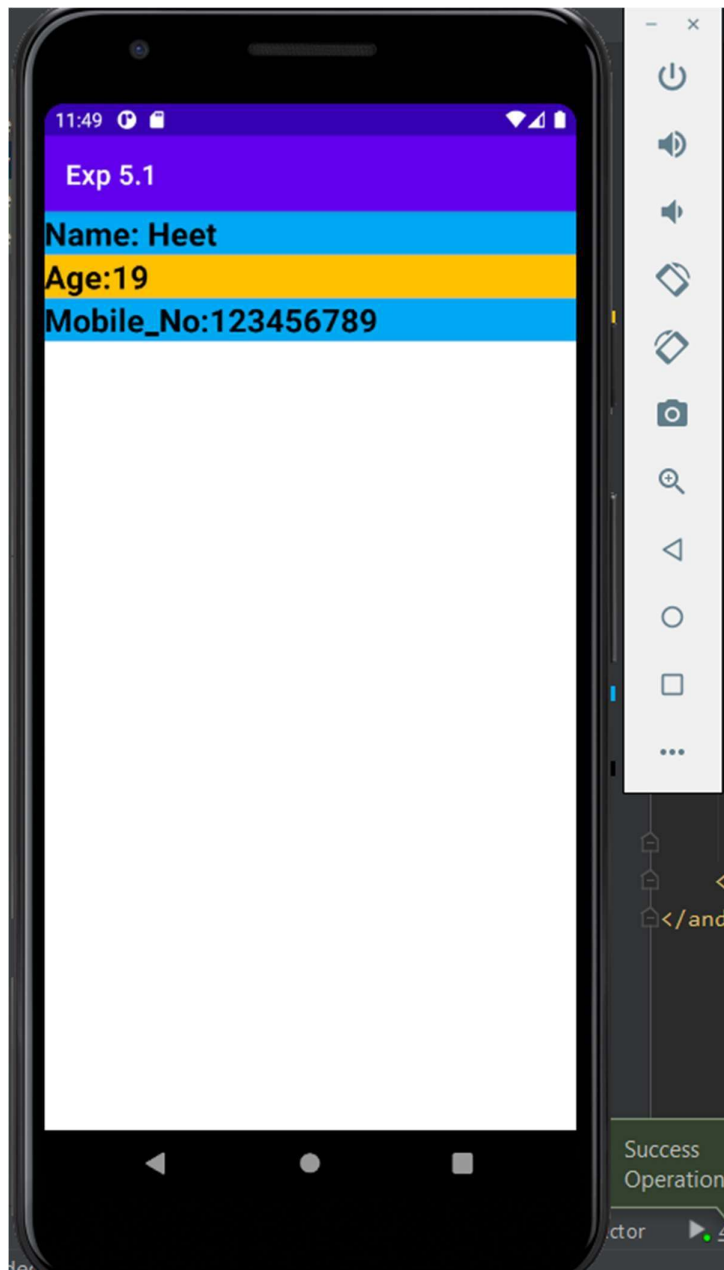
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:layout_editor_absoluteX="199dp"
    tools:layout_editor_absoluteY="362dp">
```

```
<TextView
    android:id="@+id/textView"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:background="#03A9F4"
    android:text="Name: Heet"
    android:textColor="#000000"
    android:textSize="24sp"
    android:textStyle="bold" />
```

```
<TextView
    android:id="@+id/textView2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:background="#FFC107"
    android:text="Age: 19"
    android:textColor="#000000"
    android:textSize="24sp"
    android:textStyle="bold" />
```

```
<TextView
    android:id="@+id/textView3"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:background="#03A9F4"
    android:text="Mobile_No:123456789"
    android:textColor="#000000"
    android:textSize="24sp"
    android:textStyle="bold" />
</LinearLayout>
</androidx.constraintlayout.widget.ConstraintLayout>
```



Ans 2:

```
<?xml version="1.0" encoding="utf-8"?>
<AbsoluteLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"

    android:layout_height="match_parent"
    android:background="#c7ffd8"
    tools:context=".MainActivity"
    tools:ignore="Deprecated">

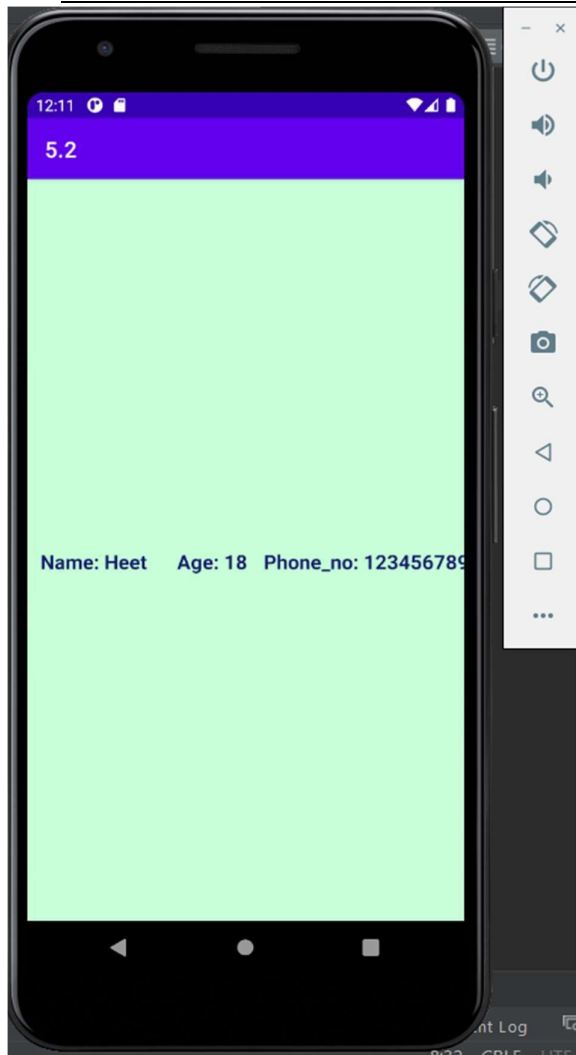
    <TextView
        android:id="@+id/textView"

        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="12dp"
        android:layout_y="338dp"
        android:text="Name: Heet"
        android:textColor="#161d6f"
        android:textSize="18sp"
        android:textStyle="bold" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="135dp"
        android:layout_y="338dp"
        android:text="Age: 18"
        android:textColor="#161d6f"
        android:textSize="18sp"
        android:textStyle="bold" />

    <TextView
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_x="213dp"
        android:layout_y="338dp"
        android:text="Phone_no: 123456789"
        android:textColor="#161d6f"
        android:textSize="18sp"
        android:textStyle="bold" />

</AbsoluteLayout>
```



XI. References / Suggestions for further Reading

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf
4. <https://developer.android.com>

XII. Assessment Scheme

Performance indicators		Weightage
Process related (10 Marks)		30%
1.	Logic Formation	10%
2.	Debugging ability	15%
3.	Follow ethical practices	5%
Product related (15 Marks)		70%
4.	Interactive GUI	20%
5.	Answer to Practical related questions	20%
6.	Expected Output	20%
7.	Timely Submission	10%
Total (25 Marks)		100%

List of student Team Members

- 1.....
- 2.....
- 3.....
- 4.....

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

Practical No. 6: Develop a program to implement frame layout, table layout and relative layout.

I. Practical Significance

To develop and place the android components accurately on the display screen, android provides various layout managers. Layout managers can be used on the simple android program too. Various layout managers can be selected as per the program requirements.

II. Relevant Program Outcomes (POs)

PO 1. Basic knowledge

PO 2. Discipline knowledge

PO 3. Experiments and practice

PO 4. Engineering tools

III. Competency and Practical Skills

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to use layout managers to indent the android components on the display screen.
2. Able to analyze different layout managers and select the appropriate layout managers as per the program requirements.

IV. Relevant Course Outcome(s)

Develop rich user Interfaces by using layouts and controls

V. Practical Outcome (PrOs)

Develop a program to implement frame layout, table layout and relative layout.

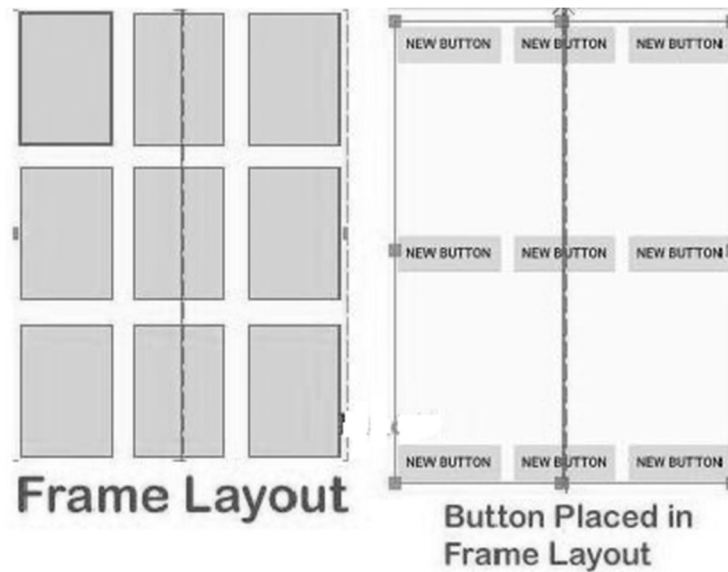
VI. Relevant Affective Domain Related Outcome(s)

1. Work collaboratively in team
2. Follow ethical Practices.

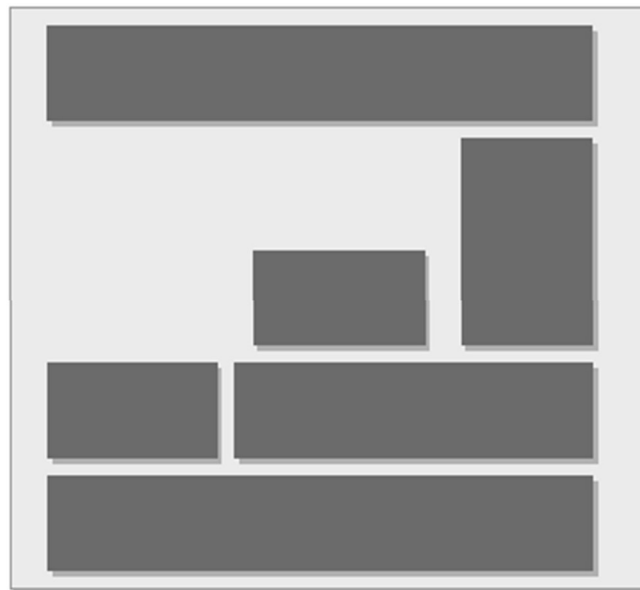
VII. Minimum Theoretical Background

1. Frame Layout:

Frame Layout is designed to block out an area on the screen to display a single item. Generally, Frame Layout should be used to hold a single child view, because it can be difficult to organize child views in a way that's scalable to different screen sizes without the children overlapping each other. You can, however, add multiple children to a Frame Layout and control their position within the Frame Layout by assigning gravity to each child, using the android: layout gravity attribute. Child views are drawn in a stack, with the most recently added child on top. The size of the Frame Layout is the size of its largest child (plus padding), visible or not (if the Frame Layout's parent permits).



2. Relative Layout:



A Relative Layout is a very powerful utility for designing a user interface because it can eliminate nested view groups and keep your layout hierarchy flat, which improves performance. If you find yourself using several nested Linear Layout groups, you may be able to replace them with a single Relative Layout.

3. Table Layout:

A Table Layout consists of a number of Table Row objects, each defining a row (actually, you can have other children, which will be explained below). Table

Layout containers do not display border lines for their rows, columns, or cells.

Each row has zero or more cells; each cell can hold one View object. The table has as many columns as the row with the most cells. A table can leave cells empty. Cells can span columns, as they can in HTML. The width of a column is defined by the row with the widest cell in that column.

Row 1		
Row 2 Column 1	Row 2 Column 2	Row 2 Column 3
Row 3		
Row 4 Column 1	Row 4 Column 2	

VIII. Resources required (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

IX. Practical related Questions

Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

1. List different attributes which can be used with any layout managers.
2. What is Grid Layout?

(Space for answers)

Ans 1:

- i. android:layout_width: This is the width of the layout.
- ii. android:layout_height: This is the height of the layout
- iii. android:layout_x: This specifies the x-coordinate of the layout.
- iv. android:layout_y: This specifies the y-coordinate of the layout.
- v. android:padding: This is the padding of the layout.
- vi. android:margin: This is the margin of the layout

Ans 2 :

A layout that places its children in a rectangular grid.

The grid is composed of a set of infinitely thin lines that separate the viewing area into cells.

Throughout the API, grid lines are referenced by grid indices. A grid with N columns has N +

1 grid indices that run from 0 through N inclusive. Regardless of how GridLayout is configured, grid index 0 is fixed to the leading edge of the container and grid index N is fixed to its trailing edge (after padding is taken into account).

X. Exercise

Note: Faculty must ensure that every group of students use different input value.

(Use blank space for answers or attach more pages if needed)

1. Write a program to display 10 students basic information in a table form using Table layout.
2. Write a program to display all the data types in object-oriented programming using Frame layout.

(Space for answers)

```
Ans 1. <?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
```

```
<TableLayout
android:layout_width="match_parent"
android:layout_height="match_parent"
android:background="#31326f"
tools:layout_editor_absoluteX="125dp"
tools:layout_editor_absoluteY="245dp">
```

```
<TableRow
android:layout_width="match_parent"
android:layout_height="match_parent">
```

```
<TextView
android:id="@+id/textView1"
android:layout_width="250dp"
android:layout_height="wrap_content"
android:layout_column="1"
android:layout_gravity="center"
android:paddingTop="25dp"
android:paddingBottom="25dp"
android:text="Name"
android:textColor="#ffc93c"
android:textSize="30sp"
android:textStyle="bold" />
```

```
<TextView
android:id="@+id/textView2"
android:layout_width="170dp"
android:layout_height="wrap_content"
android:layout_column="2"
```

```
        android:layout_gravity="center"
        android:paddingTop="25dp"
        android:paddingBottom="25dp"
        android:text="AGE"
        android:textColor="#ffc93c"
        android:textSize="30sp"
        android:textStyle="bold" />
</TableRow>

<TableRow
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/textView3"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:layout_column="1"
        android:layout_gravity="center"
        android:text="Heet"
        android:textColor="#ffc93c"
        android:textSize="30sp"
        android:textStyle="bold" />

    <TextView
        android:id="@+id/textView4"
        android:layout_width="170dp"
        android:layout_height="wrap_content"
        android:layout_column="2"
        android:layout_gravity="center"
        android:text="19"
        android:textColor="#ffc93c"
        android:textSize="30sp"
        android:textStyle="bold" />
</TableRow>

<TableRow
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/textView5"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:layout_column="1"
        android:layout_gravity="center"
        android:text="Shaun"
        android:textColor="#ffc93c"
        android:textSize="30sp"
        android:textStyle="bold" />

    <TextView
        android:id="@+id/textView6"
        android:layout_width="170dp"
```

```
        android:layout_height="wrap_content"
        android:layout_column="2"
        android:layout_gravity="center"
        android:text="19"
        android:textColor="#ffc93c"
        android:textSize="30sp"
        android:textStyle="bold" />
    </TableRow>
```

```
<TableRow
    android:layout_width="match_parent"
    android:layout_height="match_parent">
```

```
    <TextView
        android:id="@+id/textView7"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:layout_column="1"
        android:layout_gravity="center"
        android:text="Pratik"
        android:textColor="#ffc93c"
        android:textSize="30sp"
        android:textStyle="bold" />
```

```
    <TextView
        android:id="@+id/textView8"
        android:layout_width="170dp"
        android:layout_height="wrap_content"
        android:layout_column="2"
        android:layout_gravity="center"
        android:text="18"
        android:textColor="#ffc93c"
        android:textSize="30sp"
        android:textStyle="bold" />
```

```
</TableRow>
```

```
<TableRow
    android:layout_width="match_parent"
    android:layout_height="match_parent">
```

```
    <TextView
        android:id="@+id/textView9"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:layout_column="1"
        android:layout_gravity="center"
        android:text="Gaurang"
        android:textColor="#ffc93c"
        android:textSize="30sp"
        android:textStyle="bold" />
```

```
    <TextView
        android:id="@+id/textView10"
        android:layout_width="170dp"
```

```
        android:layout_height="wrap_content"
        android:layout_column="2"
        android:layout_gravity="center"
        android:text="19"
        android:textColor="#ffc93c"
        android:textSize="30sp"
        android:textStyle="bold" />
    </TableRow>
```

```
<TableRow
    android:layout_width="match_parent"
    android:layout_height="match_parent">
```

```
    <TextView
        android:id="@+id/textView11"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:layout_column="1"
        android:layout_gravity="center"
        android:text="Meet"
        android:textColor="#ffc93c"
        android:textSize="30sp"
        android:textStyle="bold" />
```

```
    <TextView
        android:id="@+id/textView12"
        android:layout_width="170dp"
        android:layout_height="wrap_content"
        android:layout_column="2"
        android:layout_gravity="center"
        android:text="19"
        android:textColor="#ffc93c"
        android:textSize="30sp"
        android:textStyle="bold" />
```

```
</TableRow>
```

```
<TableRow
    android:layout_width="match_parent"
    android:layout_height="match_parent">
```

```
    <TextView
        android:id="@+id/textView13"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:layout_column="1"
        android:layout_gravity="center"
        android:text="Naman"
        android:textColor="#ffc93c"
        android:textSize="30sp"
        android:textStyle="bold" />
```

```
    <TextView
        android:id="@+id/textView14"
        android:layout_width="170dp"
```

```
        android:layout_height="wrap_content"
        android:layout_column="2"
        android:layout_gravity="center"
        android:text="18"
        android:textColor="#ffc93c"
        android:textSize="30sp"
        android:textStyle="bold" />
    </TableRow>
```

```
<TableRow
    android:layout_width="match_parent"
    android:layout_height="match_parent">
```

```
    <TextView
        android:id="@+id/textView15"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:layout_column="1"
        android:layout_gravity="center"
        android:text="Divyesh"
        android:textColor="#ffc93c"
        android:textSize="30sp"
        android:textStyle="bold" />
```

```
    <TextView
        android:id="@+id/textView16"
        android:layout_width="170dp"
        android:layout_height="wrap_content"
        android:layout_column="2"
        android:layout_gravity="center"
        android:text="19"
        android:textColor="#ffc93c"
        android:textSize="30sp"
        android:textStyle="bold" />
```

```
</TableRow>
```

```
<TableRow
    android:layout_width="match_parent"
    android:layout_height="match_parent">
```

```
    <TextView
        android:id="@+id/textView17"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:layout_column="1"
        android:layout_gravity="center"
        android:text="Harry"
        android:textColor="#ffc93c"
        android:textSize="30sp"
        android:textStyle="bold" />
```

```
    <TextView
        android:id="@+id/textView18"
        android:layout_width="170dp"
```



```
        android:layout_height="wrap_content"
        android:layout_column="2"
        android:layout_gravity="center"
        android:text="19"
        android:textColor="#ffc93c"
        android:textSize="30sp"
        android:textStyle="bold" />
    </TableRow>

    <TableRow
        android:layout_width="match_parent"
        android:layout_height="match_parent">

        <TextView
            android:id="@+id/textView19"
            android:layout_width="250dp"
            android:layout_height="wrap_content"
            android:layout_column="1"
            android:layout_gravity="center"
            android:text="Ron"
            android:textColor="#ffc93c"
            android:textSize="30sp"
            android:textStyle="bold" />

        <TextView
            android:id="@+id/textView20"
            android:layout_width="170dp"
            android:layout_height="wrap_content"
            android:layout_column="2"
            android:layout_gravity="center"
            android:text="18"
            android:textColor="#ffc93c"
            android:textSize="30sp"
            android:textStyle="bold" />
    </TableRow>
</TableLayout>
</androidx.constraintlayout.widget.ConstraintLayout>
```



Ans 2. `<?xml version="1.0" encoding="utf-8"?>`
`<androidx.constraintlayout.widget.ConstraintLayout`
`xmlns:android="http://schemas.android.com/apk/res/android"`
`xmlns:app="http://schemas.android.com/apk/res-auto"`
`xmlns:tools="http://schemas.android.com/tools"`
`android:layout_width="match_parent"`
`android:layout_height="match_parent"`
`tools:context=".MainActivity">`

`<FrameLayout`
`android:layout_width="match_parent"`
`android:layout_height="match_parent"`
`android:background="#02475e"`
`tools:layout_editor_absoluteX="114dp"`

```
tools:layout_editor_absoluteY="242dp">

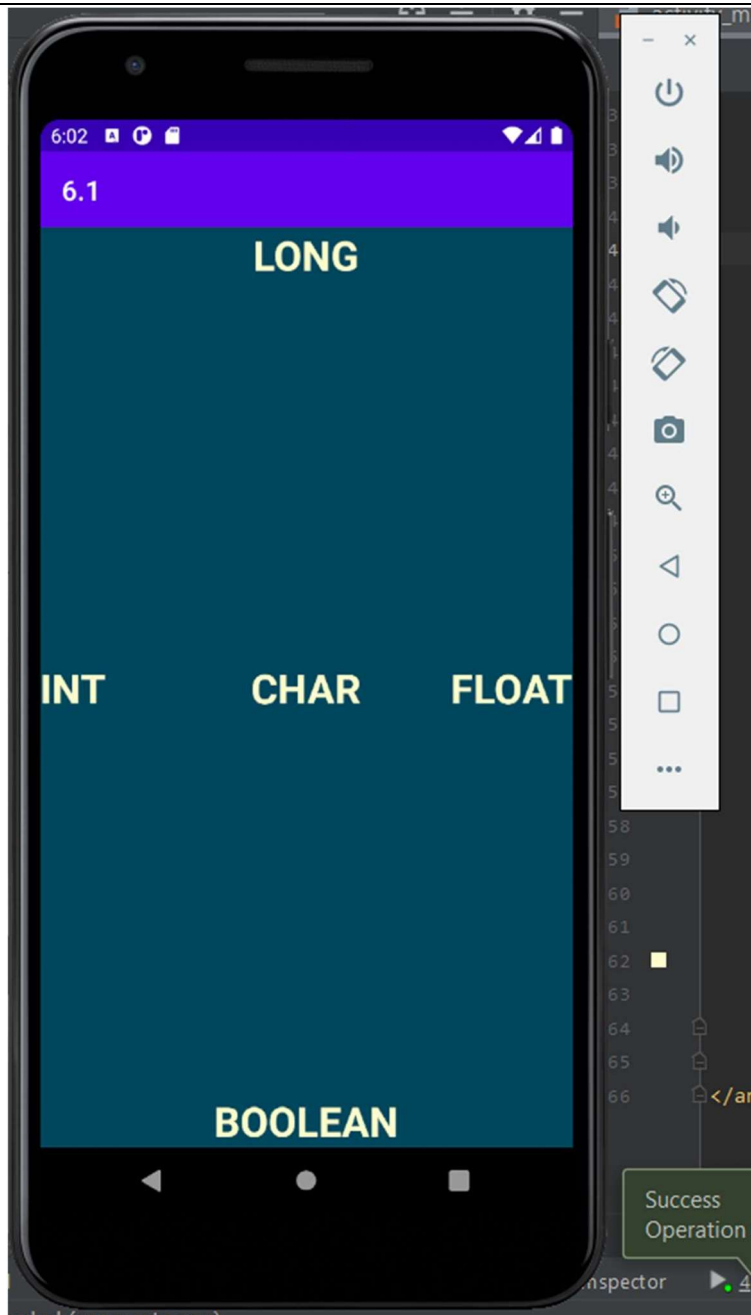
<TextView
    android:id="@+id/textView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="top|center"
    android:text="LONG"
    android:textColor="#fefecc"
    android:textSize="30sp"
    android:textStyle="bold" />

<TextView
    android:id="@+id/textView3"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="bottom|center"
    android:text="BOOLEAN"
    android:textColor="#fefecc"
    android:textSize="30sp"
    android:textStyle="bold" />

<TextView
    android:id="@+id/textView4"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="left|center"
    android:text="INT"
    android:textColor="#fefecc"
    android:textSize="30sp"
    android:textStyle="bold" />

<TextView
    android:id="@+id/textView5"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="center"
    android:text="CHAR"
    android:textColor="#fefecc"
    android:textSize="30sp"
    android:textStyle="bold" />

<TextView
    android:id="@+id/textView6"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="right|center"
    android:text="FLOAT"
    android:textColor="#fefecc"
    android:textSize="30sp"
    android:textStyle="bold" />
</FrameLayout>
</androidx.constraintlayout.widget.ConstraintLayout>
```



XI. References / Suggestions for further Reading

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf
4. <https://developer.android.com>

XII. Assessment Scheme

Performance indicators		Weightage
Process related (10 Marks)		30%
1.	Logic Formation	10%
2.	Debugging ability	15%
3.	Follow ethical practices	5%

Product related (15 Marks)		70%
4.	Interactive GUI	20%
5.	Answer to Practical related questions	20%
6.	Expected Output	20%
7.	Timely Submission	10%
Total (25 Marks)		100%

List of student Team Members

1.....

2.....

3.....

4.....

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

Practical No. 7: Develop a program to implement Text View and Edit Text.

I. Practical Significance

In this practical, UI controls in android like Text view and edit view are studied. Wherein the UI controls can be developed, used and placed on the screen using different layout managers as per the problem statement requirements.

II. Relevant Program Outcomes (POs)

PO 1. Basic knowledge

PO 2. Discipline knowledge

PO 3. Experiments and practice

PO 4. Engineering tools

PO 7. Ethics

PO 10. Life-long learning

III. Competency and Practical Skills

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to develop UI controls like Text View and Edit Text.
2. Able to test UI controls like Text View and Edit Text by checking its placing on the display screen.
3. Able to build UI controls like Text View and Edit Text, once testing is done and there are no errors.

IV. Relevant Course Outcome(s)

1. Develop rich user Interfaces by using layouts and controls.
2. Use User Interface components for android application development.

V. Practical Outcome (PrOs)

Develop a program to implement Text View and Edit Text.

VI. Relevant Affective Domain Related Outcome(s)

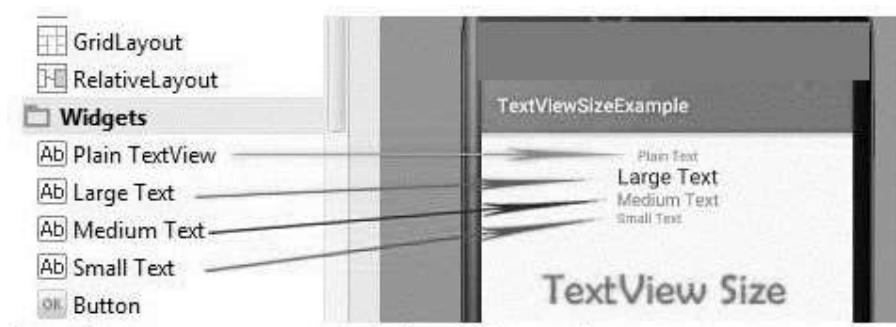
1. Work collaboratively in team
2. Follow ethical Practices.

VII. Minimum Theoretical Background

1. Text View:

In Android, Text View displays text to the user and optionally allows them to edit it programmatically. Text View is a complete text editor, however basic class is configured to not allow editing but we can edit it. View is the parent class of Text View Being a subclass of view the text view component can be used in your app's.

GUI inside a View Group, or as the content view of an activity. We can create a Text View instance by declaring it inside a layout(XML file) or by instantiating it programmatically(Java Class).



2. Edit Text:

In Android, Edit Text is a standard entry widget in android apps. It is an overlay over Text View that configures itself to be editable. Edit Text is a subclass of Text View with text editing operations. Often use Edit Text in our applications in order to provide an input or text field, especially in forms. The simplest example of Edit Text is Login or Sign-in form. Text Fields in Android Studio are basically Edit Text.

Note: An Edit Text is simply a thin extension of a Text View. An Edit Text inherits all the properties of a Text View.



VIII. Resources required (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

IX. Practical related Questions

Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

1. Which of these is not defined as a process state?
 - a. Non-visible
 - b. Visible
 - c. Foreground
 - d. Background

2. What is the name of the folder that contains the R.java file?
 - a. src
 - b. res
 - c. bin
 - d. gen

Ans 1 :

Non-visible is not defined as a process state.

Ans 2:

the name of the folder that contains the R.java file is Gen.

X. Exercise

Note: Faculty must ensure that every group of students use different input value.

(Use blank space for answers or attach more pages if needed)

1. Write a program to accept username and password from the end user using Text View and Edit Text.
2. Write a program to accept and display personal information of the student.

Ans1 <?xml version="1.0" encoding="utf-8"?>

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
```

```
<TextView
    android:id="@+id/Username"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="UserName"
    android:layout_marginTop="150dp"
    android:layout_marginLeft="40dp" />
```

```
<EditText
    android:id="@+id/User_name"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:ems="10"
    android:layout_toRightOf="@+id/Username"
```

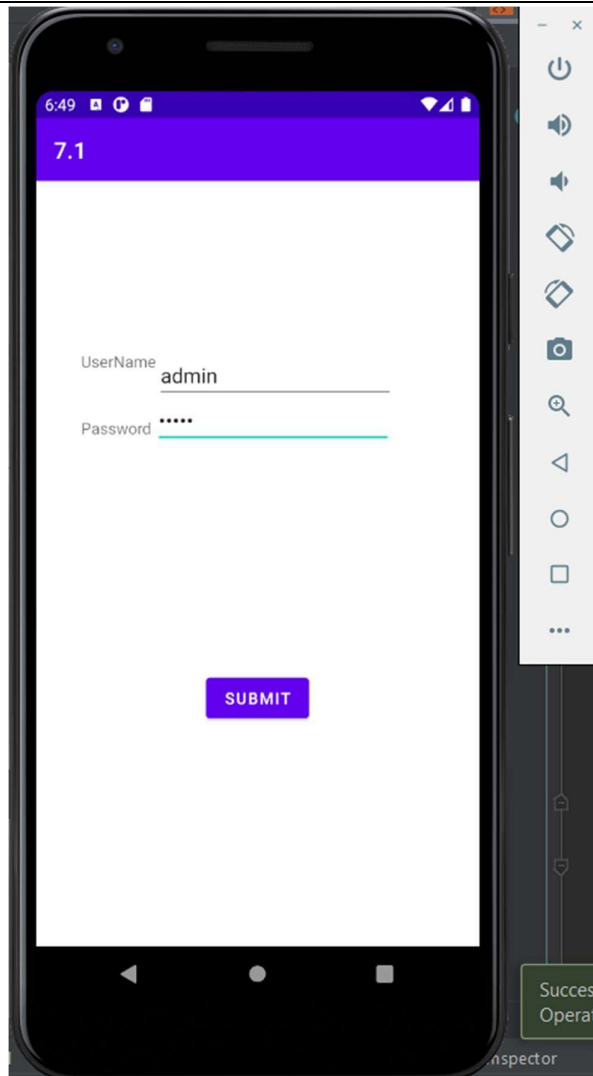


```
        android:layout_marginTop="150dp"
    />

    <TextView
        android:id="@+id/Password"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Password"
        android:layout_marginTop="40dp"
        android:layout_below="@+id/Username"
        android:layout_marginLeft="40dp"
    />

    <EditText
        android:id="@+id/Pass_word"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginLeft="2dp"
        android:layout_marginTop="190dp"
        android:layout_toRightOf="@+id/Password"
        android:ems="10"
        android:inputType="textPassword" />

    <Button
        android:id="@+id/Submit"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="true"
        android:text="Submit"
        android:layout_marginTop="200dp"
        android:layout_below="@+id/Pass_word"
    />
</RelativeLayout>
```



```

Ans 2  <?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/Name"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="30dp"
        android:text="Name" />

    <EditText
        android:id="@+id/editText_Name"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="60dp"
        android:layout_marginRight="@+id/Name"
        android:ems="10" />

```

```
<TextView
    android:id="@+id/Age"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@+id/Name"
    android:layout_margin="30dp"
    android:text="Age" />

<EditText
    android:id="@+id/editText_age"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/Name"
    android:layout_margin="60dp"
    android:layout_marginRight="@+id/Age"
    android:ems="10" />

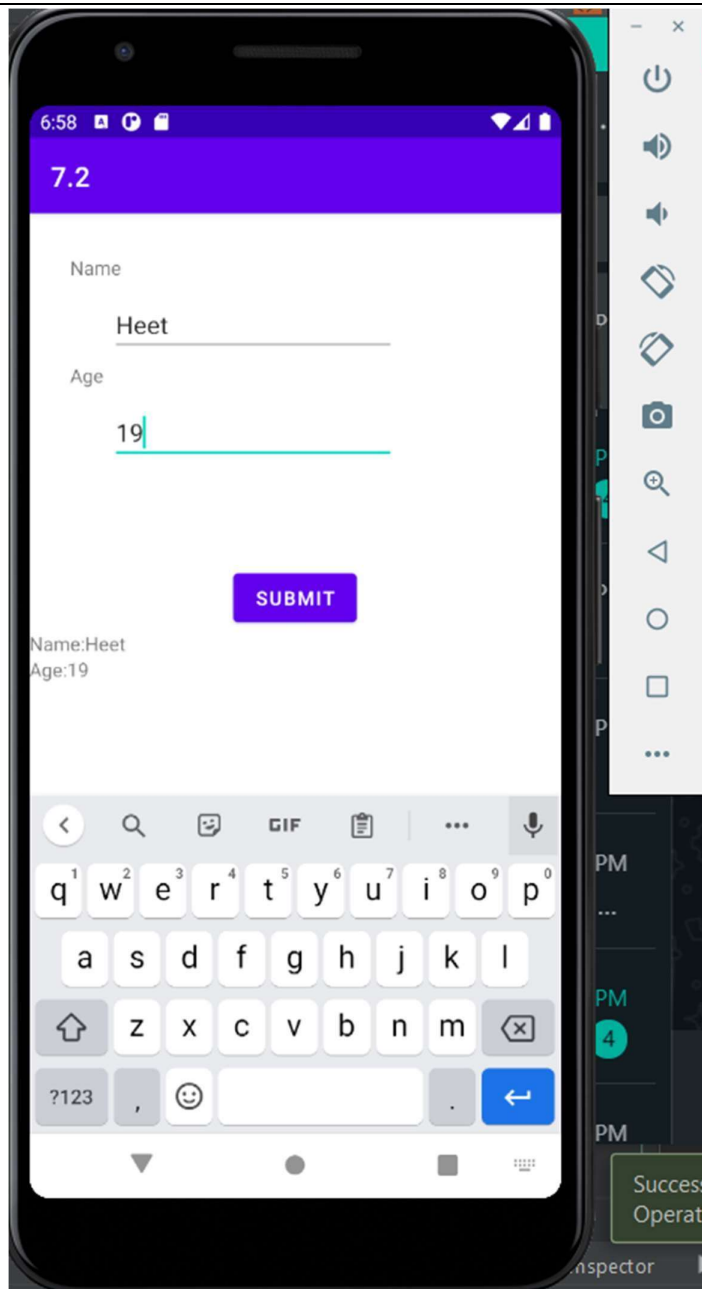
<Button
    android:id="@+id/Submit"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_centerHorizontal="true"
    android:layout_marginTop="260dp"
    android:text="Submit" />

<TextView
    android:id="@+id/Student_Name"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/Submit"
    android:visibility="invisible" />

<TextView
    android:id="@+id/Student_age"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/Student_Name"
    android:visibility="invisible" />
</RelativeLayout>
```

```
public class MainActivity extends AppCompatActivity {
    EditText Name, Age;
    String name, age;
    TextView Display_Name, Display_Age;
    Button display;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        display=findViewById(R.id.Submit);
        display.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                go();
            }
        });
    }
}
```

```
    }  
    public void go()  
    {  
        Name=findViewById(R.id.editText_Name);  
        Age=findViewById(R.id.editText_age);  
  
        name=Name.getText().toString().trim();  
        age=Age.getText().toString().trim();  
  
        Display_Name=findViewById(R.id.Student_Name);  
        Display_Age=findViewById(R.id.Student_age);  
  
        Display_Name.setVisibility(View.VISIBLE);  
        Display_Name.setText("Name:"+name);  
  
        Display_Age.setVisibility(View.VISIBLE);  
        Display_Age.setText("Age:"+age);  
    }  
}
```



XI. References / Suggestions for further Reading

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf
4. <https://developer.android.com>

XII. Assessment Scheme

Performance indicators		Weightage
Process related (10 Marks)		30%
1.	Logic Formation	10%
2.	Debugging ability	15%
3.	Follow ethical practices	5%
Product related (15 Marks)		70%
4.	Interactive GUI	20%
5.	Answer to Practical related questions	20%
6.	Expected Output	20%
7.	Timely Submission	10%
Total (25 Marks)		100%

List of student Team Members

- 1.....
- 2.....
- 3.....
- 4.....

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

Practical No. 9: Develop a program to implement Button, Image Button and Toggle Button.

I. Practical Significance

In this practical, UI controls in android like Buttons are studied. There are various types of buttons like Image button and toggle button which is studied.

II. Relevant Program Outcomes (POs)

PO 1. Basic knowledge

PO 2. Discipline knowledge

PO 3. Experiments and practice

PO 4. Engineering tools

PO 7. Ethics

PO 10. Life-long learning

III. Competency and Practical Skills

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to develop UI controls like various types of buttons.
2. Able to use buttons which handles events.
3. Able to build Passive and Active UI controls.

IV. Relevant Course Outcome(s)

1. Develop rich user Interfaces by using layouts and controls.
2. Use User Interface components for android application development.

V. Practical Outcome (PrOs)

Develop a program to implement Button, Image Button and Toggle Button.

VI. Relevant Affective Domain Related Outcome(s)

1. Work collaboratively in team
2. Follow ethical Practices.

VII. Minimum Theoretical Background

1. Buttons-

In Android, Button represents a push button. The android.widget.Button is subclass of Text View class and Compound Button is the subclass of Button class. A Push buttons can be clicked, or pressed by the user to perform an action. There are different types of buttons used in android such as Compound Button, Toggle Button, Radio Button. Button is a subclass of Text View class and compound button is the subclass of Button class. On a button we can perform different actions or events like click event, pressed event, touch event etc. Android buttons are GUI components which are sensible to taps (clicks) by the user. When the user taps/clicks on button in an Android app, the app can respond to the click/tap. These buttons can be divided into two categories: the first is Buttons with text on, and second is buttons with an image on.



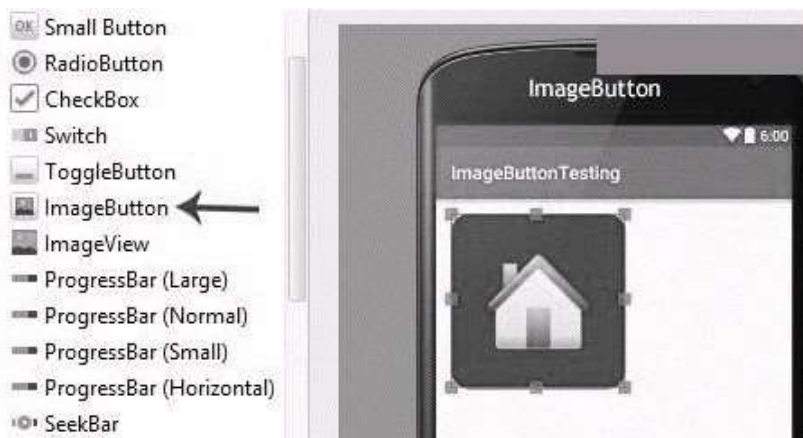
2. Types of buttons–

Buttons can be divided into two categories the first is Buttons with text on, and second is buttons with an image on.

3. Image Button –

A button with images on can contain both an image and a text. Android buttons with images on are also called Image Button. In Android, Image Button is used to display a normal button with a custom image in a button. In simple words we can say, Image Button is a button with an image that can be pressed or clicked by the users. By default it looks like a normal button with the standard button background that changes the color during different button states. An image on the surface of a button is defined within a xml (i.e. layout) by using src attribute or within java class by using setImageResource() method. We can also set an image or custom drawable in the background of the image button . Image Button has all the properties of a normal button so you can easily perform any event like click or any other event which you can perform on a normal button.

Note: Standard button background image is displayed in the background of button whenever you create an image button. To remove that image, you can define your own background image in xml by using background attribute or in java class by using setBackground() method.



4. Toggle Button–

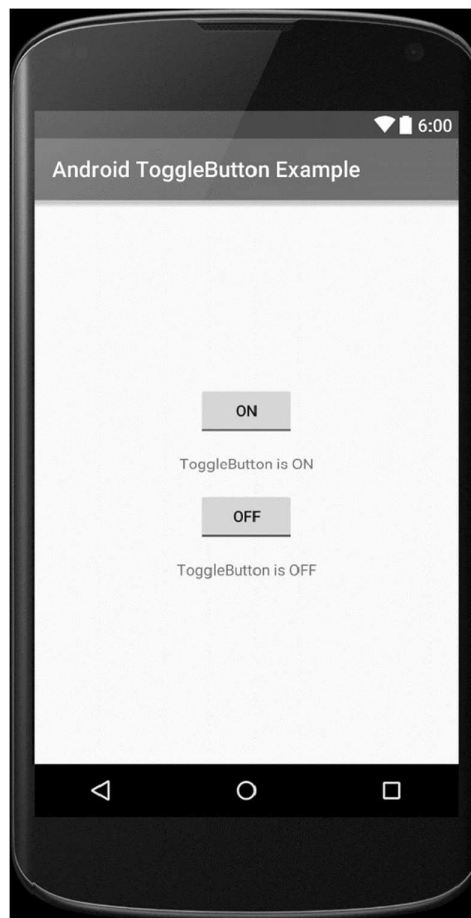
A toggle button allows the user to change a setting between two states. You can add a basic toggle button to your layout with the Toggle Button object. If you need to change a button's state yourself, you can use the Compound

Button.setChecked() or Compound Button.toggle() method. To detect when the user activates the button or switch, create a Compound Button.

OnCheckedChangeListener object and assign it to the button by calling setOnCheckedChangeListener().

It is beneficial if user have to change the setting between two states. It can be used to On/Off Sound, Wi-Fi, Bluetooth etc. By default, the android Toggle Button will be in OFF (Unchecked) state. We can change the default state of Toggle Button by using android:checked attribute. In case, if we want to change the state of Toggle Button to ON (Checked), then we need to set android:checked = “true” in our

XML layout file.



VIII. Resources required (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

IX. Practical related Questions

Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

1. Write a piece of code to set id of the button.
2. How to add image to resources file?
3. List four Android Toggle Button control attributes.

(Space for answers)

Ans 1. `<Button android:id="@+id/simpleButton" />`

X. Exercise

Note: Faculty must ensure that every group of students use different input value.

(Use blank space for answers or attach more pages if needed)

1. Write a program to create a toggle button to display ON / OFF Bluetooth on the display screen.
2. Write a program to create a simple calculator.

(Space for answers)

Ans 1.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
xmlns:android="http://schemas.android.co
m/apk/res/android"
    xmlns:app="http://schemas.android.com
/apk/res-auto"
    xmlns:tools="http://schemas.android.co
m/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ToggleButton
        android:id="@+id/toggle"
        android:layout_width="185dp"
        android:layout_height="wrap_content"

        android:layout_centerHorizontal="tru
e"

        android:layout_marginHorizontal="1
30dp"
        android:layout_marginTop="90dp"
        android:drawableStart="@drawable/s
elector"
        android:textOff="Bluetooth Off"
        android:textOn="Bluetooth ON" />
    <ImageView
        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:id="@+id/imageView"
        android:layout_marginTop="100dp"
        android:layout_below="@+id/toggle"
        android:layout_centerHorizontal="tru
e"

    />
</RelativeLayout>
```

selector.xml

```
<?xml version="1.0" encoding="utf-8"?>
<selector
xmlns:android="http://schemas.android.co
m/apk/res/android">
    <item android:state_checked="true"
        android:drawable="@drawable/bluetooth_
on"/>
    <item android:state_checked="false"
        android:drawable="@drawable/bluetooth_
off"/>
</selector>
```

Java Code

```
package com.example.togglebutton;

import
androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.ImageView;
import android.widget.ToggleButton;

public class MainActivity extends
AppCompatActivity {

    ToggleButton toggleButton;
    ImageView imageView;
    @Override
    protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_ma
in);
```

```

toggleButton=findViewById(R.id.toggleButton);
imageView=findViewById(R.id.imageView);

imageView.setImageDrawable(getResources().getDrawable(R.drawable.bluetooth_off));

toggleButton.setOnClickListener(new View.OnClickListener() {

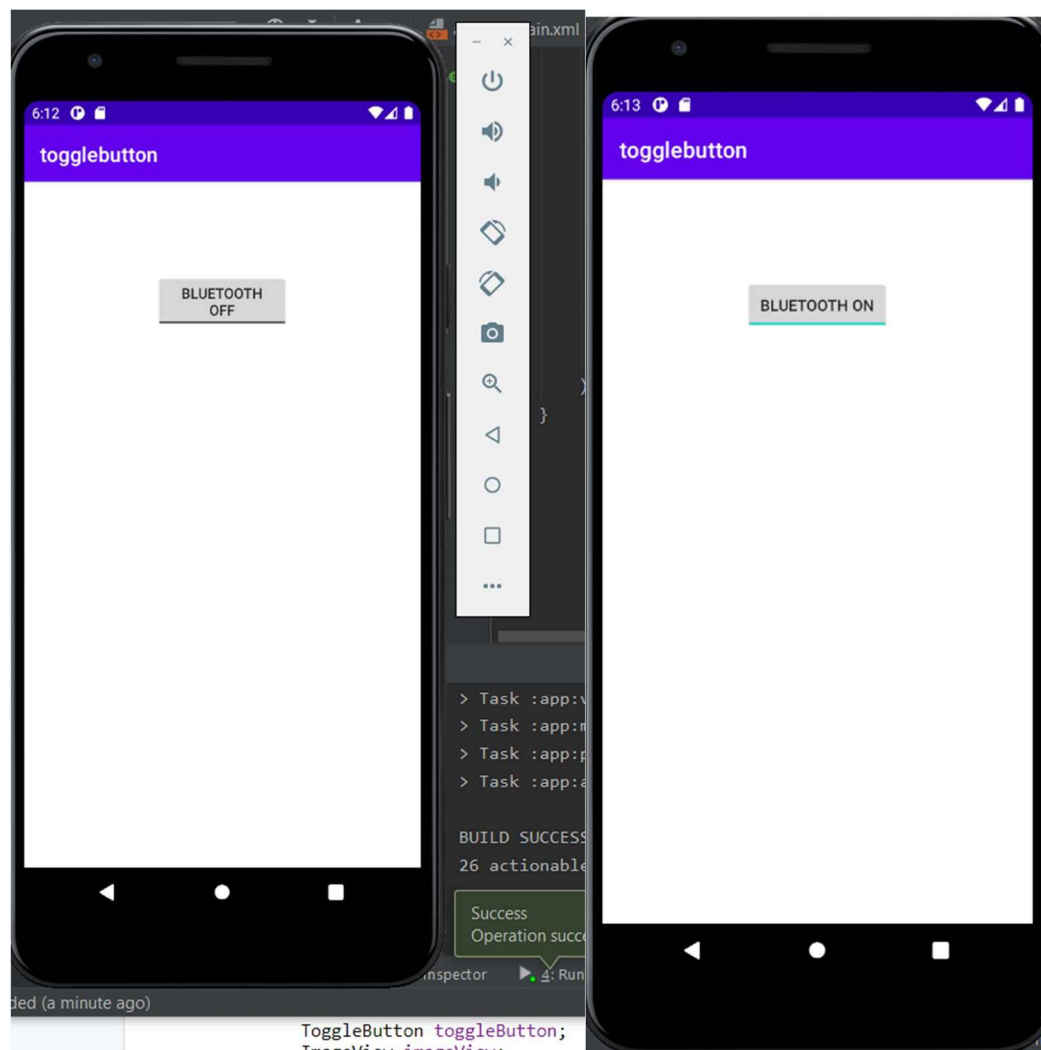
    @Override
    public void onClick(View v) {
        if (toggleButton.isChecked())
        {

```

```

            imageView.setImageDrawable(getResources().getDrawable(R.drawable.bluetooth_on));
        }
        else
        {
            imageView.setImageDrawable(getResources().getDrawable(R.drawable.bluetooth_off));
        }
    }
});
}
}

```



Ans 2.

XML code

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
xmlns:android="http://schemas.android.co
m/apk/res/android"
xmlns:app="http://schemas.android.com
/apk/res-auto"
xmlns:tools="http://schemas.android.co
m/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
```

```
    android:layout_width="match_parent"
"
    android:layout_height="match_parent"
"
    android:orientation="horizontal">
    <Button
        android:id="@+id/btn_1"
        android:layout_width="wrap_conte
nt"
        android:layout_height="wrap_cont
ent"
        android:layout_alignParentLeft="tr
ue"
        android:layout_alignParentStart="t
rue"
        android:layout_below="@+id/edTe
xt1"
        android:layout_marginTop="60dp"

        android:text="1"
        android:textSize="18sp" />
```

```
    <Button
        android:id="@+id/btn_0"
        android:layout_width="wrap_conte
nt"
        android:layout_height="wrap_cont
ent"
        android:layout_below="@+id/btn_
8"
        android:layout_toEndOf="@+id/bt
n_7"
        android:layout_toRightOf="@+id/
btn_7"
        android:text="0"
        android:textSize="18sp" />
```

```
    <Button
        android:id="@+id/btn_9"
```

```
        android:layout_width="wrap_conte
nt"
        android:layout_height="wrap_cont
ent"
        android:layout_below="@+id/btn_
6"
        android:layout_toEndOf="@+id/bt
n_5"
        android:layout_toRightOf="@+id/
btn_5"
        android:text="9"
        android:textSize="18sp" />
```

```
    <Button
        android:id="@+id/btn_8"
        android:layout_width="wrap_conte
nt"
        android:layout_height="wrap_cont
ent"
        android:layout_below="@+id/btn_
5"
        android:layout_toEndOf="@+id/bt
n_7"
        android:layout_toRightOf="@+id/
btn_7"
        android:text="8"
        android:textSize="18sp" />
```

```
    <Button
        android:id="@+id/btn_7"
        android:layout_width="wrap_conte
nt"
        android:layout_height="wrap_cont
ent"
        android:layout_alignLeft="@+id/bt
n_4"
        android:layout_alignStart="@+id/b
tn_4"
        android:layout_below="@+id/btn_
4"
        android:text="7"
        android:textSize="18sp" />
```

```
    <Button
        android:id="@+id/btn_6"
        android:layout_width="wrap_conte
nt"
        android:layout_height="wrap_cont
ent"
        android:layout_alignBaseline="@+
id/btn_5"
        android:layout_alignBottom="@+i
d/btn_5"
```

```

        android:layout_toEndOf="@+id/btn_5"
        android:layout_toRightOf="@+id/btn_5"
        android:text="6"
        android:textSize="18sp" />

<Button
    android:id="@+id/btn_5"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@+id/btn_2"
    android:layout_toEndOf="@+id/btn_4"
    android:layout_toRightOf="@+id/btn_4"
    android:text="5"
    android:textSize="18sp" />

<Button
    android:id="@+id/btn_4"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignLeft="@+id/btn_1"
    android:layout_alignStart="@+id/btn_1"
    android:layout_below="@+id/btn_1"
    android:text="4"
    android:textSize="18sp" />

<Button
    android:id="@+id/btn_3"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignBaseline="@+id/btn_2"
    android:layout_alignBottom="@+id/btn_2"
    android:layout_toEndOf="@+id/btn_2"
    android:layout_toRightOf="@+id/btn_2"
    android:text="3"
    android:textSize="18sp" />

```

```

<Button
    android:id="@+id/btn_2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignBaseline="@+id/btn_1"
    android:layout_alignBottom="@+id/btn_1"
    android:layout_toEndOf="@+id/btn_1"
    android:layout_toRightOf="@+id/btn_1"
    android:text="2"
    android:textSize="18sp" />

<Button
    android:id="@+id/btn_Add"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_above="@+id/btn_6"
    android:layout_alignParentEnd="true"
    android:layout_alignParentRight="true"
    android:backgroundTint="@android:color/darker_gray"
    android:text="+"
    android:textColor="@android:color/background_light"
    android:textSize="18sp" />

<Button
    android:id="@+id/btn_Sub"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignLeft="@+id/btn_Add"
    android:layout_alignStart="@+id/btn_Add"
    android:layout_below="@+id/btn_Add"
    android:backgroundTint="@android:color/darker_gray"
    android:text="-"
    android:textColor="@android:color/background_light"
    android:textSize="18sp" />

```

```

<Button
    android:id="@+id/btn_Mul"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignLeft="@+id/btn_Sub"
    android:layout_alignStart="@+id/btn_Sub"
    android:layout_below="@+id/btn_6"
    android:backgroundTint="@android:color/darker_gray"
    android:text="*"
    android:textColor="@android:color/background_light"
    android:textSize="18sp" />

```

```

<Button
    android:id="@+id/btn_Div"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignLeft="@+id/btn_Mul"
    android:layout_alignStart="@+id/btn_Mul"
    android:layout_below="@+id/btn_9"
    android:backgroundTint="@android:color/darker_gray"
    android:text="/"
    android:textColor="@android:color/background_light"
    android:textSize="18sp" />

```

```

<EditText
    android:id="@+id/edText1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentEnd="true"
    android:layout_alignParentLeft="true"
    android:layout_alignParentRight="true"
    android:layout_alignParentStart="true"
    android:layout_alignParentTop="true"

```

```

    android:layout_marginTop="22dp"
    android:ems="10"
    android:inputType="textPersonName"
    android:textAlignment="textEnd"
    android:textSize="24sp" />

```

```

<Button
    android:id="@+id/btn_calc"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@+id/btn_0"
    android:layout_toEndOf="@+id/btn_0"
    android:layout_toRightOf="@+id/btn_0"
    android:backgroundTint="@android:color/holo_green_light"
    android:text="="
    android:textColor="@android:color/background_light"
    android:textSize="18sp" />

```

```

<Button
    android:id="@+id/btn_dec"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@+id/btn_7"
    android:layout_toLeftOf="@+id/btn_8"
    android:layout_toStartOf="@+id/btn_8"
    android:text="."
    android:textSize="18sp" />

```

```

<Button
    android:id="@+id/btn_clear"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentEnd="true"
    android:layout_alignParentRight="true"
    android:layout_below="@+id/btn_Div"

```

```

        android:backgroundTint="@android:color/holo_blue_dark"
        android:text="clear"
        android:textColor="@android:color/background_light"
        android:textSize="18sp" />

```

```
</RelativeLayout>
```

Java Code

```

package com.example.exp9_2;

import
androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

public class MainActivity extends
AppCompatActivity {
    Button
    btn_1,btn_2,btn_3,btn_4,btn_5,btn_6,btn_
7,btn_8,btn_9,btn_0,btn_Add,btn_Sub,btn
_Mul,btn_Div,btn_calc,btn_dec,btn_clear;
    EditText ed1;

    float Value1, Value2;
    boolean mAddition, mSubtract,
mMultiplication, mDivision ;
    @Override
    protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate( savedInstanceState );
        setContentView(
R.layout.activity_main );
        btn_0 = (Button)
findViewById(R.id.btn_0);
        btn_1 = (Button)
findViewById(R.id.btn_1);
        btn_2 = (Button)
findViewById(R.id.btn_2);
        btn_3 = (Button)
findViewById(R.id.btn_3);
        btn_4 = (Button)
findViewById(R.id.btn_4);
        btn_5 = (Button)

```

```

findViewById(R.id.btn_5);
        btn_6 = (Button)
findViewById(R.id.btn_6);
        btn_7 = (Button)
findViewById(R.id.btn_7);
        btn_8 = (Button)
findViewById(R.id.btn_8);
        btn_9 = (Button)
findViewById(R.id.btn_9);
        btn_Add = (Button)
findViewById(R.id.btn_Add);
        btn_Div = (Button)
findViewById(R.id.btn_Div);
        btn_Sub = (Button)
findViewById(R.id.btn_Sub);
        btn_Mul = (Button)
findViewById(R.id.btn_Mul);
        btn_calc = (Button)
findViewById(R.id.btn_calc);
        btn_dec = (Button)
findViewById(R.id.btn_dec);
        btn_clear = (Button)
findViewById(R.id.btn_clear);
        ed1 = (EditText)
findViewById(R.id.edText1);
        btn_0.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                ed1.setText(ed1.getText()+"0");
            }
        });

        btn_1.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                ed1.setText(ed1.getText()+"1");
            }
        });

        btn_2.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                ed1.setText(ed1.getText()+"2");
            }
        });

        btn_3.setOnClickListener(new
View.OnClickListener() {
            @Override
            public void onClick(View v) {
                ed1.setText(ed1.getText()+"3");
            }
        });

```



```

    }
    });

    btn_4.setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View v) {
        ed1.setText(ed1.getText()+"4");
    }
    });

    btn_5.setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View v) {
        ed1.setText(ed1.getText()+"5");
    }
    });

    btn_6.setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View v) {
        ed1.setText(ed1.getText()+"6");
    }
    });

    btn_7.setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View v) {
        ed1.setText(ed1.getText()+"7");
    }
    });

    btn_8.setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View v) {
        ed1.setText(ed1.getText()+"8");
    }
    });

    btn_9.setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View v) {
        ed1.setText(ed1.getText()+"9");
    }
    });

    btn_dec.setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View v) {
        ed1.setText(ed1.getText()+".");
    }
    });

    btn_Add.setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View v) {
        if (ed1 == null){
            ed1.setText("");
        } else {
            Value1 =
Float.parseFloat(ed1.getText() + "");
            mAddition = true;
            ed1.setText(null);
        }
    }
    });

    btn_Sub.setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View v) {
        Value1 =
Float.parseFloat(ed1.getText() + "");
        mSubtract = true ;
        ed1.setText(null);
    }
    });

    btn_Mul.setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View v) {
        Value1 =
Float.parseFloat(ed1.getText() + "");
        mMultiplication = true ;
        ed1.setText(null);
    }
    });

    btn_Div.setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View v) {
        Value1 =
Float.parseFloat(ed1.getText()+"");
        mDivision = true ;
        ed1.setText(null);
    }
    });

```

```

        btn_calc.setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View v) {
        Value2 =
Float.parseFloat(ed1.getText() + "");

        if (mAddition == true){

            ed1.setText(Value1 + Value2
+ "");
            mAddition=false;
        }

        if (mSubtract == true){
            ed1.setText(Value1 - Value2
+ "");
            mSubtract=false;
        }

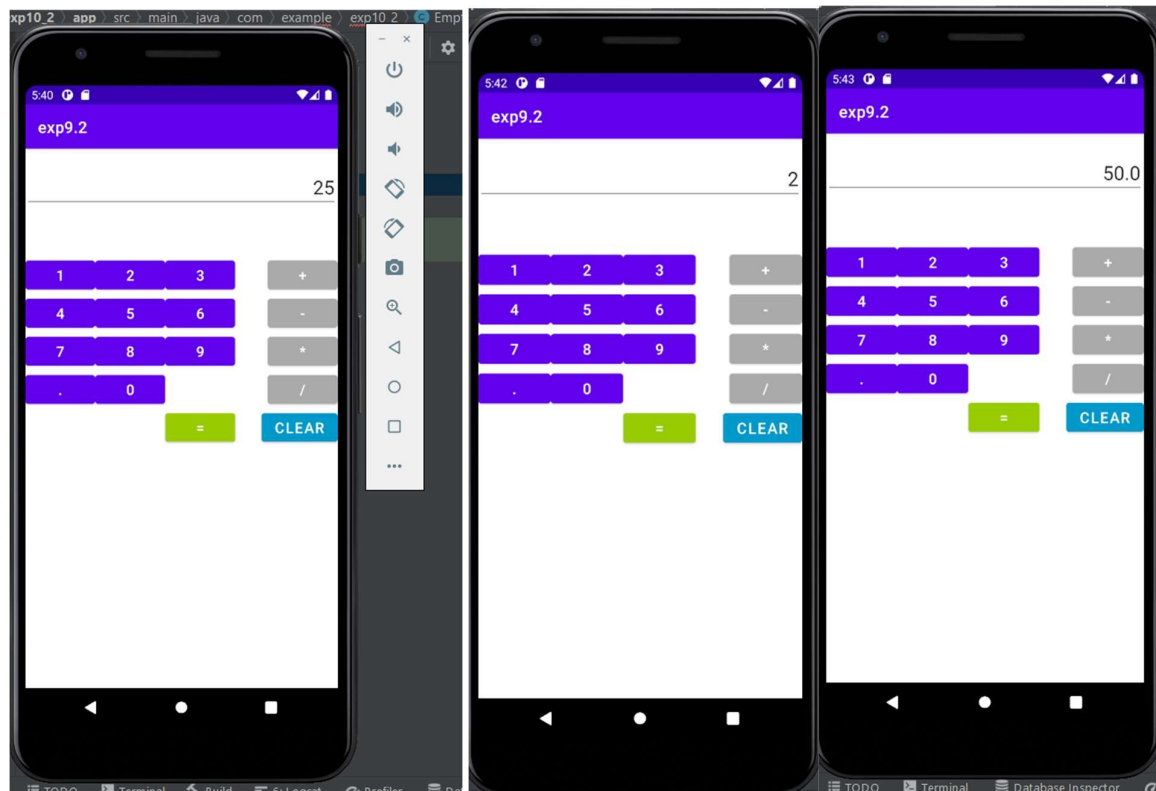
        if (mMultiplication == true){
            ed1.setText(Value1 * Value2
+ "");
            mMultiplication=false;
        }

        if (mDivision == true){
            ed1.setText(Value1 /
Value2+ "");
            mDivision=false;
        }
    }
});

    btn_clear.setOnClickListener(new
View.OnClickListener() {
    @Override
    public void onClick(View v) {
        ed1.setText("");
    }
});
}
}

```

Output



XI. References / Suggestions for further Reading

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf
4. <https://developer.android.com>

XII. Assessment Scheme

Performance indicators		Weightage
Process related (10 Marks)		30%
1.	Logic Formation	10%
2.	Debugging ability	15%
3.	Follow ethical practices	5%
Product related (15 Marks)		70%
4.	Interactive GUI	20%
5.	Answer to Practical related questions	20%
6.	Expected Output	20%
7.	Timely Submission	10%
Total (25 Marks)		100%

List of student Team Members

- 1.....
- 2.....
- 3.....
- 4.....

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

Practical No. 10: Develop a program to implement login window using above UI controls.

I. Practical Significance

In this practical, all the previous UI controls in android like Text View, Edit Text Buttons which are studied are implemented in this practical. Events are also handled on the android UI controls used in the practical.

II. Relevant Program Outcomes (POs)

PO 1. Basic knowledge

PO 2. Discipline knowledge PO 4. Engineering tools

PO 7. Ethics

PO 10. Life-long learning

III. Competency and Practical Skills

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to use the layout managers.
2. Able to develop android UI controls to create login window without using databases.

IV. Relevant Course Outcome(s)

1. Develop rich user Interfaces by using layouts and controls.
2. Use User Interface components for android application development.

V. Practical Outcome (PrOs)

Develop a program to implement Button, Image Button and Toggle Button.

VI. Relevant Affective Domain Related Outcome(s)

1. Work collaboratively in team
2. Follow ethical Practices.

VII. Minimum Theoretical Background

A login application is the screen asking your credentials to login to some particular application. You might have seen it when logging into facebook, twitter etc. Define two Text View asking username and password of the user. The password Text View must have input Type set to password. Its syntax is given below

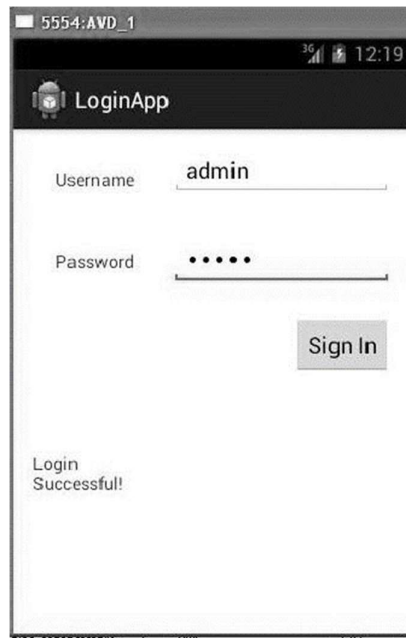
```
<EditText
    android:id = "@+id/editText2"
    android:layout_width = "wrap_content"
    android:layout_height = "wrap_content"
    android:inputType = "textPassword" />

<EditText
    android:id = "@+id/editText1"
    android:layout_width = "wrap_content"
    android:layout_height = "wrap_content"
/>
```

Define a button with login text and set its onClick Property. After that define the

```
<Button
    android:id = "@+id/button1"
    android:layout_width = "wrap_content"
    android:layout_height = "wrap_content"
    android:onClick = "login"
    android:text = "@string/Login"
/>
```

function mentioned in the onClick property in the java file.



VIII. Resources required (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

IX. Practical related Questions

Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

1. Name the file in which respective XML components can be added.
2. List all the UI components which can be used to develop login window.

(Space for answers)

1. xml file in android. The AndroidManifest.xml file contains information of your package, including components of the application such as activities, services, broadcast receivers, content providers etc.
2. . UI components used to develop login window are as follows:
 - i. TextView.
 - ii. EditText.
 - iii. Button.

X. Exercise

Note: Faculty must ensure that every group of students use different input value.

(Use blank space for answers or attach more pages if needed)

1. Write a program to create a login form for a social networking site.
2. Write a program to create a login form for student registration system

(Space for answers)

An 1.

```
package com.example.exp10_1;

import
androidx.appcompat.app.AppCo
mpatActivity;
import android.content.Intent;
import android.os.Bundle;
```

```
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import
android.widget.ToggleButton;

import android.os.Bundle;
```

	});
public class MainActivity extends	
AppCompatActivity {	}
	}
EditText username;	
String Username;	<?xml version="1.0"
Button login;	encoding="utf-8"?>
@Override	<RelativeLayout
protected void onCreate(Bundle	xmlns:android="http://schemas.a
savedInstanceState) {	ndroid.com/apk/res/android"
super.onCreate(savedInstanceState)	xmlns:app="http://schemas.andro
);	id.com/apk/res-auto"
setContentView(R.layout.activity_	xmlns:tools="http://schemas.andr
main);	oid.com/tools"
username=findViewById(R.id.Use	android:layout_width="match_pa
rname_ET);	rent"
login=findViewById(R.id.Login);	android:layout_height="match_p
	arent"
login.setOnClickListener(new	tools:context=".MainActivity">
View.OnClickListener()	
{	<TextView
@Override	android:id="@+id/Name"
public void onClick(View	
v) {	android:layout_width="wrap_con
	tent"
Username=username.getText().toS	android:layout_height="wrap_co
tring();	ntent"
	android:text="Name"
Toast.makeText(MainActivity.this	android:layout_marginTop="40d
,"Welcome	p"
"+Username,Toast.LENGTH_SH	android:layout_centerHorizontal=
ORT).show();	"true"
Intent intent=new	android:textSize="20dp"
Intent(MainActivity.this,Empty_A	/>
ctivity.class);	<EditText
//	
intent.putExtra("username",User	android:id="@+id/Name_ET"
name);	
startActivity(intent);	
}	

```
android:layout_width="match_parent"
```

```
android:layout_height="wrap_content"
    android:ems="10"
```

```
android:inputType="textPersonName"
```

```
android:layout_below="@+id/Name"
```

```
android:layout_marginTop="20dp"
/>
```

```
<TextView
    android:id="@+id/Age"
```

```
android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
    android:text="Age"
```

```
android:layout_marginTop="40dp"
```

```
android:layout_centerHorizontal="true"
```

```
android:layout_below="@+id/Name_ET"
    android:textSize="20dp"
/>
```

```
<EditText
    android:id="@+id/Age_ET"
```

```
android:layout_width="match_parent"
```

```
android:layout_height="wrap_content"
    android:ems="10"
```

```
android:inputType="textPersonName"
```

```
android:layout_marginTop="20dp"
```

```
android:layout_below="@+id/Age"
```

```
/>
<TextView
    android:id="@+id/Branch"
```

```
android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
    android:text="Branch"
```

```
android:layout_marginTop="40dp"
```

```
android:layout_centerHorizontal="true"
```

```
android:layout_below="@+id/Age_ET"
    android:textSize="20dp"
```

```
/>
<EditText
```

```
android:id="@+id/Branch_ET"
```

```
android:layout_width="match_parent"
```

```
android:layout_height="wrap_content"
    android:ems="10"
```

```
android:inputType="textPersonName"
```



```

android:layout_marginTop="20d
p"

android:layout_below="@+id/Bra
nch"
/>
<TextView
    android:id="@+id/Semester"

android:layout_width="wrap_con
tent"

android:layout_height="wrap_co
ntent"
    android:text="Semester"

android:layout_marginTop="40d
p"

android:layout_centerHorizontal=
"true"

android:layout_below="@+id/Bra
nch_ET"
    android:textSize="20dp"
/>
<EditText

android:id="@+id/Semester_ET"

android:layout_width="match_pa
rent"

android:layout_height="wrap_co
ntent"
    android:ems="10"

android:inputType="textPersonN
ame"

android:layout_marginTop="20d
p"

```

Output:

```

android:layout_below="@+id/Se
mester"
/>
<Button
    android:id="@+id/Login"

android:layout_width="wrap_con
tent"

android:layout_height="wrap_co
ntent"
    android:text="Login"

android:layout_below="@+id/Se
mester_ET"

android:layout_marginTop="40d
p"

android:layout_centerHorizontal=
"true"
/>
<TextView
    android:id="@+id/Login"

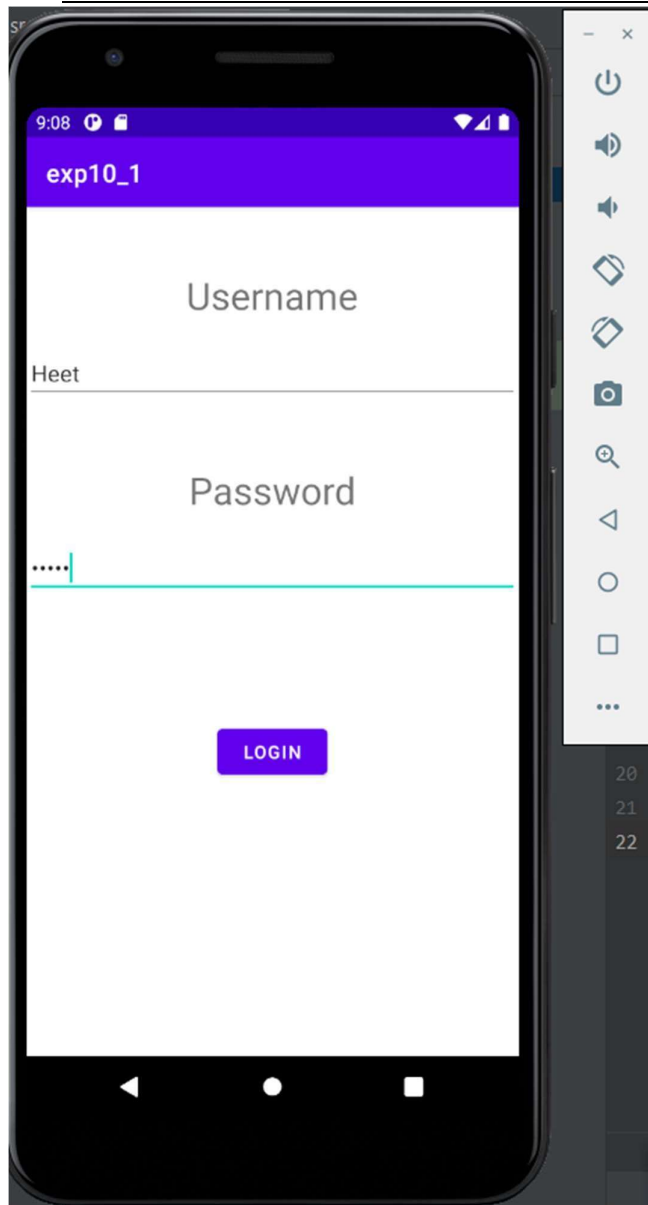
android:layout_width="wrap_con
tent"

android:layout_height="wrap_co
ntent"
    android:text="Student Login"
    android:textSize="40dp"

android:layout_centerHorizontal=
"true"

android:textColor="@color/teal_2
00"
/>
</RelativeLayout>

```



Ans 2.

XML file

```
<?xml version="1.0"
encoding="utf-8"?>
<RelativeLayout
xmlns:android="http://schemas.a
ndroid.com/apk/res/android"

xmlns:app="http://schemas.andro
id.com/apk/res-auto"

xmlns:tools="http://schemas.andr
oid.com/tools"
```

```
android:layout_width="match_pa
rent"
```

```
android:layout_height="match_p
arent"
tools:context=".MainActivity">
```

```
<TextView
android:id="@+id/Name"
```

```
android:layout_width="wrap_con
```

<pre> tent" android:layout_height="wrap_co ntent" android:text="Name" android:layout_marginTop="40d p" android:layout_centerHorizontal= "true" android:textSize="20dp" /> <EditText android:id="@+id/Name_ET" android:layout_width="match_pa rent" android:layout_height="wrap_co ntent" android:ems="10" android:inputType="textPersonN ame" android:layout_below="@+id/Na me" android:layout_marginTop="20d p" /> <TextView android:id="@+id/Age" android:layout_width="wrap_con tent" android:layout_height="wrap_co ntent" android:text="Age" android:layout_marginTop="40d p" </pre>	<pre> android:layout_centerHorizontal= "true" android:layout_below="@+id/Na me_ET" android:textSize="20dp" /> <EditText android:id="@+id/Age_ET" android:layout_width="match_pa rent" android:layout_height="wrap_co ntent" android:ems="10" android:inputType="textPersonN ame" android:layout_marginTop="20d p" android:layout_below="@+id/Age " /> <TextView android:id="@+id/Branch" android:layout_width="wrap_con tent" android:layout_height="wrap_co ntent" android:text="Branch" android:layout_marginTop="40d p" android:layout_centerHorizontal= "true" android:layout_below="@+id/Age _ET" </pre>
---	--

```

        android:textSize="20dp"

    />
    <EditText

    android:id="@+id/Branch_ET"

    android:layout_width="match_pa
    rent"

    android:layout_height="wrap_co
    ntent"

        android:ems="10"

    android:inputType="textPersonN
    ame"

    android:layout_marginTop="20d
    p"

    android:layout_below="@+id/Bra
    nch"

    />
    <TextView
        android:id="@+id/Semester"

    android:layout_width="wrap_con
    tent"

    android:layout_height="wrap_co
    ntent"

        android:text="Semester"

    android:layout_marginTop="40d
    p"

    android:layout_centerHorizontal=
    "true"

    android:layout_below="@+id/Bra
    nch_ET"

        android:textSize="20dp"
    />
    <EditText

```

```

    android:id="@+id/Semester_ET"

    android:layout_width="match_pa
    rent"

    android:layout_height="wrap_co
    ntent"

        android:ems="10"

    android:inputType="textPersonN
    ame"

    android:layout_marginTop="20d
    p"

    android:layout_below="@+id/Se
    mester"

    />
    <Button
        android:id="@+id/Login"

    android:layout_width="wrap_con
    tent"

    android:layout_height="wrap_co
    ntent"

        android:text="Login"

    android:layout_below="@+id/Se
    mester_ET"

    android:layout_marginTop="40d
    p"

    android:layout_centerHorizontal=
    "true"

    />
    <TextView
        android:id="@+id/Login"

    android:layout_width="wrap_con
    tent"

    android:layout_height="wrap_co

```

```

    intent"
        android:text="Student Login"
        android:textSize="40dp"

    android:layout_centerHorizontal=
    "true"

    android:textColor="@color/teal_2
    00"
    />
</RelativeLayout>

```

Java File:

```

package com.example.exp10_2;

import
androidx.appcompat.app.AppCo
mpatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import
android.widget.ToggleButton;

import android.os.Bundle;

public class MainActivity extends
AppCompatActivity {

    EditText
    name_et,age_et,branch_et,sem_et
    ;
    String
    Username,Age,Branch,Semester;
    Button login;

    @Override
    protected void onCreate(Bundle
    savedInstanceState) {

    super.onCreate(savedInstanceState

```

```

    e);

    setContentView(R.layout.activity_
    main);

    name_et=findViewById(R.id.Nam
    e_ET);

    age_et=findViewById(R.id.Age_E
    T);

    branch_et=findViewById(R.id.Br
    anch_ET);

    sem_et=findViewById(R.id.Semes
    ter_ET);
    login
    =findViewById(R.id.Login);

    login.setOnClickListener(new
    View.OnClickListener()
    {
        @Override
        public void onClick(View
    v) {

    Username=name_et.getText().toSt
    ring();

    Age=age_et.getText().toString();

    Branch=branch_et.getText().toStr
    ing();

    Semester=sem_et.getText().toStrin
    g();

    Toast.makeText(MainActivity.this
    ,"Welcome "+Username
    ,Toast.LENGTH_SHORT).show()
    ;

    Toast.makeText(MainActivity.this
    ,"Age "+Age
    ,Toast.LENGTH_SHORT).show()

```

```

;

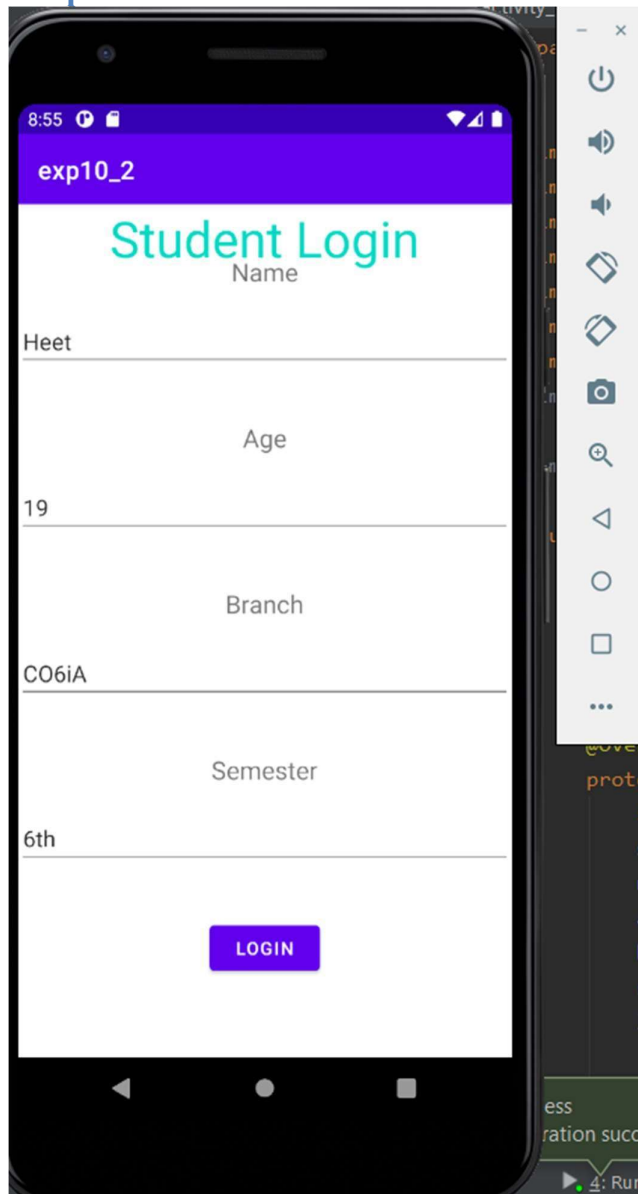
Toast.makeText(MainActivity.this
,"Branch "+Branch
,Toast.LENGTH_SHORT).show()
;

Toast.makeText(MainActivity.this
,"Semester "+Semester
,Toast.LENGTH_SHORT).show()
;

Intent intent=new
Intent(MainActivity.this,Empty_Activity.class);
//
intent.putExtra("username",Username);
startActivity(intent);

}
});
}
}

```

Output:

XI. References / Suggestions for further Reading

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf
4. <https://developer.android.com>

XII. Assessment Scheme

Performance indicators		Weightage
Process related (10 Marks)		30%
1.	Logic Formation	10%
2.	Debugging ability	15%
3.	Follow ethical practices	5%
Product related (15 Marks)		70%
4.	Interactive GUI	20%
5.	Answer to Practical related questions	20%
6.	Expected Output	20%
7.	Timely Submission	10%
Total (25 Marks)		100%

List of student Team Members

- 1.....
- 2.....
- 3.....
- 4.....

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

Practical No. 12: Develop a program to implement Radio Button and Radio Group.

I. Practical Significance

Radio Buttons are used when we need to select only one item from a list of presented items. If Radio Buttons are in Radio Group, when one Radio Button within a group is selected, all others are automatically deselected.

II. Relevant Program Outcomes (POs)

PO 2. Discipline knowledge

PO 3. Experiments and practice

PO 4. Engineering tools

III. Competency and Practical Skills

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to develop an application using Radio Button and Radio Group controls.

IV. Relevant Course Outcome(s)

1. Develop rich user Interfaces by using layouts and controls.
2. Use User Interface components for android application development.

V. Practical Outcomes (PrOs)

Use Develop a program to implement Radio Button and Radio Group.

VI. Relevant Affective Domain Related Outcome(s)

1. Work collaboratively in team
2. Follow ethical practices

VII. Minimum Theoretical Background

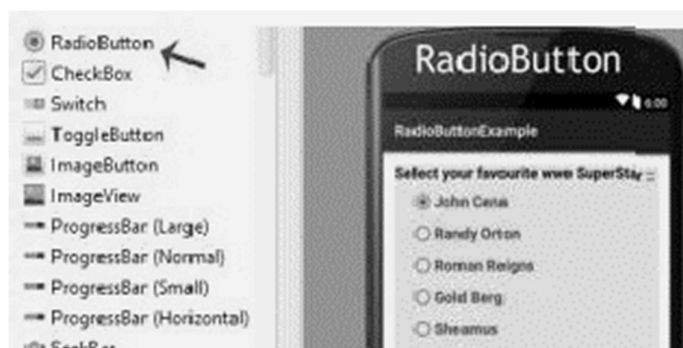
Radio Button is generally used with Radio Group. **Radio Group** is a set of radio buttons, marking one radio button as checked makes all other radio buttons as unchecked. A radio button consists of two states – checked and unchecked. Clicking an unchecked button changes its state to “checked” state and “unchecked” for the previously selected radio button. To toggle a checked state to unchecked state, we need to choose another item.

Following are the important attributes related to Radio Group control.

1. **android:checkedButton** : This is the id of child radio button that should be checked by default within this radio group.
2. **android:orientation** : This property on the Radio group defines the orientation to position its child view consisting of Radio Buttons.
Following are the few methods of radio button:

1. **check(id)**: This sets the selection to the radio button whose identifier is passed in parameter. -1 is used as the selection identifier to clear the selection.
2. **clearCheck()** : It clears the selection. When the selection is cleared, no radio button in this group is selected and `getCheckedRadioButtonId()` returns null.
3. **getCheckedRadioButtonId()** : It returns the identifier of the selected radio button in this group. If its empty selection, the returned value is -1.

4. **setOnCheckedChangeListener()** : This registers a callback to be invoked when the checked radio button changes in this group. We must supply instance of Radio Group. OnCheckedChangeListener to setOnCheckedChangeListener() method



VIII. Resources used (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

IX. Practical related Questions

Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

- Write xml tag to create a Radio button.
- Write the purpose of Radio Button
- List different methods of Radio Button

(Space for answers)

- ```
<RadioButton android:id="@+id/radio_pirates"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:text="@string/pirates"
 android:onClick="onRadioButtonClicked"/>
```
- Radio buttons allow the user to select one option from a set. You should use radio buttons for optional sets that are mutually exclusive if you think that the user needs to see all available options side-by-side.

### 3. Methods of Radio Button:

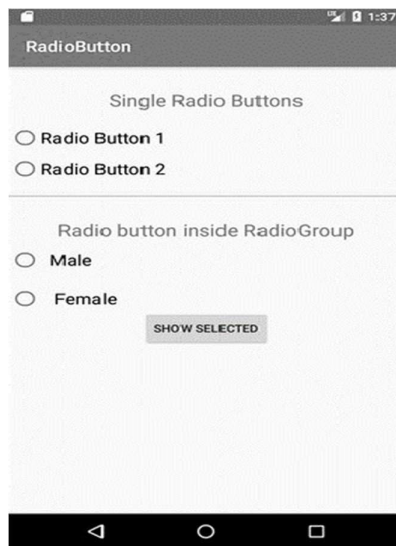
- i. isChecked()
- ii. setChecked()
- iii. getText()
- iv. setText()

### X. Exercise

**Note:** Faculty must ensure that every group of students use different examples.

(Use blank space for answers or attach more pages if needed)

1. Write a program to show the following output. First two radio buttons are without using radio group and next two radio buttons are using radio group. Note



## (Space for answers)

Ans 1.

XML Code:

```

<?xml version="1.0"
encoding="utf-8"?>
<RelativeLayout
xmlns:android="http://schemas.
as.android.com/apk/res/andro
id"

xmlns:app="http://schemas.a
ndroid.com/apk/res-auto"

xmlns:tools="http://schemas.
android.com/tools"

android:layout_width="matc
h_parent"

android:layout_height="matc
h_parent"

tools:context=".MainActivity"
">

 <TextView

android:id="@+id/textv"

android:layout_width="wrap
_content"

android:layout_height="wrap
_content"
 android:text="Single
Radio Buttons"

android:textColor="#000000"
"

android:visibility="visible"

android:layout_centerHorizo
ntal="true"
 android:textSize="20dp"

android:layout_marginTop="
20dp"
 />
 <RadioButton

```

```

android:id="@+id/radioButt
on1"

android:layout_width="wrap
_content"

android:layout_height="wrap
_content"
 android:text="Radio
Button 1"

android:layout_marginTop="
80dp"
 android:textSize="40dp"
 />
 <RadioButton

android:id="@+id/radioButt
on2"

android:layout_width="wrap
_content"

android:layout_height="wrap
_content"
 android:text="Radio
Button 2"

android:layout_marginTop="
40dp"

android:layout_below="@+id
/radioButton1"
 android:textSize="40dp"
 />
 <RadioGroup

android:layout_below="@+id
/radioButton2"

android:layout_width="matc
h_parent"

android:layout_height="200d
p"

android:layout_marginTop="

```

```

50dp"

android:id="@+id/radioGroup"
">

 <TextView

android:id="@+id/textvr"

android:layout_width="wrap
_content"

android:layout_height="wrap
_content"
 android:text="Radio
button inside RadioGroup"

android:textSize="20dp"

android:textColor="#000000"
"

android:visibility="visible"

android:layout_marginTop="
25dp"
 />
 <RadioButton

android:id="@+id/male"

android:layout_width="wrap
_content"

android:layout_height="wrap
_content"
 android:text="Male"

android:checked="true"

android:textSize="40dp"
 />
 <RadioButton

android:id="@+id/female"

android:layout_width="wrap
_content"

android:layout_height="wrap
_content"

android:text="Female"

```

```

android:checked="true"

android:textSize="40dp"
 />
 </RadioGroup>
 <Button

android:id="@+id/Select"

android:layout_width="wrap
_content"

android:layout_height="wrap
_content"
 android:text="Select"

android:layout_centerHorizo
ntal="true"

android:layout_below="@+id
/radioGroup"
 />
</RelativeLayout>

```

**Java Code:**

```

package com.example.exp12;

import
androidx.appcompat.app.App
CompatActivity;

import android.os.Bundle;
import android.view.View;
import
android.widget.Button;
import
android.widget.RadioButton;
import
android.widget.RadioGroup;
import android.widget.Toast;

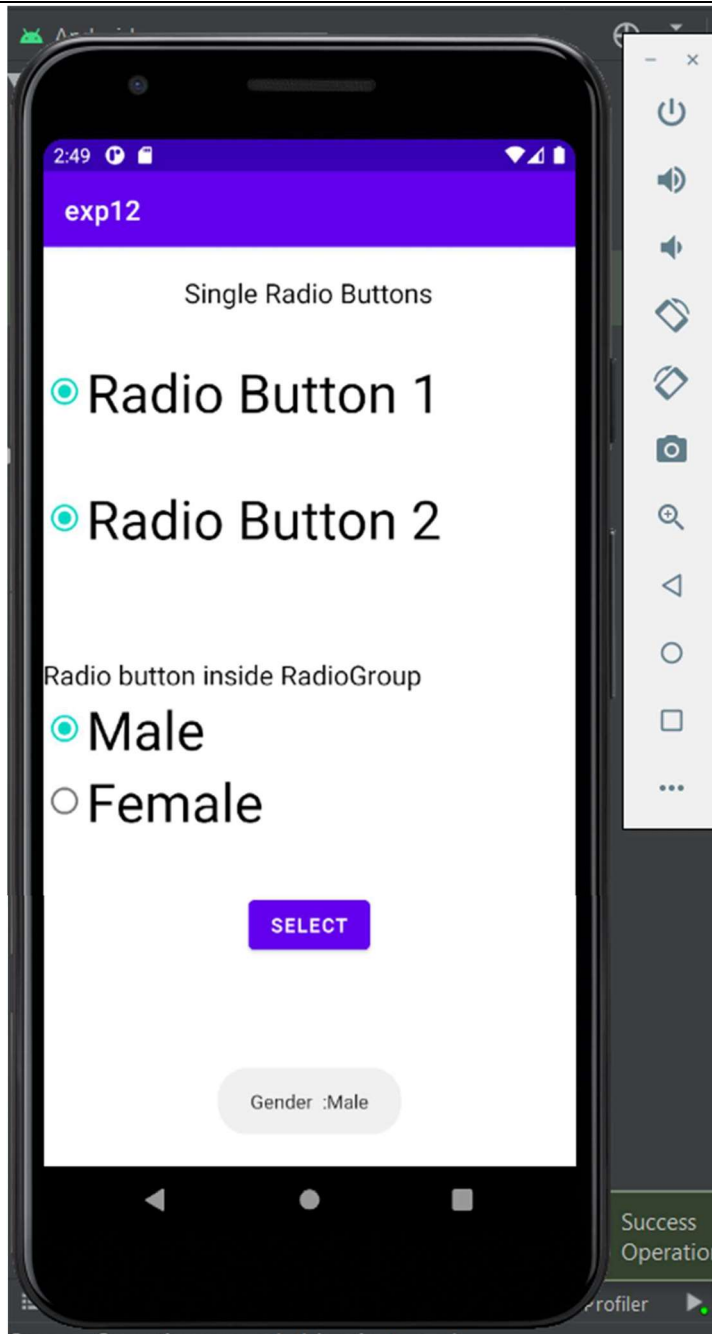
public class MainActivity
extends AppCompatActivity {

 RadioButton
 radgender,rad1,rad2;
 RadioGroup radSelect;
 String gender;
 Button select;

 @Override
 protected void
 onCreate(Bundle

```

|                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> savedInstanceState) {  super.onCreate(savedInstanceState);  setContentView(R.layout.activity_main);  radSelect=findViewById(R.id.radioGroup);  rad1=findViewById(R.id.radioButton1);  rad2=findViewById(R.id.radioButton2);  select=findViewById(R.id.Select);  select.setOnClickListener(     new View.OnClickListener() {         public         void onClick(View v) {  select();          }     } ); } public void select() </pre> | <pre> {     int radioID= radSelect.getCheckedRadioButtonId();  radgender=findViewById(radioID);  gender=radgender.getText().toString().trim();     if(rad1.isChecked())     {         Toast.makeText(this, "Radio button 1 is selected", Toast.LENGTH_SHORT).show();     }     if(rad2.isChecked())     {         Toast.makeText(this, "Radio button 2 is selected", Toast.LENGTH_SHORT).show();     }     Toast.makeText(this, "Gender :" + gender, Toast.LENGTH_SHORT).show(); } } </pre> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



**XI. References / Suggestions for further Reading**

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. [https://www.tutorialspoint.com/android/android\\_advanced\\_tutorial.pdf](https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf)
4. <https://developer.android.com>

**XII. Assessment Scheme**

| Performance indicators            |                                       | Weightage   |
|-----------------------------------|---------------------------------------|-------------|
| <b>Process related (10 Marks)</b> |                                       | <b>30%</b>  |
| 1.                                | Logic Formation                       | 10%         |
| 2.                                | Debugging ability                     | 15%         |
| 3.                                | Follow ethical practices              | 5%          |
| <b>Product related (15 Marks)</b> |                                       | <b>70%</b>  |
| 4.                                | Interactive GUI                       | 20%         |
| 5.                                | Answer to Practical related questions | 20%         |
| 6.                                | Expected Output                       | 20%         |
| 7.                                | Timely Submission                     | 10%         |
| <b>Total (25 Marks)</b>           |                                       | <b>100%</b> |

***List of students/Team Members***

1. ....
2. ....
- 3.....
- 4.....

| Marks Obtained         |                        |            | Dated signature<br>of<br>Teacher |
|------------------------|------------------------|------------|----------------------------------|
| Process<br>Related(10) | Product<br>Related(15) | Total (25) |                                  |
|                        |                        |            |                                  |

## **Practical No. 13: Develop a program to implement Progress Bar**

### **I. Practical Significance**

Progress bars are used to show progress of a task. For example, when you are uploading or downloading something from the internet, it is better to show the progress of download/upload to the user. In android there is a class called Progress Dialog that allows you to create progress bar.

### **II. Relevant Program Outcomes (POs)**

PO 2. Discipline knowledge

PO 3. Experiments and practice

PO 4. Engineering tools

### **III. Competency and Skills**

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to develop an application using Progress bar.

### **IV. Relevant Course Outcome(s)**

1. Develop rich user Interfaces by using layouts and controls.
2. Use User Interface components for android application development.

### **V. Practical Outcomes (PrOs)**

Develop a program to implement Progress Bar

### **VI. Relevant Affective Domain Related Outcome(s)**

1. Work collaboratively in team
2. Follow ethical practices

### **VII. Minimum Theoretical Background**

A user interface element that indicates the progress of an operation. For a visual overview of the difference between determinate and indeterminate progress modes, see Progress & activity. Display progress bars to a user in a non-interruptive way.

Progress bar supports two modes to represent progress: determinate and indeterminate.

#### **Indeterminate Progress**

Use indeterminate mode for the progress bar when you do not know how long an operation will take. Indeterminate mode is the default for progress bar and shows a cyclic animation without a specific amount of progress indicated.

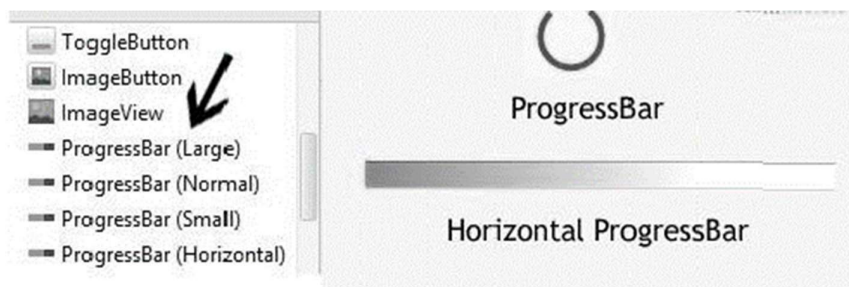
#### **Determinate Progress**

Use determinate mode for the progress bar when you want to show that a specific quantity of progress has occurred. For example, the percent remaining of a file being retrieved, the amount records in a batch written to database, or the percent remaining of an audio file that is playing.

**Progress Dialog** is a class that allows you to create progress bar. In order to do this, you need to instantiate an object of this class. Its syntax is.

ProgressDialog dialog=new ProgressDialog(this);





### VIII. Resources required (Additional)

| Sr. No. | Instrument /Object                                               | Specification | Quantity | Remarks                               |
|---------|------------------------------------------------------------------|---------------|----------|---------------------------------------|
| 1       | Android enabled smartphone / Android version supporting emulator | 2 GB RAM      | 1        | Data cable is mandatory for emulators |

### IX. Practical related Questions

*Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.*

1. State different methods to update the percentage of progress displayed.
2. Write an xml tag for the determinate progress bar.
3. List different progress bar styles provide by the system.

(Space for answers)

**Ans1)** You can update the percentage of progress displayed by using the **setProgress(int)** method, or by calling **incrementProgressBy(int)** to increase the current progress completed by a specified amount. By default, the progress bar is full when the progress value reaches 100. You can adjust this default by setting the **android:max** attribute.

**Ans 2)** `<ProgressBar  
 android:id="@+id/indeterminateBar"  
 android:layout_width="wrap_content"  
 android:layout_height="wrap_content"  
 />`

**Ans 3)** XML attributes

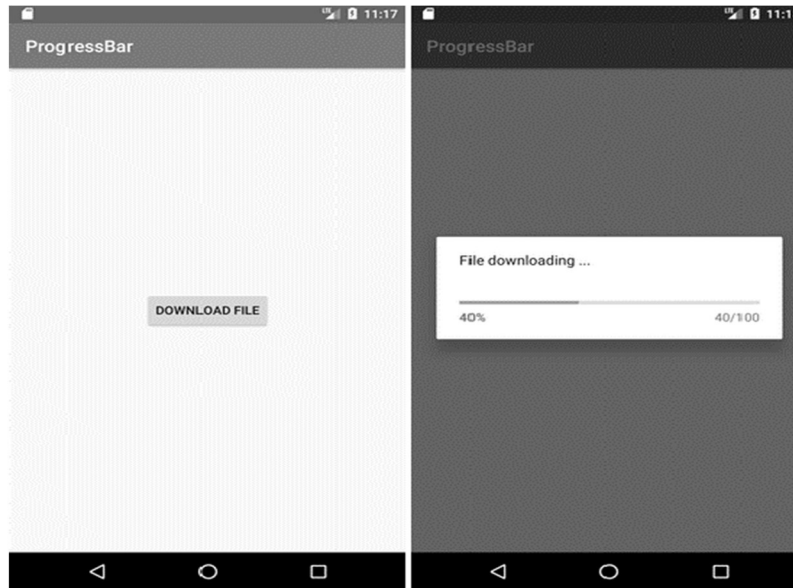
1. **android:indeterminateTintMode**
  2. **android:interpolator**
  3. **android:max**
  4. **android:maxHeight**
  5. **android:maxLength**
  6. **android:min**
- android:minHeight**

**X. Exercise**

**Note:** Below given are few sample questions for reference. Teachers must design different questions for practice.

(Use blank space for answers or attach more pages if needed)

1. Write a program to display circular progress bar.
2. Write a program to show the following output.



(Space for answers)

Ans 1:

**XML CODE**

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
 xmlns:android="http://schemas.android.com/apk/res/android"
 xmlns:app="http://schemas.android.com/apk/res-auto"
 xmlns:tools="http://schemas.android.com/tools"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 tools:context=".MainActivity">

 <Button
 android:id="@+id/btnDownloadFile"
 android:layout_width="wrap_content"
 "
 android:layout_height="wrap_content"
 "
 android:text="Download File"
 android:layout_centerInParent="true"
 />

</RelativeLayout>
```

**JAVA CODE**

```
package com.example.progprogress;

import
androidx.appcompat.app.AppCompatActivity;
import android.app.ProgressDialog;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.os.Bundle;

public class MainActivity extends
AppCompatActivity {
 Button btnDownloadFile;
 ProgressDialog progressDialog;
 @Override
 protected void onCreate(Bundle
savedInstanceState) {
 super.onCreate(savedInstanceState);
```

```

 setContentView(R.layout.activity_main);
 btnDownloadFile =
 findViewById(R.id.btnDownloadFile);
 btnDownloadFile.setOnClickListener(
 new View.OnClickListener() {
 @Override
 public void onClick(View v) {
 progressDialog = new
 ProgressDialog (MainActivity.this);
 progressDialog.setProgressStyle(
 ProgressDialog.STYLE_HORIZONTAL);
 progressDialog.setTitle("File
 Downloading");
 progressDialog.setProgress(100)
 ;
 Thread t = new Thread(new
 Runnable() {
 @Override
 public void run() {
 int progress = 0;

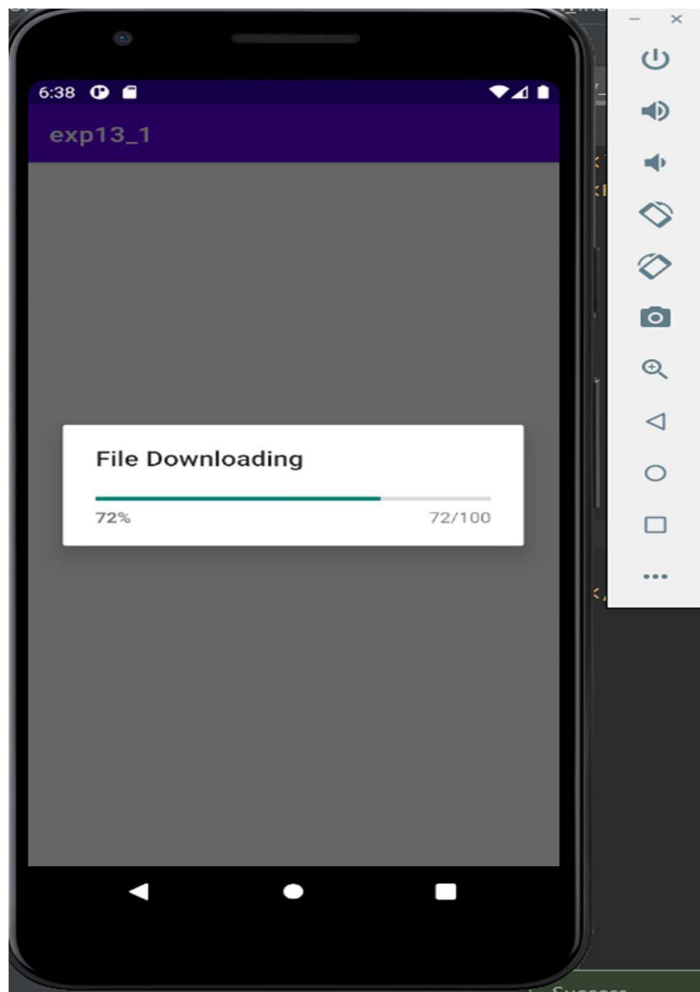
```

```

 while (progress <=100){
 try {
 progressDialog.setProgress(
 progress);
 progress++;
 Thread.sleep(200);
 } catch (Exception ex){
 }
 }
 progressDialog.dismiss();
 }
 });
 t.start();
 progressDialog.show();
 }
 }
 }
 }
}

```

## OUTPUT



**XI. References / Suggestions for further Reading**

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. [https://www.tutorialspoint.com/android/android\\_advanced\\_tutorial.pdf](https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf)
4. <https://developer.android.com>

**XII. Assessment Scheme**

Performance indicators		Weightage
<b>Process related (10 Marks)</b>		<b>30%</b>
1.	Logic Formation	10%
2.	Debugging ability	15%
3.	Follow ethical practices	5%
<b>Product related (15 Marks)</b>		<b>70%</b>
4.	Interactive GUI	20%
5.	Answer to Practical related questions	20%
6.	Expected Output	20%
7.	Timely Submission	10%
<b>Total (25 Marks)</b>		<b>100%</b>

**List of students/Team Members**

1. ....
2. ....
- 3.....
- 4.....

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

## **Practical No. 14: Develop a program to implement List View, Grid View, Image View and Scroll View.**

### **I. Practical Significance**

A View occupies a rectangular area on the screen and is responsible for drawing and event handling. View is the base class for widgets, which are used to create interactive UI components (buttons, text fields, etc.). The View Group subclass is the base class for layouts, which are invisible containers that hold other Views (or other ViewGroups) and define their layout properties.

### **II. Relevant Program Outcomes (POs)**

PO 1. Basic knowledge

PO 2- Discipline knowledge

PO 3- Experiments and practice

PO 4- Engineering tools

### **III. Competency and Skills**

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to develop an application using list view
2. Able to develop an application using grid view
3. Able to develop an application using image view
4. Able to develop an application using scroll view

### **IV. Relevant Course Outcome(s)**

1. Develop rich user Interfaces by using layouts and controls.
2. Use User Interface components for android application development.

### **V. Practical Outcome (PrOs)**

Develop a program to implement List View, Grid View, Image View and Scroll View.

### **VI. Relevant Affective Domain Related Outcome(s)**

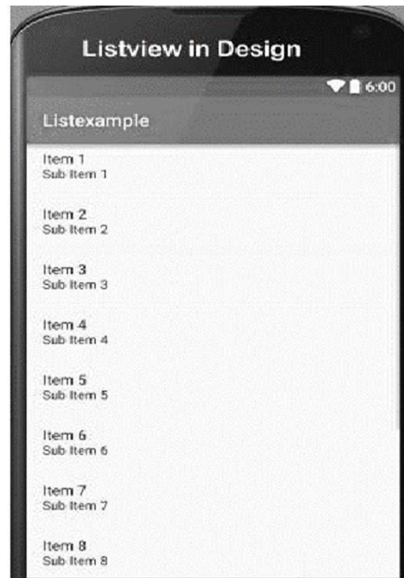
1. Work collaboratively in team
2. Follow ethical practices

### **VII. Minimum Theoretical Background**

#### **List View**

List of scrollable items can be displayed in Android using List View. It helps you to displaying the data in the form of a scrollable list. Users can then select any list item by clicking on it. List View is default scrollable so we do not need to use scroll View or anything else with List View.

List View is widely used in android applications. A very common example of List View is your phone contact book, where you have a list of your contacts displayed in aList View and if you click on it then user information is displayed.



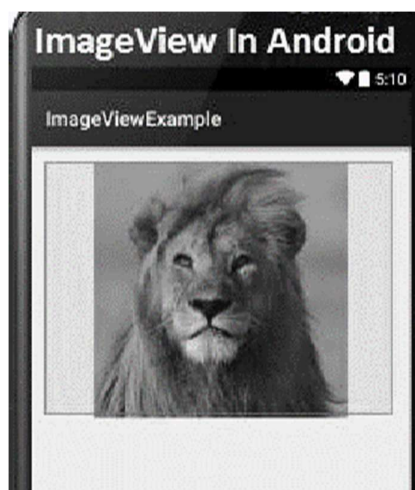
### Grid View

In android Grid View is a view group that display items in two dimensional scrolling grid (rows and columns), the grid items are not necessarily predetermined but they are automatically inserted to the layout using a List Adapter. Users can then select any grid item by clicking on it. Grid View is default scrollable so we don't need to use Scroll View or anything else with Grid View.



### Image View

In Android, Image View class is used to display an image file in application. Image file is easy to use but hard to master in Android, because of the various screen sizes in Android devices. An android is enriched with some of the best UI design widgets that allows us to build good looking and attractive UI based application.



### Scroll Vie

In android scroll View can hold only one direct child. This means that, if you have complex layout with more views (Buttons, Text Views or any other view) then you must enclose them inside another standard layout like Table Layout, Relative

Layout or Linear Layout. You can specify layout\_width and layout\_height to adjust width and height of screen. You can specify height and width in dp (density pixel) or px (pixel). Then after enclosing them in a standard layout, enclose the whole layout in scroll View to make all the element or views scrollable.



### VIII. Resources required (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

**IX. Practical related Questions**

*Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.*

1. List all attributes of Image View.
2. Write steps to add following string array to grid view.  
static final String [] example= new String {“A”, “B”, “C”, “D”, “E”};
3. Describe android:stretchMode attribute of Grid view in detail.

**(Space for answers)**

**Ans 1)** All the attributes of Image view are as follows:

1. android:adjustViewBounds
2. android:baseline
3. android:baselineAlignBottom
4. android:cropToPadding
5. android:maxHeight
6. android:maxLength
7. android:scaleType
8. android:src.
9. android:tint
10. android:tintMode

**Ans 2)** Steps are as follows:

1. Create a main layout.
2. Paste the following xml and java code.

**activity\_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<GridView xmlns:android="http://schemas.android.com/apk/res/android"
 android:id="@+id/gridView"
 android:layout_width="fill_parent"
 android:layout_height="fill_parent"
 android:columnWidth="50dp"
 android:gravity="center"
 android:numColumns="auto_fit"
 android:stretchMode="columnWidth" >
```



```
</GridView>
```

### ActivityMain.java

```
import android.app.Activity;

import android.os.Bundle;
import android.widget.AdapterView;
import android.widget.AdapterView.OnItemClickListener;
import android.widget.ArrayAdapter;
import android.widget.GridView;
import android.widget.TextView;
import android.widget.Toast;
import android.view.View;

public class MainActivity extends Activity {
 GridView grid;
 static final String[] letters = new String[] {
 "A", "B", "C", "D", "E",
 "F", "G", "H", "I", "J",
 "K", "L", "M", "N", "O",
 "P", "Q", "R", "S", "T",
 "U", "V", "W", "X", "Y", "Z"};

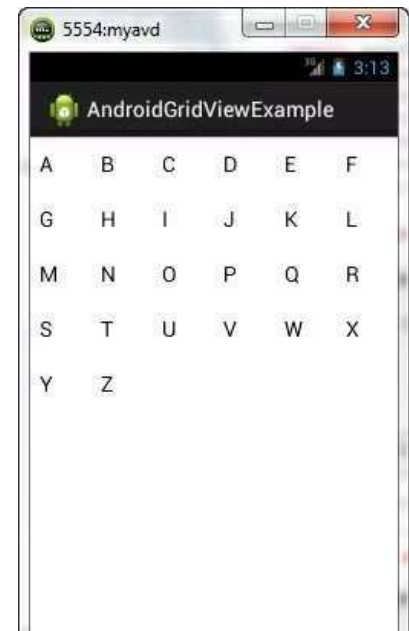
 @Override
 public void onCreate(Bundle savedInstanceState) {
 super.onCreate(savedInstanceState);
 setContentView(R.layout.main);

 grid = (GridView) findViewById(R.id.gridView);

 ArrayAdapter adapter = new ArrayAdapter(this,
 android.R.layout.simple_list_item_1, letters);

 grid.setAdapter(adapter);

 grid.setOnItemClickListener(new OnItemClickListener() {
 public void onItemClick(AdapterView parent, View v, int position, long id)
```



```

 Toast.makeText(getApplicationContext(),
 ((TextView) v).getText(), Toast.LENGTH_SHORT).show();
 }
});

 }
}

```

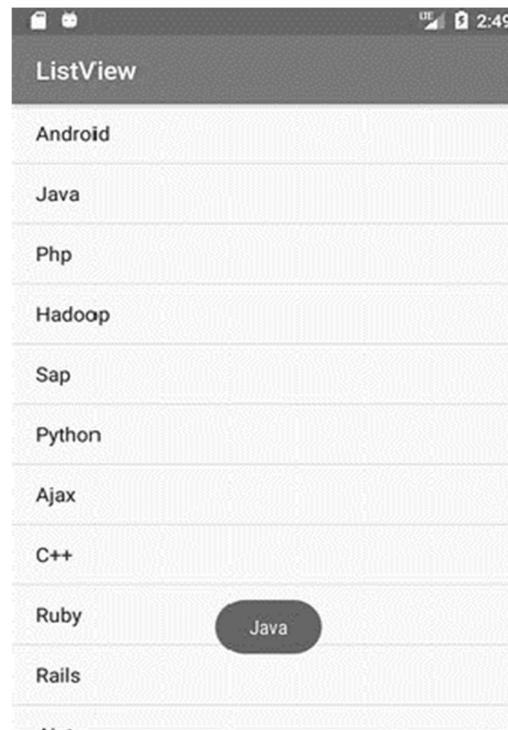
### 3. Run the application

**Ans 3)** android:stretchMode : Defines how columns should stretch to fill the available empty space, if any. This must be either of the values : none : Stretching is disabled. spacingWidth : The spacing between each column is stretched.

#### X. Exercise

(Use blank space provide for answers or attached more pages if needed)

1. Write a program to show the following output. Use appropriate view for the same.



2. Write a program to display an image using Image View and a button named as "Change Image". Once you click on button another image

should get displayed.

3. Write a program to display 15 buttons using grid view.
4. Write a program to display a text view using vertical scroll view.

**(Space for answers)**

Ans

#### XML code

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
xmlns:android="http://schemas.android.co
m/apk/res/android"
xmlns:app="http://schemas.android.com
/apk/res-auto"
xmlns:tools="http://schemas.android.co
m/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

<ListView
 android:layout_width="match_parent"
"
 android:layout_height="match_parent"
"
 android:entries="@array/Language_li
st"
/>
```

```
</RelativeLayout>
```

#### strings.xml

```
<resources>
 <string
name="app_name">Exp14_1</string>
 <array name="Language_list">
 <item>Android</item>
 <item>Java</item>
 <item>PHP</item>
 <item>Hadoop</item>
 <item>Sap</item>
 <item>Phython</item>
 <item>Ajax</item>
 <item>C++</item>
 <item>Puby</item>
 <item>Rails</item>
 </array>
</resources>
```

#### Java Code

```
package com.example.exp14_1;
```

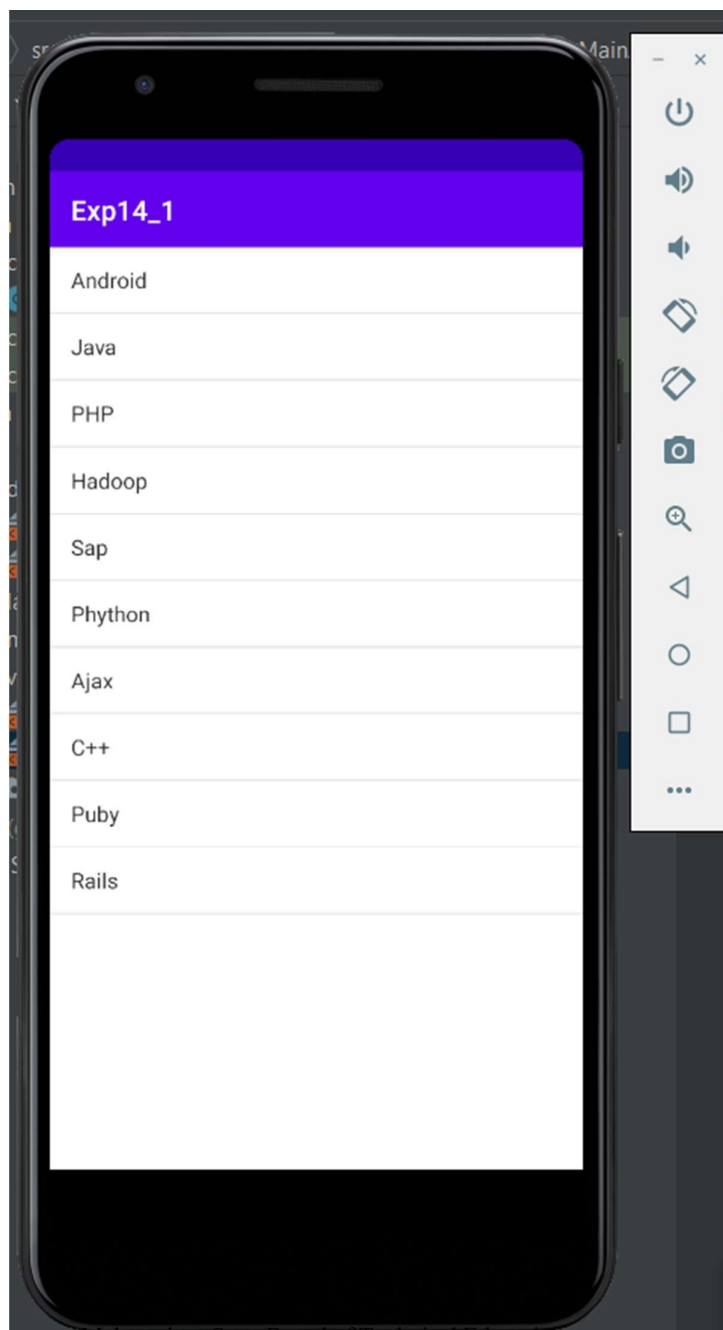
```
import
androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

public class MainActivity extends
AppCompatActivity {

 @Override
 protected void onCreate(Bundle
savedInstanceState) {
 super.onCreate(savedInstanceState);
 setContentView(R.layout.activity_ma
in);
 }
}
```

Output:



Ans 2:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
 android:layout_width="fill_parent"
 android:layout_height="fill_parent"
 android:background="#458"
 android:orientation="vertical" >

 <ImageView
 android:id="@+id/imageView1"
 android:layout_width="match_parent"
 android:layout_height="347dp" />

 <Button
 android:id="@+id/button1"
 android:layout_width="200dp"
 android:layout_height="58dp"
 android:gravity="center"
 android:onClick="mess"
 android:text="First Image "
 android:textSize="20sp" />
 <Button
 android:id="@+id/button2"
 android:layout_width="218dp"
 android:layout_height="63dp"
 android:onClick="mess"
 android:text="Second Image"
 android:textSize="20sp" />
</LinearLayout>
```

### **ActivityMain.java**

```
package com.example.exp14_2;
```

```
import
android.os.Bundle;
import
android.view.View;
import
android.widget.ImageView;
import
android.widget.TextView;
import android.app.Activity;
public class MainActivity extends Activity {
 @Override
 protected void onCreate(Bundle savedInstanceState) {
 super.onCreate(savedInstanceState);
 setContentView(R.layout.activity_main);
 }
 public void mess(View v)
 {
 //initialize image view object
 ImageView im=(ImageView)findViewById(R.id.imageView1);
 //get clicked button id from view object
 switch(v.getId())
 {
 case R.id.button1:
 //if button1 is clicked than set image1
 im.setImageResource(R.drawable.myi
 mage); break;
 case R.id.button2:
 //if button2 is clicked than set image2
 im.setImageResource(R.drawable.myi
```

```

 image2); break;
 }
}
}

```



Ans 3

```

<?xml version="1.0" encoding="utf-8"?>
<GridView xmlns:android="http://schemas.android.com/apk/res/android"
 xmlns:app="http://schemas.android.com/apk/res-auto"
 xmlns:tools="http://schemas.android.com/tools"
 android:id="@+id/activity_grid"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 android:background="@android:color/black"
 android:columnWidth="90dp"
 android:gravity="center"
 android:horizontalSpacing="50dp"
 android:listSelector="@color/white"

```

```
 android:numColumns="3"
 android:padding="20dp"
 android:stretchMode="columnWidth"
 android:verticalSpacing="10dp"
 tools:context=".MainActivity" />
```

### **MainActivity.java**

```
package com.example.exp14_3;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.GridView;
public class MainActivity extends AppCompatActivity {
 @Override
 protected void onCreate(Bundle savedInstanceState) {
 super.onCreate(savedInstanceState);
 setContentView(R.layout.activity_main);
 GridView gridview = (GridView) findViewById(R.id.activity_grid);
 gridview.setAdapter(new ButtonAdapter(this));
 }
}
```

### **ButtonAdapter.java**

```
package com.example.exp14_3;
import android.content.Context;
import android.view.View;
import android.view.ViewGroup;
import android.widget.BaseAdapter;
import android.widget.Button;
import android.widget.Toast;
public class ButtonAdapter extends BaseAdapter
{
 private Context mContext;
 private int btn_id;
 private int total_btns = 15;
```



```
public ButtonAdapter(Context context) {
 this.mContext = context;
}
@Override
public int getCount() {
 return total_btns;
}
@Override
public Object getItem(int i) {
 return null;
}
@Override
public long getItemId(int i) {
 return 0;
}
@Override
public View getView(final int i, View view, ViewGroup viewGroup)
{
 Button btn;
 if (view == null) {
 btn = new Button(mContext);
 btn.setText(" Button " + (++btn_id));
 btn.setLayoutParams (new ViewGroup.LayoutParams(300,300));
 } else {
 btn = (Button) view;
 }
 btn.setOnClickListener(new View.OnClickListener()
 {
 @Override
 public void onClick(View v)
 {
```

```

 Toast.makeText(v.getContext(), "Button #" +
 (i + 1), Toast.LENGTH_SHORT).show();
 }
});
return btn;
}
}

```

**4) Ans:****XML File:**

```

<?xml version="1.0" encoding="utf-8"?>
<ScrollView
 xmlns:android="http://schemas.android.com/apk/res/android"
 xmlns:tools="http://schemas.android.com/tools"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 tools:context=".Exp_14_3">
 <RelativeLayout
 android:layout_width="match_parent"
 android:layout_height="wrap_content">
 <TextView
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:id="@+id/textView1"
 android:layout_centerHorizontal="true"
 android:layout_marginTop="30dp"
 android:text="Scroll down"
 android:textSize="20dp"/>
 <TextView
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:id="@+id/textView2"
 android:layout_centerHorizontal="true"
 android:layout_marginTop="1000dp"
 android:text="You have scrolled too far..."
 android:textSize="20dp"/>
 </RelativeLayout>
</ScrollView>

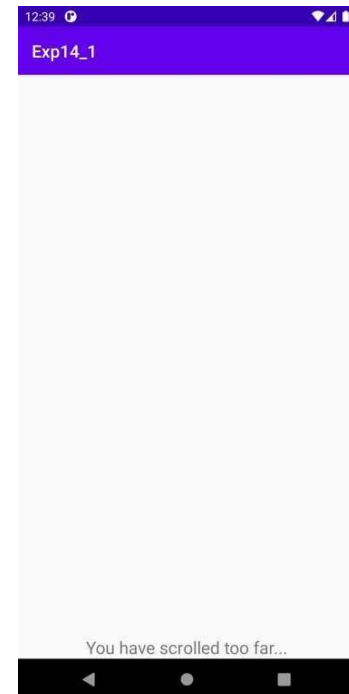
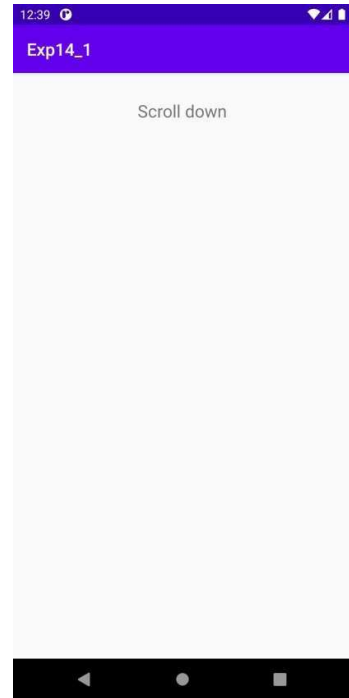
```

**Java File:**

```

package com.example.exp14_4;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class MainActivity extends AppCompatActivity {
 @Override
 protected void onCreate(Bundle savedInstanceState) {
 super.onCreate(savedInstanceState);
 setContentView(R.layout.activity_main);
 }
}

```



## **Practical No. 17: Develop a program to create an activity**

### **I. Practical Significance**

An activity represents a single screen with a user interface. For example, an email application might have one activity that shows a list of new emails, another activity to compose an email, and one for reading emails.

### **II. Relevant Program Outcomes (POs)**

PO 2- Discipline knowledge

PO 3- Experiments and practice

PO 4- Engineering tools practice

PO 10- Life-long learning

### **III. Competency and Skills**

“Create simple Android applications”

This practical is expected to develop the following skills

1. Create an activity to load all the UI components.

### **IV. Relevant Course Outcome(s)**

1. Develop rich user Interfaces by using layouts and controls.
2. Use User Interface components for android application development.

### **V. Practical Outcomes (PrOs)**

Develop a program to create an activity

### **VI. Relevant Affective Domain Related Outcome(s)**

1. Work collaboratively in team
2. Follow ethical practices

### **VII. Minimum Theoretical Background**

An activity is the single screen in android. It is like window or frame of Java. By the help of activity, you can place all your UI components or widgets in a single screen.

If an application has more than one activity, then one of them should be marked as the activity that is presented when the application is launched.

As in C, C++ or Java programming language program starts from main () function, android system initiates its program within an Activity starting with a call on onCreate() method. Android **Activity** class is the subclass of ContextThemeWrapper class. An activity class loads all the UI component using the XML file available in res/layout folder of the project. Following statement loads UI components from res/layout/activity\_main.xmlfile:

```
setContentView(R.layout.activity_main);
```

To write our own activity the new activity must be the derived from Activity class as given below

```
public class MainActivity extends Activity
{
 @Override
 public void onCreate(Bundle savedInstanceState) {
```

```

 super.onCreate(savedInstanceState);
 setContentView(R.layout.activity_main);
 /*Code goes here*/
 }
 /** Called when the activity is about to
 become visible. */ @Override
 protected void onStart() { super.onStart();
 /*Code goes here*/
 }
 /** Called when the activity has
 become visible. */ @Override
 protected void onResume() { super.onResume();
 /*Code goes here*/
 }
 /** Called when another activity is
 taking focus. */ @Override
 protected void
 onPause() {
 super.onPause();
 /*Code goes here*/
 }
 /** Called when the activity is no
 longer visible. */ @Override
 protected void onStop() { super.onStop();
 /*Code goes here*/
 }
 /** Called just before the activity is
 destroyed. */ @Override
 public void onDestroy() { super.onDestroy();
 /*Code goes here*/
 }
}

```

### VIII. Resources required (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks
1	Android enabled smartphone / Android versionsupporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

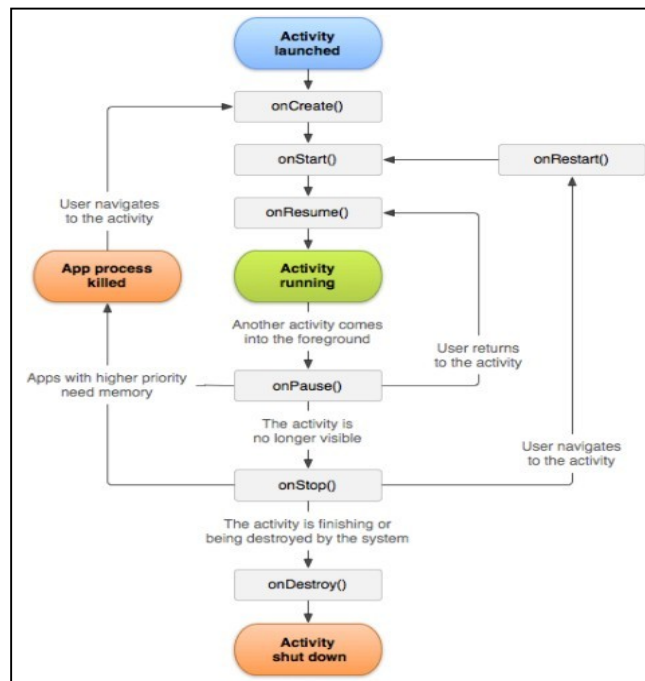
### IX. Practical related Questions

Note: Below given are few sample questions for reference. Teachers mustdesign more such questions to ensure the achievement of identified CO.

1. Draw the activity life cycle diagram.
2. Give the hierarchy of directory structure where you store activity file.
3. Write difference between onStop() and onDestroy() methods, also between onPause() and onResume()methods.

(Space for answers)

Ans 1:



Ans 2: All graphics, strings, layouts, and other resource files are stored in the resource file hierarchy under the res directory.

Ans 3:

Once `onStop()` is called then `onRestart()` can be called. `onDestroy()` is last in the order after `onStop()`. `onDestroy()` is called just before an activity is destroyed and after that it is gone it is not possible to resurrect this.

`onPause()` is called when the activity is still in the screen, once the method execution is completed then the activity loses focus. So, `onPause()` is logically before `onStop()`. From `onPause()` it is possible to call `onResume()` but it is not possible once `onStop()` is called.

## X. Exercise

(Use blank space provide for answers or attached more pages if needed)

1. Write a program to create a HelloWorld Activity using all lifecycles methods to display messages using Log.d.

**(Space for answers)**

Ans1:

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/andro
id"
 xmlns:app="http://schemas.android.com/apk/res-
auto"
 xmlns:tools="http://schemas.android.com/tools"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 tools:context=".MainActivity">

</androidx.constraintlayout.widget.ConstraintLayout>
```

Java code

```
package com.example.exp_17_1;

import
androidx.appcompat.app.AppCompatActivity;
import android.util.Log;
import android.os.Bundle;

public class MainActivity extends

AppCompatActivity { @Override
protected void onCreate(Bundle savedInstanceState) {
 super.onCreate(savedInstanceState);
 setContentView(R.layout.activity_main);
 Log.d("lifecycle","onCreate invoked by meena
talele");
}

@Override
```

```
protected void onStart() {
 super.onStart();

 Log.i("lifecycle","onStart
 invoked");
}

@Override
protected void onResume(){
 super.onResume();
 Log.v("lifecycle","onResume
 invoked");
}

@Override
protected void onPause(){
 super.onPause();
 Log.d("lifecycle","onPause
 invoked");
}

@Override
protected void onStop(){
 super.onStop();
 Log.d("lifecycle","onStop
 invoked");
}

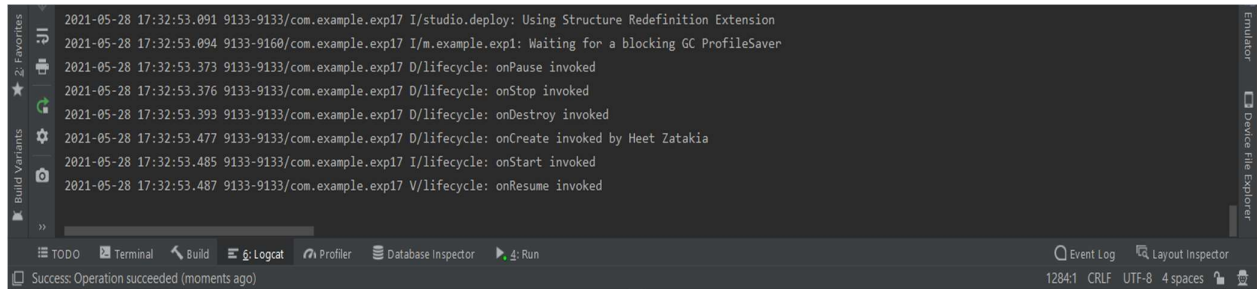
@Override
protected void onRestart(){
 super.onRestart();
 Log.d("lifecycle","onRestart
 invoked");
}

@Override
protected void onDestroy(){
 super.onDestroy();
 Log.d("lifecycle","onDestroy
 invoked");
}
```

## //DONE BY Heet Zatakia

}

### Output:



### XI. References / Suggestions for further Reading

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. [https://www.tutorialspoint.com/android/android\\_advanced\\_tutorial.pdf](https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf)
4. <https://developer.android.com>

### XII. Assessment Scheme

Performance indicators		Weightage
<b>Process related (10 Marks)</b>		<b>30%</b>
1.	Logic Formation	10%
2.	Debugging ability	15%
3.	Follow ethical practices	5%
<b>Product related (15 Marks)</b>		<b>70%</b>
4.	Interactive GUI	20%
5.	Answer to Practical related questions	20%
6.	Expected Output	20%
7.	Timely Submission	10%
<b>Total (25 Marks)</b>		<b>100%</b>



**List of student Team Members**

1 Gaurang Jadhav

2 Shweta Gupta

3 Heet Zatakia

Marks Obtained			Dated signature of Teacher
Process Related(15)	Product Related(10)	Total (25)	

## **Practical No.18: Develop a program to implement new activity using explicit intent and implicit intent.**

### **I. Practical Significance**

Android Intent is the message that is passed between components such as activities, content providers, broadcast receivers, services etc. It facilitates communication between different android components

### **II. Relevant Program Outcomes (POs)**

PO 1- Basic knowledge:

PO 2- Discipline knowledge

PO 3- Experiments and practice

PO 4- Engineering tools practice

### **III. Competency and Skills**

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Create an activity using implicit intent.
2. Create an activity and call another activity using explicit intent.

### **IV. Relevant Course Outcome(s)**

1. Develop rich user Interfaces by using layouts and controls.
2. Use User Interface components for android application development.

### **V. Practical Outcomes (PrOs)**

Develop a program to implement new activity using explicit intent and implicit intent.

### **VI. Relevant Affective Domain Related Outcome(s)**

1. Work collaboratively in team
2. Follow ethical practices

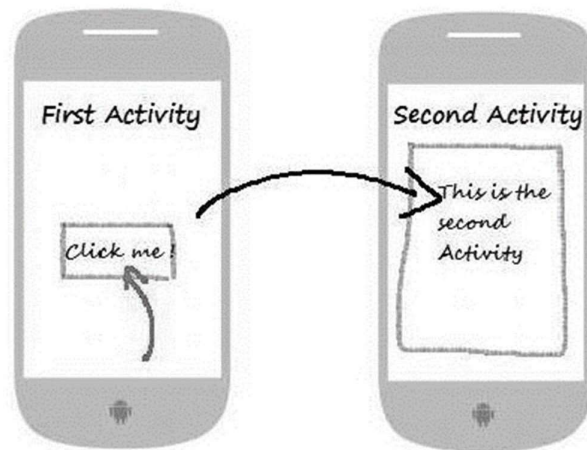
### **VII. Minimum Theoretical Background**

Android application components can connect to other Android applications. This connection is based on a task description represented by an Intent object.

Intents are asynchronous messages which allow application components to request functionality from other Android components. Intents allow you to interact with components from the same applications as well as with components contributed by other applications. For example, an activity can start an external activity for taking a picture.

Mostly Intents are used for:

- a. For Launching an Activity
- b. To start a New Service
- c. For Broadcasting Messages
- d. To Display a list of contacts in List View



### Types of intents:

There are two types of intents

- a. Implicit Intent
- b. Explicit Intent

#### Implicit Intent:

The implicit intent is the intent where instead of defining the exact components, you define the action that you want to perform for different activities.

#### Syntax:

```
Intent i=new Intent();
i.setAction(Intent.ACTION_SEND);
```

#### Explicit Intent:

An explicit intent is an Intent where you explicitly define the component that needs to be called by the Android System. An explicit intent is one that you can use to launch a specific app component, such as a particular activity or service in your app.

#### Syntax:

```
Intent I = new
Intent(getApplicationContext(),NextActivity.class);
I.putExtra("value1" , "This value for Next Activity");
I.putExtra("value2" , "This value for Next Activity");
```

### VIII. Resources required (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

**IX. Practical related Questions**

Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

1. List different methods used in Intent.
2. Write an intent to display the phone dialer with the given number filled in.

(Space for answers)

Ans 1:

Methods	Description
<b>Context.startActivity()</b>	This is to launch a new activity or get an existing activity to be action.
<b>Context.startService()</b>	This is to start a new service or deliver instructions for an existence service.
<b>Context.sendBroadcast()</b>	This is to deliver the message to broadcast receivers.

Ans 2:

```
Intent i = new Intent(Intent.ACTION_DIAL, u); Context.startActivity(i)
```

**X. Exercise**

(Solve any one of the following. Use blank space provide for answers or attached more pages if needed)

1. Write a program to create a text field and a button “Navigate”. When you enter “www.google.com” and press navigate button it should open google page.
2. Write a program to create button “Start Dialer”. When u click on this button it should open the phone dialer.
3. Write a program to create two screens. First screen will take one number input from user. After click on Factorial button, second screen will open and it should display factorial of the same number. Also specify which type of intent you will use in this case.

(Space for answers)

## Java Code:

```

package com.example.exp18_1;
import androidx.appcompat.app.AppCompatActivity; import
android.content.Intent;
import android.net.Uri;
import android.os.Bundle; import
android.view.View; import
android.widget.Button; import
android.widget.EditText;

public class MainActivity extends AppCompatActivity {
 EditText editText;
 Button btnNavigate; String
 text;
 @Override
 protected void onCreate(Bundle savedInstanceState) {
 super.onCreate(savedInstanceState);
 setContentView(R.layout.activity_main);

 btnNavigate=findViewById(R.id.btnNavigate);
 btnNavigate.setOnClickListener(new View.OnClickListener()
 {
 @Override
 public void onClick(View v) {
 editText=findViewById(R.id.edittext);
 text=editText.getText().toString().trim();
 if(text.equals("www.youtube.com"))
 {
 go();
 }
 }
 });
 }
 public void go()
 {
 Uri uri=Uri.parse("https://www.youtube.com"); Intent
 intent=new Intent(Intent.ACTION_VIEW,uri);
 startActivity(intent);
 }
}

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
 xmlns:app="http://schemas.android.com/apk/res-auto"
 xmlns:tools="http://schemas.android.com/tools"
 android:layout_width="match_parent" android:layout_height="match_parent"
 tools:context=".MainActivity">

 <EditText android:id="@+id/edittext"
 android:layout_width="match_parent"
 android:layout_height="wrap_content"
 android:layout_centerInParent="true"

```

```
android:hint="Enter URL"
```

```
/>
```

```
<Button
```

```
 android:id="@+id/btnNavigate"
```

```
 android:layout_width="wrap_content"
```

```
 android:layout_height="wrap_content"
```

```
 android:text="Navigate"
```

```
 android:layout_below="@+id/edittext"
```

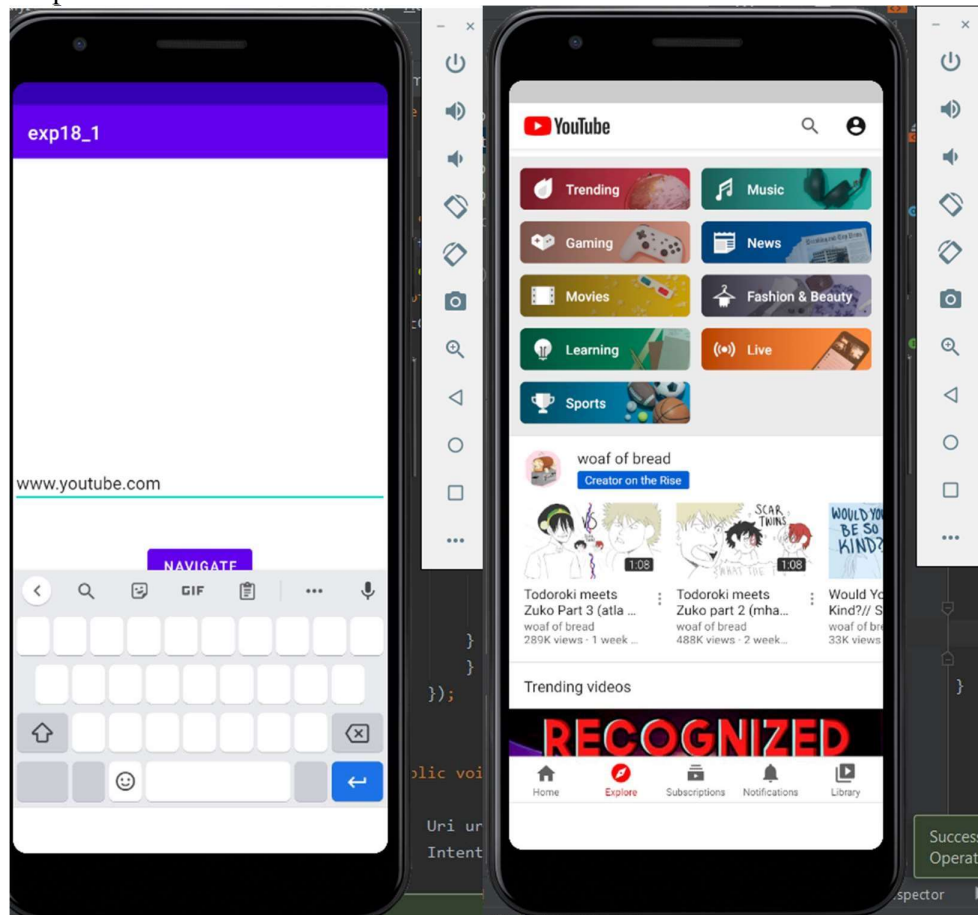
```
 android:layout_centerHorizontal="true"
```

```
 android:layout_marginTop="40dp"
```

```
/>
```

```
</RelativeLayout>
```

Output:



Ans 2:

```
<?xml version="1.0" encoding="utf-8"?>
```

```
 <RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
 xmlns:app="http://schemas.android.com/apk/res-auto"
```

```
 xmlns:tools="http://schemas.android.com/tools" android:layout_width="match_parent"
```

```
 android:layout_height="match_parent"
```

```
 tools:context=".MainActivity">
```

```
<Button
 android:id="@+id/btnOpenDialer" android:layout_width="wrap_content"
 android:layout_height="wrap_content" android:text="Open Dialer"
 android:layout_centerInParent="true"
/>

</RelativeLayout>
```

## Java

```
package com.example.exp18_2;
```

```
import androidx.appcompat.app.AppCompatActivity; import android.content.Intent;
import android.view.View;
import android.widget.Button; import android.os.Bundle;

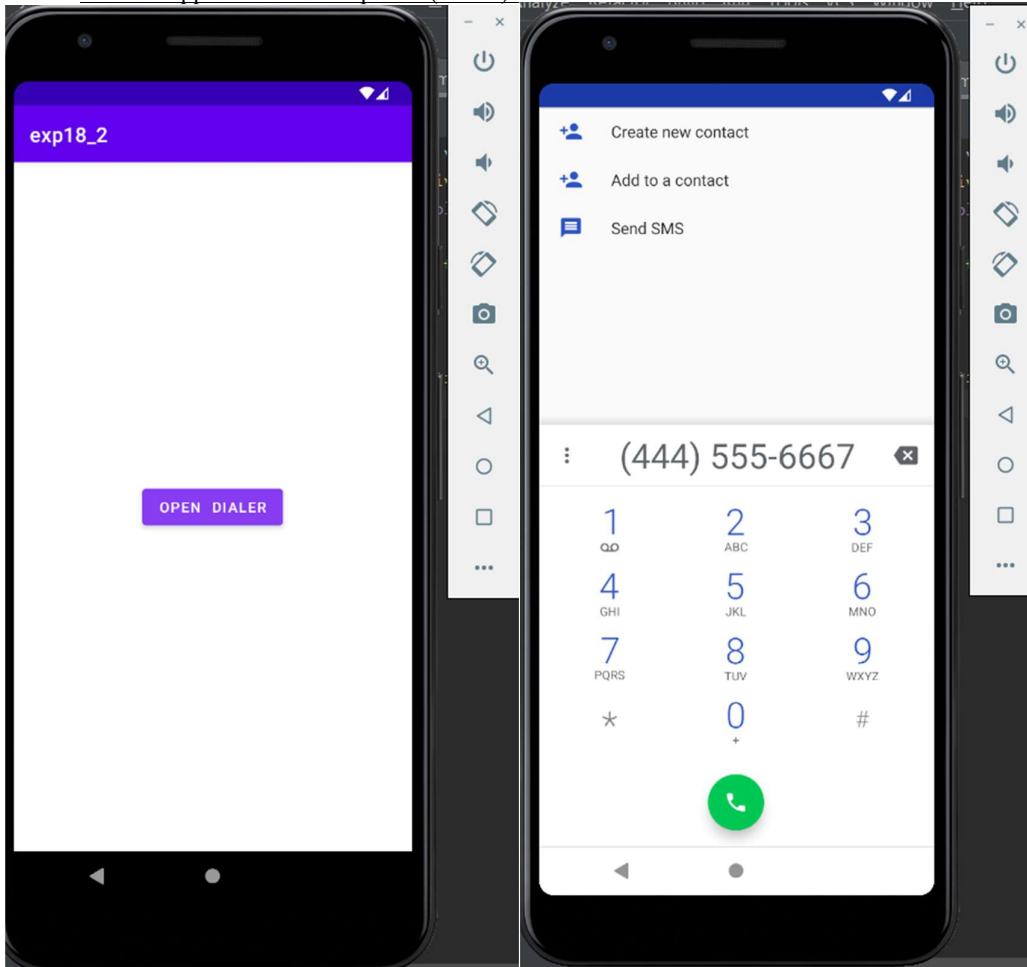
public class MainActivity extends AppCompatActivity { Button btnOenDialer;
 @Override
 protected void onCreate(Bundle savedInstanceState) {
 super.onCreate(savedInstanceState); setContentView(R.layout.activity_main);
 btnOenDialer=findViewById(R.id.btnOpenDialer); btnOenDialer.setOnClickListener(new
 View.OnClickListener() {

 @Override

 public void onClick(View v) { Intent intent=new
 Intent((Intent.ACTION_DIAL));
 startActivity(intent);

 }

 }
);
 }
}
```

**Ans 3) activity\_main.xml**

```
<xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/relativeLayout"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
<EditText
android:id="@+id/editText"
android:layout_width="315dp"
android:layout_height="53dp"
android:layout_centerHorizontal="true"
android:layout_centerVertical="true"
android:fontFamily="@font/poppins_medium"
android:hint="Enter a number"
android:inputType="number"
android:textSize="20sp"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintLeft_toLeftOf="parent"
app:layout_constraintRight_toRightOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.371"/>
```



```
<Button
android:id="@+id/btnFact"
android:layout_width="195dp"
android:layout_height="43dp"
android:layout_marginTop="84dp"
android:fontFamily="@font/poppins_medium"
android:text="Factorial"
app:layout_constraintLeft_toLeftOf="parent"
app:layout_constraintRight_toRightOf="parent"
app:layout_constraintTop_toBottomOf="@+id/editText"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

### MainActivity.java

```
package com.example.exp18_2;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
public class MainActivity extends AppCompatActivity
{EditText editText;
Button btnFact;
int fact = 1;
int num=0;
@Override
protected void onCreate(Bundle savedInstanceState)
{super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
editText = findViewById(R.id.editText);
btnFact = findViewById(R.id.btnFact);
btnFact.setOnClickListener(new View.OnClickListener()
{@Override
public void onClick(View v)
{go();
}
});
}
public void go(){
num = Integer.parseInt(editText.getText().toString().trim());
for(int i = 1; i <= num; i++){
fact = fact * i;
}
navigate();
}
public void navigate(){
Intent intent = new Intent(MainActivity.this,EmptyActivity.class);
intent.putExtra("Fact",fact);
intent.putExtra("Num",num);
startActivity(intent);
}
}
```

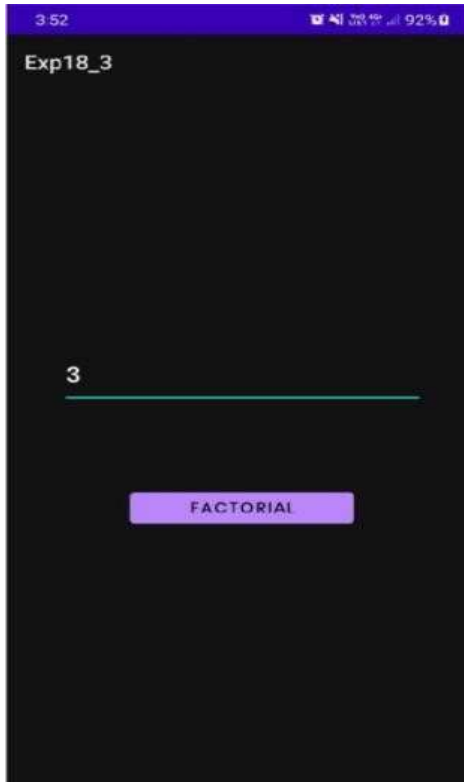
**activity\_empty.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
 xmlns:app="http://schemas.android.com/apk/res-auto"
 xmlns:tools="http://schemas.android.com/tools"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 tools:context=".EmptyActivity">
 <TextView
 android:id="@+id/textview1"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_centerHorizontal="true"
 android:layout_marginTop="250dp"
 android:fontFamily="@font/poppins_medium"
 android:text=""
 android:textColor="#FFFFFF"
 android:textSize="24sp"/>
 <TextView
 android:id="@+id/textview2"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_below="@+id/textview1"
 android:layout_centerHorizontal="true"
 android:layout_marginTop="20dp"
 android:fontFamily="@font/poppins_medium"
 android:text=""
 android:textColor="#FFFFFF"
 android:textSize="24sp" />
</RelativeLayout>
```

**EmptyActivity.java**

```
package com.example.exp18_2;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.widget.TextView;
public class EmptyActivity extends AppCompatActivity
{TextView textView1,textView2;
int num,fact;
@Override
protected void onCreate(Bundle savedInstanceState)
{super.onCreate(savedInstanceState);
setContentView(R.layout.activity_empty);
Intent intent = getIntent();
num = intent.getIntExtra("Num",0);
fact = intent.getIntExtra("Fact",0);
textView1 = findViewById(R.id.textview1);
textView2 = findViewById(R.id.textview2);
textView1.setText("Number = "+ num);
textView2.setText("Factorial = "+ fact);
}
```

}



### I. References / Suggestions for furtherReading

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. [https://www.tutorialspoint.com/android/android\\_advanced\\_tutorial.pdf](https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf)
4. <https://developer.android.com>

### II. Assessment Scheme

Performance indicators		Weightage
<b>Process related (10 Marks)</b>		<b>30%</b>
1.	Logic Formation	10%
2.	Debugging ability	15%
3.	Follow ethical practices	5%
<b>Product related (15 Marks)</b>		<b>70%</b>
4.	Interactive GUI	20%
5.	Answer to Practical related questions	20%
6.	Expected Output	20%
7.	Timely Submission	10%
<b>Total (25 Marks)</b>		<b>100%</b>

**List of student Team Members**

4 Gaurang Jadhav

5 Shweta Gupta

6 Heet Zatakia

Marks Obtained			Dated signature of Teacher
Process Related(15)	Product Related(10)	Total (25)	

## **Practical No. 21: Develop a program to implement broadcast receiver**

### **I. Practical Significance**

Broadcast Receivers simply respond to broadcast messages from other applications or from the system. For example, applications can also initiate broadcasts to let other applications know that some data has been downloaded to the device and is available for them to use, so this is broadcast receiver who will intercept this communication and will initiate appropriate action.

### **II. Relevant Program Outcomes (POs)**

PO1- Basic knowledge

PO2- Discipline knowledge

PO3- Experiments and practice

PO4- Engineering tools

### **III. Competency and Practical Skills**

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to Create the Broadcast Receiver.
2. Able to Register Broadcast Receiver.

### **IV. Relevant Course Outcome(s)**

1. Interpret features of Android operating system.
2. Configure Android environment and development tools.
3. Develop rich user Interfaces by using layouts and controls.

### **V. Practical Outcome (PrOs)**

Develop a program to implement broadcast receiver.

### **VI. Relevant Affective Domain Related Outcome(s)**

1. Work collaboratively in team
2. Follow ethical Practices.

### **VII. Minimum Theoretical Background**

#### **Creating the Broadcast Receiver:**

A broadcast receiver is implemented as a subclass of Broadcast Receiver class and overriding the onReceive() method where each message is received as an Intent object parameter.

#### **Registering Broadcast Receiver:**

An application listens for specific broadcast intents by registering a broadcast receiver in AndroidManifest.xml file. Consider we are going to register MyReceiver for system generated event ACTION\_BOOT\_COMPLETED which is fired by the system once the Android system has completed the boot process.



### VIII. Resources required (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

### IX. Practical related Questions

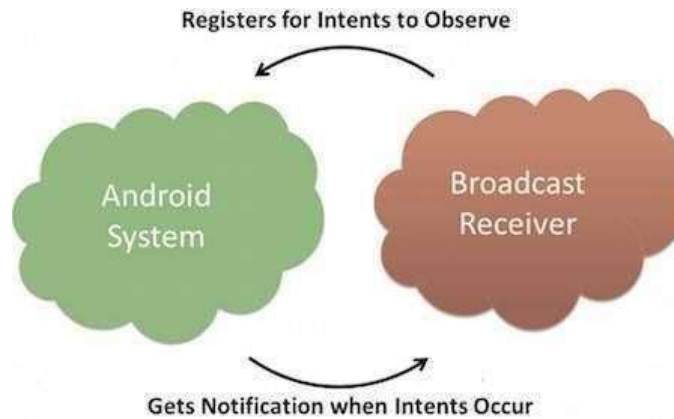
**Note:** Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

1. Differentiated between Activity Intent and Broadcasting Intent.
2. Draw Broadcast Receivers Lifecycle.
3. List the System Events related to Broadcast Receivers.

(Space for answers)

**Ans 1)** Broadcasting Intents are different from the Intents used to start an Activity or a Service (discussed in previous Android Tutorials). The intent used to start an Activity makes changes to an operation the user is interacting with, so the user is aware of the process. However, in case of broadcasting intent, the operation runs completely in the background, and is therefore invisible to the user.

**Ans 2)** Broadcast Receivers Lifecycle



**Ans 3)** System Events related to Broadcast Receivers are as follows:

1. android.intent.action.BATTERY\_CHANGED
2. android.intent.action.BATTERY\_LOW
3. android.intent.action.BATTERY\_OKAY
4. android.intent.action.BOOT\_COMPLETED
5. android.intent.action.BUG\_REPORT
6. android.intent.action.CALL
7. android.intent.action.CALL\_BUTTON

**X. Exercise**

**Note: Faculty must ensure that every group of students use different input value.**

(Use blank space for answers or attach more pages if needed)

1. Write a program to demonstrate all the system broadcast messages.

**(Space for answers)**

**ActivityMain.xml**

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/a
 pk/res/android"
 xmlns:app="http://schemas.android.com/apk/res-auto"
 xmlns:tools="http://schemas.android.com/tools"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 tools:context=".MainActivity">

 <Button
 android:id="@+id/button"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_marginStart="97dp"
 android:layout_marginTop="341dp"
 android:text="Broadcasting button"
 app:layout_constraintStart_toStartOf="parent"
 app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

**MainActivity.java**

```

package com.example.exp21;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.content.IntentFilter;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity
{
 Button button;
 MyReceiver myReceiver = new MyReceiver();
 IntentFilter filter = new IntentFilter();
 @Override
 protected void onCreate(Bundle savedInstanceState)
 {
 super.onCreate(savedInstanceState);
 setContentView(R.layout.activity_main);
 button = findViewById(R.id.button);
 filter.addAction(Intent.ACTION_AIRPLANE_MODE_CHANGED);
 button.setOnClickListener(new View.OnClickListener() {
 @Override
 public void onClick(View v)
 {
 MainActivity.this.registerReceiver(myReceiver, filter);
 }
 });
 }

 @Override
 protected void onDestroy()
 {
 super.onDestroy();
 MainActivity.this.unregisterReceiver(myReceiver);
 }
}

```

**MyReceiver.java**

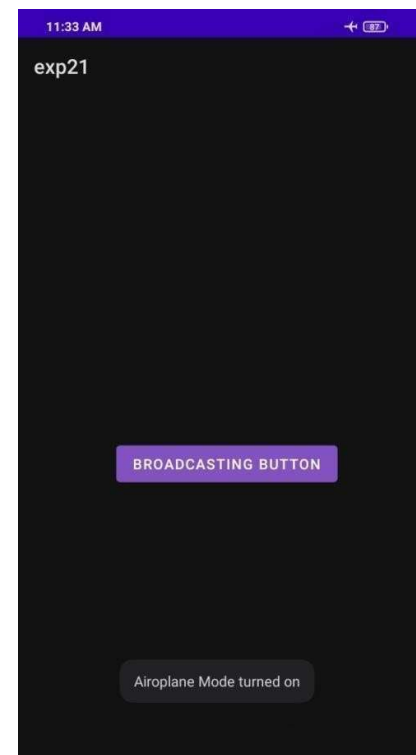
```

package com.example.exp21;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.widget.Toast;

public class MyReceiver extends BroadcastReceiver
{
 public MyReceiver() {
 super();
 }

 @Override
 public void onReceive(Context context, Intent intent)
 {
 String intentAction = intent.getAction();
 if (intentAction != null) {
 String toastMessage = "unknown intent action";

```





```
 switch(intentAction){
 case Intent.ACTION_AIRPLANE_MODE_CHANGED:
 boolean status = intent.getBooleanExtra("state",true);
 if(status){
 toastMessage = "Airoplane Mode turned on";
 }else {
 toastMessage = "Airoplane Mode turned off";
 }
 break;
 }
 Toast.makeText(context,toastMessage,Toast.LENGTH_SHORT).show();
 }
}
```

**AndroidManifest.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
 package="com.example.broadcast_messages">
 <uses-permission android:name="android.permission.ACCESS_NETWORK_STATE"/>
 <application
 android:allowBackup="true"
 android:icon="@mipmap/ic_launcher"
 android:label="@string/app_name"
 android:roundIcon="@mipmap/ic_launcher_round"
 android:supportsRtl="true"
 android:theme="@style/Theme.Exp21">
 <activity android:name=".MainActivity">
 <intent-filter>
 <action android:name="android.intent.action.MAIN" />

 <category android:name="android.intent.category.LAUNCHER" />
 </intent-filter>
 </activity>
 <receiver android:name=".MyReceiver"
 android:exported="false">
 </receiver>

 </application>

</manifest>
```

**XI. References / Suggestions for further Reading**

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. [https://www.tutorialspoint.com/android/android\\_advanced\\_tutorial.pdf](https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf)
4. <https://developer.android.com>

**XII. Assessment Scheme**

Performance indicators		Weightage
<b>Process related (10 Marks)</b>		<b>30%</b>
1.	Logic Formation	10%
2.	Debugging ability	15%
3.	Follow ethical practices	5%
<b>Product related (15 Marks)</b>		<b>70%</b>
4.	Interactive GUI	20%
5.	Answer to Practical related questions	20%
6.	Expected Output	20%
7.	Timely Submission	10%
<b>Total (25 Marks)</b>		<b>100%</b>

**List of student Team Members**

1 Gaurang Jadhav

2 Shweta Gupta

3 Heet Zatakia

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

## **Practical No. 22: Develop a program to implement sensors.**

### **I. Practical Significance**

Most of the android devices have built-in sensors that measure motion, orientation, and various environmental condition. Android allows us to get the raw data from these sensors and use it in our application.

### **II. Relevant Program Outcomes (POs)**

PO2- Discipline knowledge

PO3- Experiments and practice

PO4- Engineering tools

PO5- The engineer and society

PO7- Ethics

### **III. Competency and Practical Skills**

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to get a list of sensors supported by the device.
2. Able to demonstrate the use of Sensor Manager class.

### **IV. Relevant Course Outcome(s)**

1. Interpret features of Android operating system.
2. Configure Android environment and development tools.

### **V. Practical Outcome (PrOs)**

Develop a program to implement sensors.

### **VI. Relevant Affective Domain Related Outcome(s)**

1. Work collaboratively in team
2. Follow ethical Practices.

### **VII. Minimum Theoretical Background**

The android platform supports three broad categories of sensors. Motion Sensors, Environmental sensors, Position sensors. Some of the sensors are hardware based and some are software-based sensors. Whatever the sensor is, android allows us to get the raw data from these sensors and use it in our application. For this, android provides us with some classes. Android provides Sensor Manager and Sensor classes to use the sensors in our application. In order to use sensors, first thing you need to do is to instantiate the object of SensorManager class.

Example: `SensorManager sMgr;`

`sMgr = (SensorManager)this.getSystemService(SENSOR_SERVICE);`

The next thing you need to do is to instantiate the object of Sensor class by calling the `getDefaultSensor()` method of the SensorManager class. Its syntax is given below:

`Sensor light;`

`light = sMgr.getDefaultSensor(Sensor.TYPE_LIGHT);`

Once that sensor is declared, you need to register its listener and override two methods which are onAccuracyChanged and onSensorChanged. Its syntax is as follows:

```
sMgr.registerListener(this, light, SensorManager.SENSOR_DELAY_NORMAL);
public void onAccuracyChanged(Sensor sensor, int accuracy)
{
}
public void onSensorChanged(SensorEvent event) { }
```

#### Methods:

1. getDefaultSensor(int type) :- This method get the default sensor for a given type.  
Explain methods
2. getOrientation(float[] R, float[] values) :- This method returns a description of the current primary clip on the clipboard but not a copy of its data.
3. getInclination(float[] I) :- This method computes the geomagnetic inclination angle in radians from the inclination matrix.
4. registerListener(SensorListener listener, int sensors, int rate) :-This method registers a listener for the sensor
5. unregisterListener(SensorEventListener listener, Sensor sensor) :-This method unregisters a listener for the sensors with which it is registered.
6. getOrientation(float[] R, float[] values) :-This method computes the device's orientation based on the rotation matrix.
7. getAltitude(float p0, float p) :-This method computes theAltitude in meters from the atmospheric pressure and the pressure at sea-level.

#### VIII. Resources required (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

#### IX. Practical related Questions

**Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.**

1. List the best practices for accessing and using sensors.
2. Differentiate between Sensor Class and Sensor Manager Class.

(Space for answers)

**Ans 1. Best practice for accessing and using sensors are:**

**i. Unregister sensor listeners:** Always make sure to disable sensors you don't need, especially when your activity is paused.

**ii. Don't block the onSensorChanged() method:** Sensor data can change at a high rate, which means the system may call the onSensorChanged(SensorEvent) method quite often. As a best practice, you should do as little as possible within the onSensorChanged

(SensorEvent) method so you don't block it.

**iii. Avoid using deprecated methods or sensor types:** Several methods and constants have been deprecated. In particular, the TYPE\_ORIENTATION sensor type has been deprecated.

**iv. Verify sensors before you use them:** Always verify that a sensor exists on a device before you attempt to acquire data from it. Don't assume that a sensor exists simply because it's a frequently-used sensor.

**v. Choose sensor delays carefully:** Sensors can provide data at very high rates. Allowing the system to send extra data that you don't need wastes system resources and uses battery power.

**Ans 2. i. Sensor Class:** The android.hardware.Sensor class provides methods to get information about the sensor such as sensor name, sensor type, sensor resolution, sensor type etc.

**ii. Sensor Manager Class:** The android.hardware.SensorManager class provides methods :

- to get sensor instance,
- to access and list sensors,
- to register and unregister sensor listeners etc.

You can get the instance of SensorManager by calling the method getSystemService() and passing the SENSOR\_SERVICE constant in it.

```
SensorManager sm = (SensorManager)getSystemService(SENSOR_SERVICE);
```

### Exercise

**Note: Faculty must ensure that every group of students use different input value.**

(Use blank space for answers or attach more pages if needed)

3. Write a program to changes the background color when device is shuffled.
4. Write a program to display the list of sensors supported by the mobile device.

**(Space for answers)**

**Ans 3) activity\_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:id="@+id/Layout"
tools:context=".MainActivity">
</androidx.constraintlayout.widget.ConstraintLayout>
```

**MainActivity.java**

```

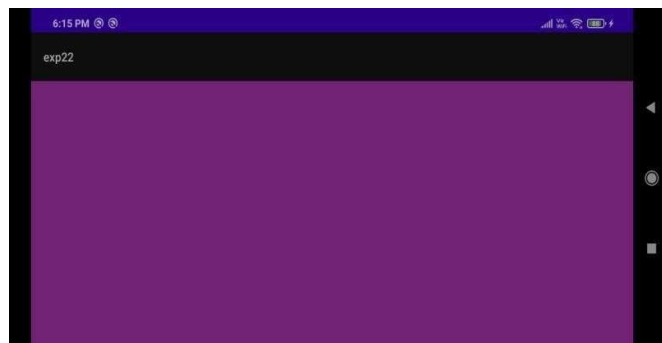
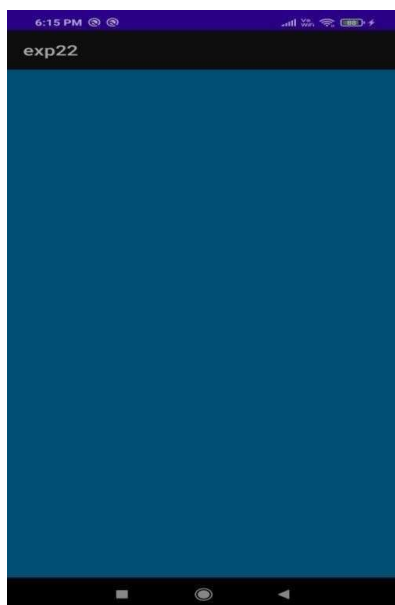
package com.example.exp22;

import androidx.appcompat.app.AppCompatActivity;
import androidx.constraintlayout.widget.ConstraintLayout;
import android.content.res.Configuration;
import android.graphics.Color;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity
{ConstraintLayout constraintLayout;

@Override
protected void onCreate(Bundle savedInstanceState)
{super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
constraintLayout = findViewById(R.id.Layout);
if(getResources().getConfiguration().orientation ==
Configuration.ORIENTATION_PORTRAIT){
constraintLayout.setBackgroundColor(Color.parseColor("#2cbdf2"));
}
if(getResources().getConfiguration().orientation ==
Configuration.ORIENTATION_LANDSCAPE){ constraintLayout.setBackgroundColor(Color.parse
Color"#f083f2");
}}}

```



**Ans 4) activity\_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
<ListView
android:id="@+id/listView1"
android:layout_width="match_parent"
android:layout_height="match_parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

**MainActivity.java**

```
package com.example.exp22_1;
import androidx.annotation.RequiresApi;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Context;
import android.hardware.Sensor;
import android.hardware.SensorManager;
import android.os.Build;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import java.util.List;
public class MainActivity extends AppCompatActivity
{SensorManager smm;
List<Sensor> sensor;
ListView lv;
@RequiresApi(api = Build.VERSION_CODES.M)
@Override
protected void onCreate(Bundle savedInstanceState)
{super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
smm = (SensorManager) getSystemService(Context.SENSOR_SERVICE);
lv = (ListView) findViewById (R.id.listView1);
sensor = smm.getSensorList(Sensor.TYPE_ALL);
lv.setAdapter(new ArrayAdapter<Sensor>(this,android.R.layout.simple_list_item_1,
sensor));
}
}
```

A screenshot of an Android application interface. At the top, a status bar shows the time as 6:23 PM, signal strength, Wi-Fi, and battery icons. The app's title bar is purple and contains the text 'exp22\_1'. The main content area has a black background with white text displaying a list of sensors and their properties in JSON-like format. The sensors listed include an Accelerometer, Magnetometer, Rotation Vector, Gyroscope, two Ambient Light Sensors (Non-wakeup and Wakeup), two Proximity Sensors (Non-wakeup and Wakeup), and a Gravity sensor. Each entry provides details like name, vendor, version, type, maxRange, resolution, power, and minDelay.

```
exp22_1

{Sensor name="sc7a20 Accelerometer
Non-wakeup", vendor="SLCompany",
version=742, type=1, maxRange=78.4532,
resolution=0.03824594, power=0.011,
minDelay=2500}
{Sensor name="mmc56x3x Magnetometer
Non-wakeup", vendor="memsic",
version=20120408, type=2, maxRange=3000.0,
resolution=0.0976, power=1.0, minDelay=10000}
{Sensor name="Rotation Vector Non-wakeup",
vendor="qualcomm", version=1, type=3,
maxRange=1.0, resolution=0.01, power=1.415,
minDelay=10000}
{Sensor name="gyro Gyroscope Non-wakeup",
vendor="Voltafield", version=261, type=4,
maxRange=38.39611, resolution=0.0026179168,
power=0.02, minDelay=10000}
{Sensor name="stk_stk3a5x Ambient Light Sensor
Non-wakeup", vendor="sensortek", version=200,
type=5, maxRange=4096.0, resolution=0.54,
power=0.09, minDelay=0}
{Sensor name="stk_stk3a5x Ambient Light Sensor
Wakeup", vendor="sensortek", version=200, type=5,
maxRange=4096.0, resolution=0.54, power=0.09,
minDelay=0}
{Sensor name="stk_stk3a5x Proximity Sensor
Non-wakeup", vendor="sensortek", version=200,
type=8, maxRange=5.0, resolution=0.0, power=0.1,
minDelay=0}
{Sensor name="stk_stk3a5x Proximity Sensor
Wakeup", vendor="sensortek", version=200,
type=8, maxRange=5.0, resolution=0.0, power=0.1,
minDelay=0}
{Sensor name="gravity Non-wakeup",
vendor="qualcomm", version=1, type=9,
maxRange=156.99008, resolution=0.01,
power=0.515, minDelay=10000}
```



**X. References / Suggestions for further Reading**

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. [https://www.tutorialspoint.com/android/android\\_advanced\\_tutorial.pdf](https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf)
4. <https://developer.android.com>

**XI. Assessment Scheme**

Performance indicators		Weightage
<b>Process related (10 Marks)</b>		<b>30%</b>
1.	Logic Formation	10%
2.	Debugging ability	15%
3.	Follow ethical practices	5%
<b>Product related (15 Marks)</b>		<b>70%</b>
4.	Interactive GUI	20%
5.	Answer to Practical related questions	20%
6.	Expected Output	20%
7.	Timely Submission	10%
<b>Total (25 Marks)</b>		<b>100%</b>

**List of student Team Members**

1 Gaurang Jadhav

2 Shweta Gupta

3 Heet Zatakia

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

## **Practical No. 23: Develop a program to build Camera.**

### **I. Practical Significance:**

The Android framework includes support for various cameras and camera features available on devices, allowing you to capture pictures and videos in your application.

### **II. Relevant Program Outcomes (POs)**

PO2- Discipline knowledge

PO3- Experiments and practice

PO4. Engineering tools.

PO5.The engineer and society.

PO7- Ethics

### **III. Competency and Practical Skills**

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to Launch the existing camera application.
2. Able to use the camera API to integrate the camera in any application.

### **IV. Relevant Course Outcome(s)**

1. Configure Android environment and development tools.
2. Develop rich user Interfaces by using layouts and controls.

### **V. Practical Outcome (PrOs)**

Develop a program for Camera

### **VI. Relevant Affective Domain Related Outcome(s)**

1. Work collaboratively in team
2. Follow ethical Practices.

### **VII. Minimum Theoretical Background**

Camera can be used in your application in following ways.

1. Using existing android camera application in our application
2. Directly using Camera API provided by android in our application

You will use `MediaStore.ACTION_IMAGE_CAPTURE` to launch an existing camera application installed on your phone. Its syntax is given below:

```
Intent intent = new
```

```
Intent(android.provider.MediaStore.ACTION_IMAGE_CAPTURE);
```

We will be using the camera API to integrate the camera in our application. First you will need to initialize the camera object using the static method provided by the api called `Camera.open()`. Its syntax is:

```
Camera object = null;
```

```
object = Camera.open();
```

#### **Methods:**

1. `startActivityForResult(Intent intent, int requestCode, Bundle options)` It starts an activity, but can take extra bundle of options with it.
2. `startActivityFromChild(Activity child, Intent intent, int requestCode)` It launches

the activity when your activity is child of any other activity.

3. `startActivityFromChild(Activity child, Intent intent, int requestCode, Bundle options)` It work same as above, but it can take extra values in the shape of bundle with it.
4. `startActivityFromFragment(Fragment fragment, Intent intent, int requestCode)` It launches activity from the fragment you are currently inside.
5. `startActivityFromFragment(Fragment fragment, Intent intent, int requestCode, Bundle options)` It not only launches the activity from the fragment, but can take extra values with it.

### VIII. Resources required (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

### IX. Practical related Questions

**Note:** Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

1. List all the methods related to camera class.
2. Explain the method that is used to detect the face.

**(Space for answers)**

**Ans 1)** List of all methods related to camera class are as follows:

1. `addCallbackBuffer(byte[] callbackBuffer)`
2. `autoFocus(Camera.AutoFocusCallback cb)`
3. `cancelAutoFocus()`
4. `enableShutterSound(boolean enabled)`
5. `getCameraInfo(int cameraId, Camera.CameraInfo cameraInfo)`
6. `getNumberOfCameras()`
7. `getParameters()`
8. `lock()`
9. `open()`
10. `open(int cameraId)`
11. `reconnect()`
12. `release()`
13. `setAutoFocusMoveCallback(Camera.AutoFocusMoveCallback cb)`
14. `setDisplayOrientation(int degrees)`
15. `setErrorCallback(Camera.ErrorCallback cb)`

**Ans 2)** startFaceDetection() method is used to detect the face.

**X. Exercise**

**Note: Faculty must ensure that every group of students use different input value. (Solve anyone of the following. Use blank space for answers or attach more pages if needed)**

1. Write a program to capture an image and display it using image view.
2. Write a program to record a video using various camera methods.

**(Space for answers)**

**Ans 1) activity\_main.xml**

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
 xmlns:app="http://schemas.android.com/apk/res-auto"
 xmlns:tools="http://schemas.android.com/tools"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 tools:context=".MainActivity"
 android:orientation="vertical">
 <ImageView
 android:id="@+id/imageView"
 android:layout_width="fill_parent"
 android:layout_height="403dp"
 android:layout_centerHorizontal="true" />
 <Button
 android:id="@+id/button"
 android:layout_width="match_parent"
 android:layout_height="73dp"
 android:fontFamily="@font/poppins_medium"
 android:text="Click here to capture image using camera"/>
</LinearLayout>
```

**MainActivity.java**

```
package com.example.exp23;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import android.Manifest;
```

```
import android.content.Intent;
import android.content.pm.PackageManager;
import android.graphics.Bitmap;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity
{
 Button button ;
 ImageView imageView ;
 public static final int RequestPermissionCode = 1 ;
 @Override
 protected void onCreate(Bundle savedInstanceState)
 {
 super.onCreate(savedInstanceState);
 setContentView(R.layout.activity_main);
 button = (Button)findViewById(R.id.button);
 imageView = (ImageView)findViewById(R.id.imageView);
 EnableRuntimePermission();
 button.setOnClickListener(new View.OnClickListener()
 {
 @Override
 public void onClick(View view)
 {
 Intent intent = new
 Intent(android.provider.MediaStore.ACTION_IMAGE_CAPTURE);
 startActivityForResult(intent, 7);
 }
 });
 }
 protected void onActivityResult(int requestCode, int resultCode, Intent data)
 {
 super.onActivityResult(requestCode, resultCode, data);
 if (requestCode == 7 && resultCode == RESULT_OK)
 {
 Bitmap bitmap = (Bitmap) data.getExtras().get("data");
 imageView.setImageBitmap(bitmap);
 }
 }
 public void EnableRuntimePermission(){
```

---

```

if (ActivityCompat.shouldShowRequestPermissionRationale(MainActivity.this,Manifest.permission.CAMERA))
{
 Toast.makeText(MainActivity.this,"CAMERA permission allows us to
 Access CAMERA app", Toast.LENGTH_LONG).show();
} else {
 ActivityCompat.requestPermissions(MainActivity.this,new
 String[] {Manifest.permission.CAMERA}, RequestPermissionCode);
}
}

@Override
public void onRequestPermissionsResult(int RC, String per[], int[] PResult)
{switch (RC) {
 case RequestPermissionCode:
 if (PResult.length > 0 && PResult[0] == PackageManager.PERMISSION_GRANTED) {
 Toast.makeText(MainActivity.this,"Permission Granted, Now your
 application can access CAMERA.", Toast.LENGTH_LONG).show();
 } else {
 Toast.makeText(MainActivity.this,"Permission Canceled, Now your
 application cannot access CAMERA.", Toast.LENGTH_LONG).show();
 }
 break;
 }
}
}

```

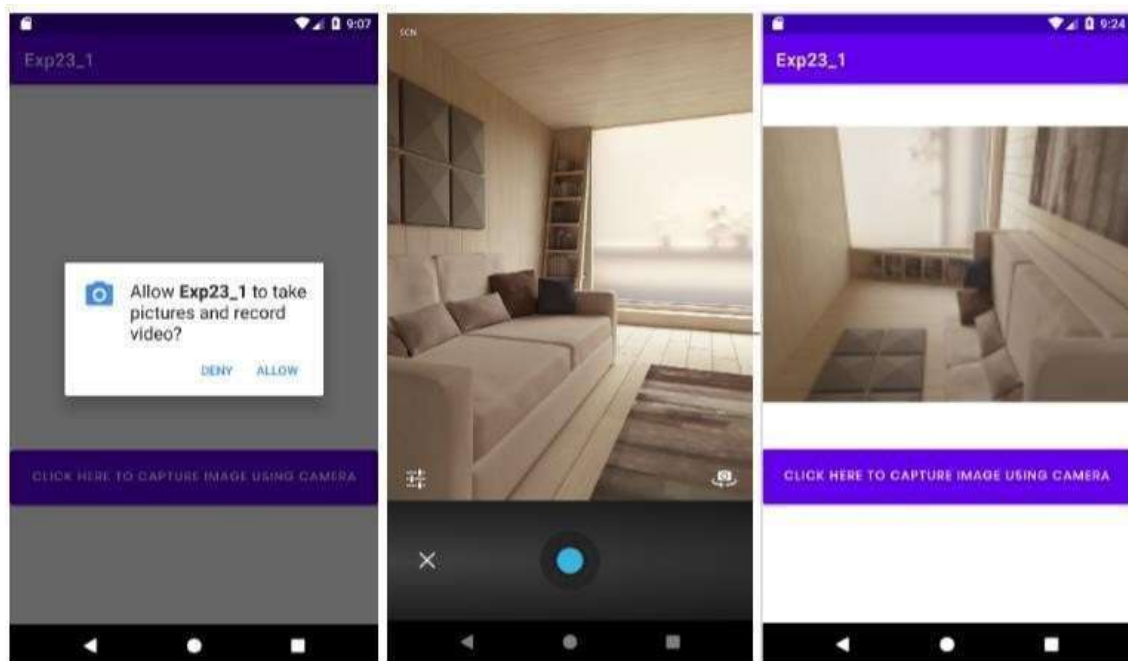
### **AndroidManifest.xml**

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
 package="com.example.exp23">
 <uses-permission android:name="android.permission.CAMERA" />
 <uses-feature
 android:name="android.hardware.camera2"
 android:required="true"/>
 <application
 android:allowBackup="true"

```

```
android:icon="@mipmap/ic_launcher"
android:label="@string/app_name"
android:roundIcon="@mipmap/ic_launcher_round"
android:supportsRtl="true"
android:theme="@style/Theme.Exp23">
<activity android:name=".MainActivity">
<intent-filter>
<action android:name="android.intent.action.MAIN" />
<category android:name="android.intent.category.LAUNCHER"/>
</intent-filter>
</activity>
<meta-data
android:name="preloaded_fonts"
android:resource="@array/preloaded_fonts"/>
</application>
</manifest>
```



**Ans 2) activity\_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:app="http://schemas.android.com/apk/res-auto"
 xmlns:tools="http://schemas.android.com/tools"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 xmlns:android="http://schemas.android.com/apk/res/android"
 tools:context=".MainActivity"
 android:orientation="vertical">
 <Button
 android:layout_width="250dp"
 android:layout_height="wrap_content"
 android:text="Capture Video"
 android:onClick="captureVideo"
 android:layout_gravity="center_horizontal"
 android:layout_marginTop="150dp"
 />
 <Button
 android:layout_width="250dp"
 android:layout_height="wrap_content"
 android:text="Play Video"
 android:onClick="playVideo"
 android:layout_gravity="center_horizontal"
 android:layout_marginTop="25dp"
 />
</LinearLayout>
```

**MainActivity.java**

```
package com.example.exp23_2;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.provider.MediaStore;
import android.view.View;
public class MainActivity extends AppCompatActivity
{private static int VIDEO_REQUEST = 101;
private Uri videoUri = null;
@Override
protected void onCreate(Bundle savedInstanceState)
{super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
}
public void captureVideo(View view) {
Intent videoIntent = new Intent(MediaStore.ACTION_VIDEO_CAPTURE);
if(videoIntent.resolveActivity(getPackageManager())!=null)
{ startActivityForResult(videoIntent,VIDEO_REQUEST);
}
}
public void playVideo(View view) {
Intent playIntent = new Intent(this,VideoPlayActivity.class);
playIntent.putExtra("videoUri", videoUri.toString());
startActivity(playIntent);
}
```



```

}
@Override
protected void onActivityResult(int requestCode, int resultCode, Intent data)
{super.onActivityResult(requestCode, resultCode, data);
if (requestCode == VIDEO_REQUEST && resultCode == RESULT_OK)
{videoUri = data.getData();
}
}
}
}

```

**activity\_play\_video.xml**

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".VideoPlayActivity">
<VideoView
android:layout_width="match_parent"
android:layout_height="match_parent"
android:layout_margin="35dp"
android:id="@+id/videoView"
/>
</RelativeLayout>

```

**VideoPlayActivity.java**

```

package com.example.exp23_2;
import androidx.appcompat.app.AppCompatActivity;
import android.net.Uri;
import android.os.Bundle;
import android.widget.VideoView;
public class VideoPlayActivity extends AppCompatActivity
{private VideoView mVideoView;
@Override
protected void onCreate(Bundle savedInstanceState)
{super.onCreate(savedInstanceState);
setContentView(R.layout.activity_video_play);
mVideoView = findViewById(R.id.videoView);
Uri videoUri = Uri.parse(getIntent().getExtras().getString("videoUri"));
mVideoView.setVideoURI(videoUri);
mVideoView.start();
}
}

```

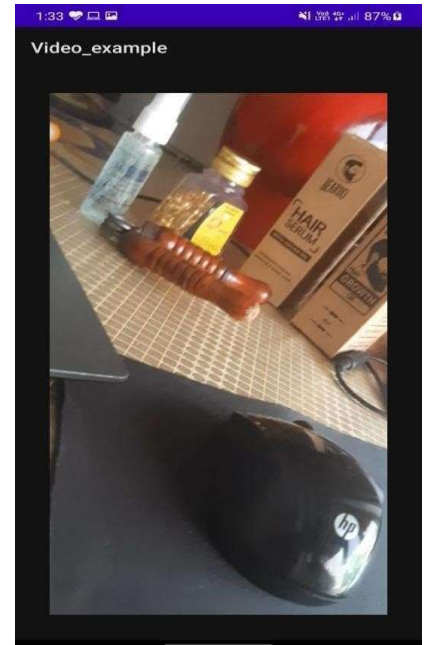
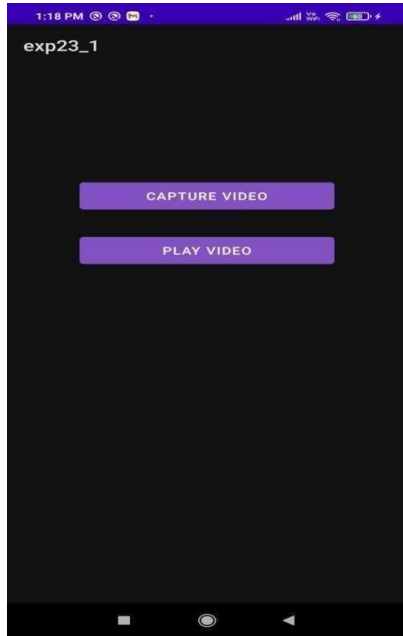
**AndroidManifest.xml**

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
package="com.example.exp23_2">
<uses-feature
android:name="android.hardware.camera2"
android:required="true" />
<application
android:allowBackup="true"
android:icon="@mipmap/ic_launcher"
android:label="@string/app_name"
android:roundIcon="@mipmap/ic_launcher_round"

```

```
android:supportsRtl="true"
android:theme="@style/Theme.Exp23_2">
<activity android:name=".VideoPlayActivity"></activity>
<activity android:name=".MainActivity">
<intent-filter>
<action android:name="android.intent.action.MAIN" />
<category android:name="android.intent.category.LAUNCHER" />
</intent-filter>
</activity>
</application>
</manifest>
```



**XI. References / Suggestions for further Reading**

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. [https://www.tutorialspoint.com/android/android\\_advanced\\_tutorial.pdf](https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf)
4. <https://developer.android.com>

**XII. Assessment Scheme**

Performance indicators		Weightage
<b>Process related (10 Marks)</b>		<b>30%</b>
1.	Logic Formation	10%
2.	Debugging ability	15%
3.	Follow ethical practices	5%
<b>Product related (15 Marks)</b>		<b>70%</b>
4.	Interactive GUI	20%
5.	Answer to Practical related questions	20%
6.	Expected Output	20%
7.	Timely Submission	10%
<b>Total (25 Marks)</b>		<b>100%</b>

**List of student Team Members**

1 Gaurang Jadhav

2 Shweta Gupta

3 Heet Zatakia

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

## **Practical No. 24: Develop a program for providing Bluetooth connectivity.**

### **I. Practical Significance**

Bluetooth is a way to send or receive data between two different devices. Android platform includes support for the Bluetooth framework that allows a device to wirelessly exchange data with other Bluetooth devices.

### **II. Relevant Program Outcomes (POs)**

PO2- Discipline knowledge

PO3- Experiments and practice

PO4- Engineering tools.

PO7- Ethics

### **III. Competency and Practical Skills**

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to communicate with Bluetooth hardware.
2. Able to integrate Bluetooth in any application.

### **IV. Relevant Course Outcome(s)**

1. Configure Android environment and development tools.
2. Develop rich user Interfaces by using layouts and controls.
3. Use User Interface components for android application development.

### **V. Practical Outcome (PrOs)**

Develop a program for Bluetooth.

### **VI. Relevant Affective Domain Related Outcome(s)**

1. Work collaboratively in team.
2. Follow ethical Practices.

### **VII. Minimum Theoretical Background**

Android provides Bluetooth API to perform these different operations.

1. Scan for other Bluetooth devices
2. Get a list of paired devices.
3. Connect to other devices through service discovery.

Android provides Bluetooth Adapter class to communicate with Bluetooth. Create an object of this class by calling the static method `getDefaultAdapter()`. Its syntax is given below.

```
private BluetoothAdapter BA;
BA = BluetoothAdapter.getDefaultAdapter();
```

In order to enable the Bluetooth of your device, call the intent with the following

Bluetooth constant `ACTION_REQUEST_ENABLE`. Its syntax is.

```
Intent turnOn = new Intent(BluetoothAdapter.ACTION_REQUEST_ENABLE);
startActivityForResult(turnOn, 0);
```

Once you enable the Bluetooth, you can get a list of paired devices by calling `getBondedDevices()` method. It returns a set of Bluetooth devices. Its syntax is.

```
private
Set<BluetoothDevice>pairedDevices;
pairedDevices =
BA.getBondedDevices();
```

### VIII. Resources required (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

### IX. Practical related Questions

**Note:** Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

1. Name the methods which are used to enable and disable Bluetooth adapter.
2. Explain the purpose of `ACTION_REQUEST_DISCOVERABLE` Constant.
3. List the uses of `setName(String name)` method.

(Space for answers)

**Ans 1)** `enable()` and `disable()` methods are used to enable and disable Bluetooth adapters.

**Ans 2)** `ACTION_REQUEST_DISCOVERABLE` constant is used for turn on discovery of bluetooth.

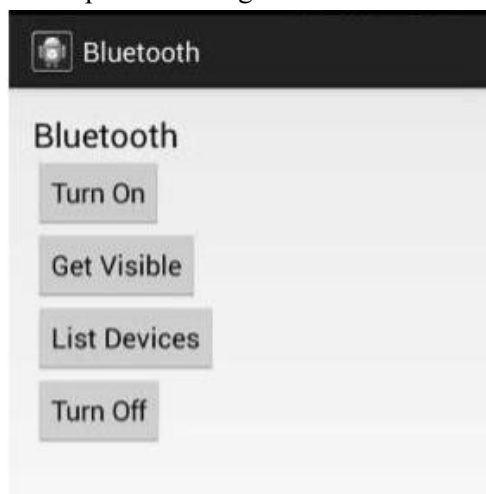
**Ans 3)** The `setName()` method provided by `java.lang.Thread` class is used to change the name of the thread and `setName` assigns a name to an object.

### X. Exercise

**Note:** Faculty must ensure that every group of students use different input value.

(Use blank space for answers or attach more pages if needed)

1. Write a program to turn on, get visible, list devices and turn off Bluetooth with the help of following GUI.



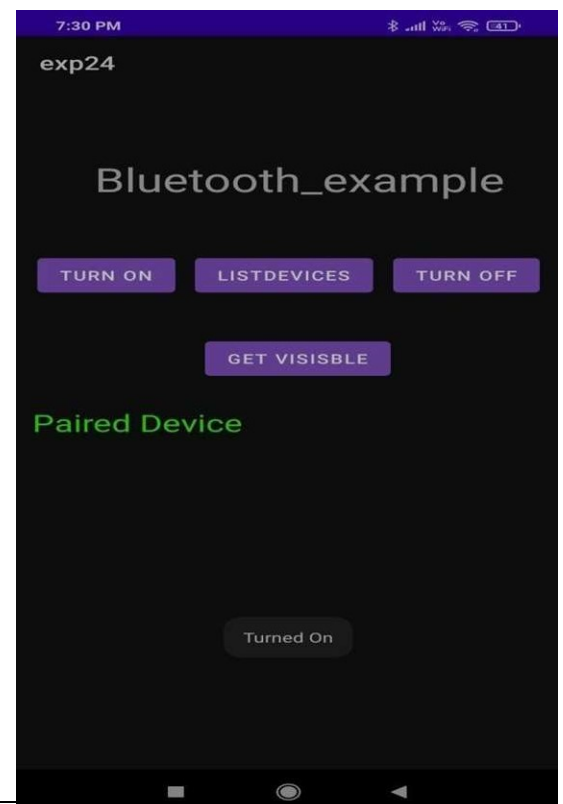
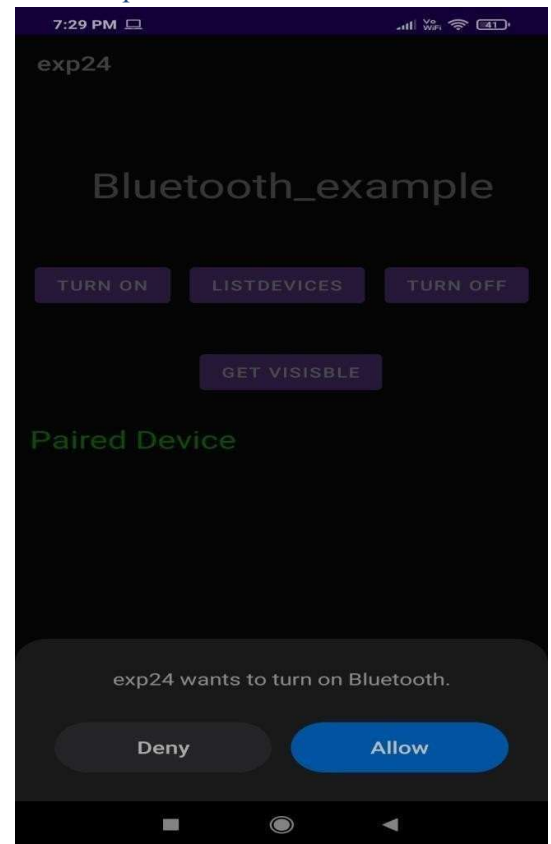
**Activity\_main.xml**

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/a
pk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/layout"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:transitionGroup="true"
tools:context=".MainActivity">

<TextView
android:id="@+id/textView"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginStart="56dp"
android:layout_marginTop="72dp"
android:text="Bluetooth_example"
android:textSize="35dp"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent" />
<Button
android:id="@+id/button"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="54dp"
android:text="Turn On"
app:layout_constraintEnd_toStartOf="@+id/button2"
app:layout_constraintHorizontal_bias="0.5"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toBottomOf="@+id/textView"
android:clickable="true"
android:onClick="on"
/>
<Button
android:id="@+id/button2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="54dp"
android:text="ListDevices"
app:layout_constraintEnd_toStartOf="@+id/button3"
app:layout_constraintHorizontal_bias="0.5"
app:layout_constraintStart_toEndOf="@+id/button"
app:layout_constraintTop_toBottomOf="@+id/textView"
android:onClick="list"/>
<Button
android:id="@+id/button3"
android:layout_width="wrap_content"
android:layout_height="wrap_content"

```



```

android:layout_marginTop="54dp"
android:text="Turn Off"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.5"
app:layout_constraintStart_toEndOf="@+id/button2"
app:layout_constraintTop_toBottomOf="@+id/textView"
android:onClick="off"/>
<Button
android:id="@+id/button4"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginStart="136dp"
android:layout_marginTop="40dp"
android:text="Get Visible"

app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toBottomOf="@+id/button2"
android:onClick="visible"/>
<ListView
android:id="@+id/listView"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="84dp"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toBottomOf="@+id/button4" />
<TextView
android:id="@+id/textView2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginStart="11dp"
android:layout_marginTop="27dp"
android:text="Paired Device"
android:textColor="#ff34ff06"
android:textSize="25dp"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toBottomOf="@+id/button4" />
</androidx.constraintlayout.widget.ConstraintLayout>

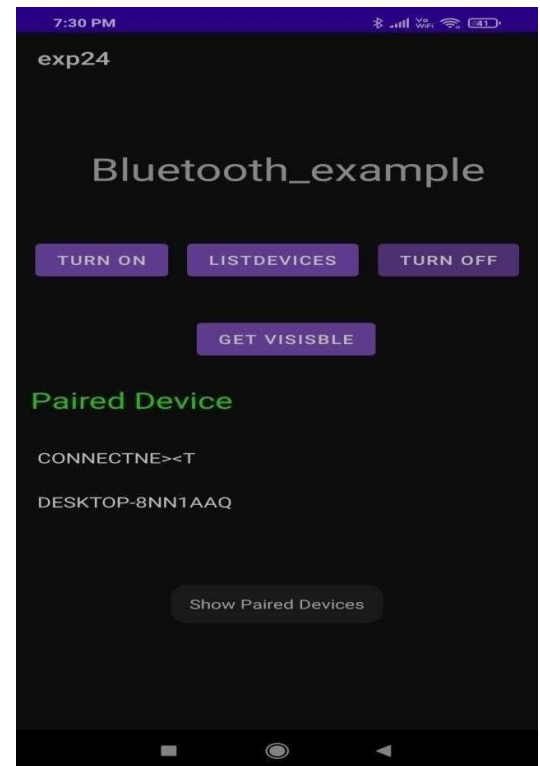
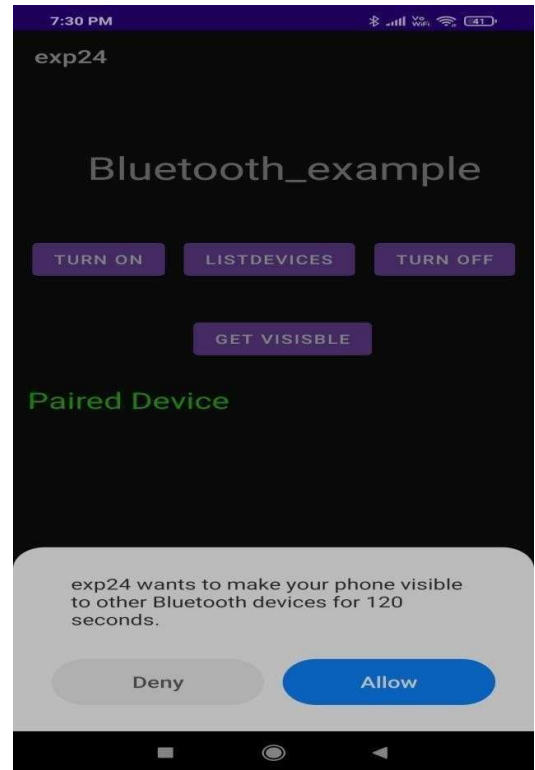
```

### MainActivity.java

```

package com.example.exp24;
import androidx.appcompat.app.AppCompatActivity;
import android.bluetooth.BluetoothAdapter;
import android.bluetooth.BluetoothDevice;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.ListView;
import android.widget.Toast;
import java.util.ArrayList;
import java.util.Set;

```



```

public class MainActivity extends AppCompatActivity
{
 Button b1,b2,b3,b4;
 private BluetoothAdapter BA;
 private Set<BluetoothDevice>pairedDevices;
 ListView lv;
 @Override
 protected void onCreate(Bundle savedInstanceState)
 {
 super.onCreate(savedInstanceState);
 setContentView(R.layout.activity_main);
 b1 = findViewById(R.id.button);
 b2 = findViewById(R.id.button2);
 b3 = findViewById(R.id.button3);
 b4 = findViewById(R.id.button4);
 BA = BluetoothAdapter.getDefaultAdapter();
 lv = findViewById(R.id.listView);
 }
 public void on(View
 v){if(!BA.isEnabled()
){
 Intent turnOn = new Intent(BluetoothAdapter.ACTION_REQUEST_ENABLE);
 startActivityForResult(turnOn,0);
 Toast.makeText(getApplicationContext(),"Turned On", Toast.LENGTH_LONG).show();
 }else {
 Toast.makeText(getApplicationContext(),"Already On", Toast.LENGTH_LONG).show();
 }
 }
 public void off(View
 v){BA.disable();
 Toast.makeText(getApplicationContext(),"Turned Off", Toast.LENGTH_LONG).show();
 }
 public void visible(View v){
 Intent getVisible = new Intent(BluetoothAdapter.ACTION_REQUEST_DISCOVERABLE);
 startActivityForResult(getVisible,0);
 }
 public void list(View
 v){ pairedDevices =
 BA.getBondedDevices();ArrayList list =
 new ArrayList();
 for(BluetoothDevicebt : pairedDevices) list.add(bt.getName());
 Toast.makeText(getApplicationContext(),"Show Paired Devices",
 Toast.LENGTH_LONG).show();
 final ArrayAdapter adapter = new ArrayAdapter(this,
 android.R.layout.simple_list_item_1,list);
 lv.setAdapter(adapter);
 }
}

```

### Android\_Manifest.xml

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
 package="com.example.exp24">
 <uses-permission android:name="android.permission.BLUETOOTH"/>
 <uses-permission android:name="android.permission.BLUETOOTH_ADMIN"/>

```



```

<application
 android:allowBackup="true"
 android:icon="@mipmap/ic_launcher"
 android:label="@string/app_name"
 android:roundIcon="@mipmap/ic_launcher_round"
 android:supportRtl="true"
 android:theme="@style/Theme.Exp24">
 <activity android:name=".MainActivity">
 <intent-filter>
 <action android:name="android.intent.action.MAIN" />

 <category android:name="android.intent.category.LAUNCHER" />
 </intent-filter>
 </activity>
</application>
</manifest>

```

## XI. References / Suggestions for further Reading

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. [https://www.tutorialspoint.com/android/android\\_advanced\\_tutorial.pdf](https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf)
4. <https://developer.android.com>

## XII. Assessment Scheme

Performance indicators		Weightage
<b>Process related (10 Marks)</b>		<b>30%</b>
1.	Logic Formation	10%
2.	Debugging ability	15%
3.	Follow ethical practices	5%
<b>Product related (15 Marks)</b>		<b>70%</b>
4.	Interactive GUI	20%
5.	Answer to Practical related questions	20%
6.	Expected Output	20%
7.	Timely Submission	10%
<b>Total (25 Marks)</b>		<b>100%</b>

### List of student Team Members

- 1 Gaurang Jadhav
- 2 Shweta Gupta
- 3 Heet Zatakia

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

---

## **Practical No. 24: Develop a program for providing Bluetooth connectivity.**

### **I. Practical Significance**

Bluetooth is a way to send or receive data between two different devices. Android platform includes support for the Bluetooth framework that allows a device to wirelessly exchange data with other Bluetooth devices.

### **II. Relevant Program Outcomes (POs)**

PO2- Discipline knowledge

PO3- Experiments and

practice PO4- Engineering  
tools.

PO7- Ethics

### **III. Competency and Practical Skills**

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to communicate with Bluetooth hardware.
2. Able to integrate Bluetooth in any application.

### **IV. Relevant Course Outcome(s)**

1. Configure Android environment and development tools.
2. Develop rich user Interfaces by using layouts and controls.
3. Use User Interface components for android application development.

### **V. Practical Outcome (PrOs)**

Develop a program for Bluetooth.

### **VI. Relevant Affective Domain Related Outcome(s)**

1. Work collaboratively in a team.
2. Follow ethical Practices.

### **VII. Minimum Theoretical Background**

Android provides Bluetooth API to perform these different operations.

1. Scan for other Bluetooth devices
2. Get a list of paired devices.
3. Connect to other devices through service discovery.

Android provides Bluetooth Adapter class to communicate with Bluetooth. Create an object of this class by calling the static method `getDefaultAdapter()`. Its syntax is given below.

```
private BluetoothAdapter BA;
```

```
BA = BluetoothAdapter.getDefaultAdapter();
```

In order to enable the Bluetooth of your device, call the intent with the following Bluetooth constant `ACTION_REQUEST_ENABLE`. Its syntax is.

```
Intent turnOn = new
```

```
Intent(BluetoothAdapter.ACTION_REQUEST_ENABLE);
```

```
startActivityForResult(turnOn, 0);
```

Once you enable the Bluetooth, you can get a list of paired devices by calling `getBondedDevices()` method. It returns a set of Bluetooth devices. Its syntax is.

```
private
```

```
Set<BluetoothDevice>pairedDevi
```

```
ces; pairedDevices =
```

```
BA.getBondedDevices();
```

#### VIII. Resources required (Additional)

Sr . No .	Instrume nt /Object	Specificati on	Quantit y	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

#### IX. Practical related Questions

*Note: Below given are a few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.*

1. Name the methods which are used to enable and disable Bluetooth adapters.
2. Explain the purpose of `ACTION_REQUEST_DISCOVERABLE` Constant.
3. List the uses of the `setName(String name)` method.

(Space for answers)

**Ans 1) enable() and disable()** methods are used to enable and disable Bluetooth adapters.

**Ans 2) ACTION\_REQUEST\_DISCOVERABLE** constant is used for turn on discovery of bluetooth.

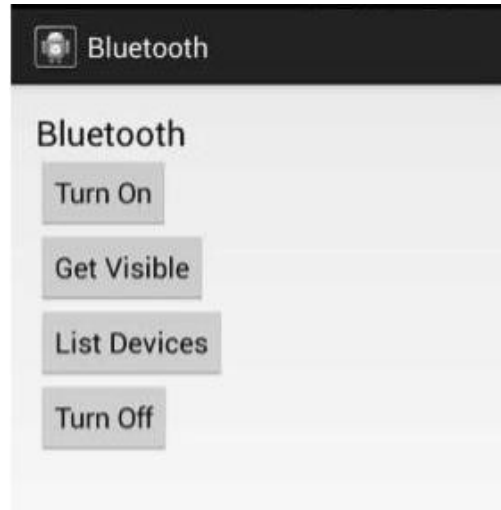
**Ans 3)** The `setName()` method provided by `java.lang.Thread` class is used to change the name of the thread and `setName` assigns a name to an object.

## X. Exercise

*Note: Faculty must ensure that every group of students use different input value.*

(Use blank space for answers or attach more pages if needed)

1. Write a program to turn on, get visible, list devices and turn off Bluetooth with the help of following GUI.



(Space for answers)

### Ans 1) activity\_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
 xmlns:android="http://schemas.android.com/apk/res/android"
 xmlns:app="http://schemas.android.com/apk/res-auto"
 xmlns:tools="http://schemas.android.com/tools"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 tools:context=".MainActivity"
 android:transitionGroup="true"

 android:id="@+id/relativeLayout">

 <TextView
 android:id="@+id/textview"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_marginTop="32dp"
 android:text="Bluetooth Example"
 android:textSize="35dp"
 app:layout_constraintLeft_toLeftOf="parent"
 app:layout_constraintRight_toRightOf="parent"
```

```
app:layout_constraintTop_toTopOf="parent" />
```

```
<Button
```

```
 android:id="@+id/button"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_marginStart="16dp"
 android:layout_marginLeft="16dp"
 android:layout_marginTop="28dp"
 android:layout_toStartOf="@+id/imageView"
 android:layout_toLeftOf="@+id/imageView"
 android:clickable="true"
 android:onClick="on"
 android:text="Turn On"
 app:layout_constraintStart_toStartOf="parent"
 app:layout_constraintTop_toBottomOf="@+id/textview" />
```

```
<Button
```

```
 android:id="@+id/button2"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_below="@+id/textview"
 android:layout_alignBottom="@+id/button"
 android:layout_centerHorizontal="true"
 android:layout_marginStart="144dp"
 android:layout_marginLeft="144dp"
 android:layout_marginTop="24dp"
 android:onClick="visible"

 android:text="Get visible"
 app:layout_constraintStart_toStartOf="parent"
 app:layout_constraintTop_toBottomOf="@+id/button3" />
```

```
<Button
```

```
 android:id="@+id/button3"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_marginStart="136dp"
 android:layout_marginLeft="136dp"
 android:layout_marginTop="28dp"
```

```
 android:onClick="list"
 android:text="List devices"
 app:layout_constraintStart_toStartOf="parent"
 app:layout_constraintTop_toBottomOf="@+id/textview" />
```

<Button

```
 android:id="@+id/button4"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_below="@+id/button"
 android:layout_alignParentStart="true"
 android:layout_alignParentLeft="true"
 android:layout_marginTop="28dp"
 android:layout_marginEnd="16dp"
 android:layout_marginRight="16dp"
 android:onClick="off"
 android:text="turn off"
 app:layout_constraintEnd_toEndOf="parent"
 app:layout_constraintTop_toBottomOf="@+id/textview" />
```

<ListView

```
 android:id="@+id/listView"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_below="@+id/textView2"
 android:layout_alignStart="@+id/button"
 android:layout_alignLeft="@+id/button"
 android:layout_alignParentBottom="true"
 app:layout_constraintStart_toStartOf="@+id/button"
 app:layout_constraintTop_toBottomOf="@+id/textView2" />
```

<TextView

```
 android:id="@+id/textView2"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_below="@+id/button4"
 android:layout_alignStart="@+id/listView"
 android:layout_alignLeft="@+id/listView"

 android:layout_marginStart="24dp"
```

```

 android:layout_marginLeft="24dp"
 android:layout_marginTop="24dp"
 android:text="Paired devices:"
 android:textColor="#ff34ff06"
 android:textSize="25dp"
 app:layout_constraintStart_toStartOf="parent"
 app:layout_constraintTop_toBottomOf="@+id/button2" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

### **MainActivity.java**

```

package com.example.exp24;
import android.app.Activity;
import android.bluetooth.BluetoothAdapter;
import android.bluetooth.BluetoothDevice;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.ListView;
import android.widget.Toast;
import java.util.ArrayList;
import java.util.Set;
public class MainActivity extends Activity {
 Button b1,b2,b3,b4;
 private BluetoothAdapter BA;
 private Set<BluetoothDevice>pairedDevices;

 ListView lv;
 @Override
 protected void onCreate(Bundle savedInstanceState) {
 super.onCreate(savedInstanceState);
 setContentView(R.layout.activity_main);
 b1 = (Button) findViewById(R.id.button);
 b2=(Button)findViewById(R.id.button2);
 b3=(Button)findViewById(R.id.button3);
 b4=(Button)findViewById(R.id.button4);
 BA = BluetoothAdapter.getDefaultAdapter();

 lv = (ListView)findViewById(R.id.listView);
 }
}

```



```

 }
 public void on(View v){
 if (!BA.isEnabled()) {
 Intent turnOn = new
Intent(BluetoothAdapter.ACTION_REQUEST_ENABLE);
 startActivityForResult(turnOn, 0);
 Toast.makeText(getApplicationContext(), "Turned
on",Toast.LENGTH_LONG).show();
 } else {
 Toast.makeText(getApplicationContext(), "Already on",
Toast.LENGTH_LONG).show();
 }
 }
 public void off(View v){
 BA.disable();
 Toast.makeText(getApplicationContext(), "Turned off"
,Toast.LENGTH_LONG).show();
 }
 public void visible(View v){
 Intent getVisible = new
Intent(BluetoothAdapter.ACTION_REQUEST_DISCOVERABLE);
 startActivityForResult(getVisible, 0);
 }
 public void list(View v){
 pairedDevices = BA.getBondedDevices();
 ArrayList list = new ArrayList();
 for(BluetoothDevice bt : pairedDevices) list.add(bt.getName());
 Toast.makeText(getApplicationContext(), "Showing Paired

Devices",Toast.LENGTH_SHORT).show();
 final ArrayAdapter adapter = new
ArrayAdapter(this,android.R.layout.simple_list_item_1, list);
 lv.setAdapter(adapter);
 }
}

AndroidManifest.xml
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"

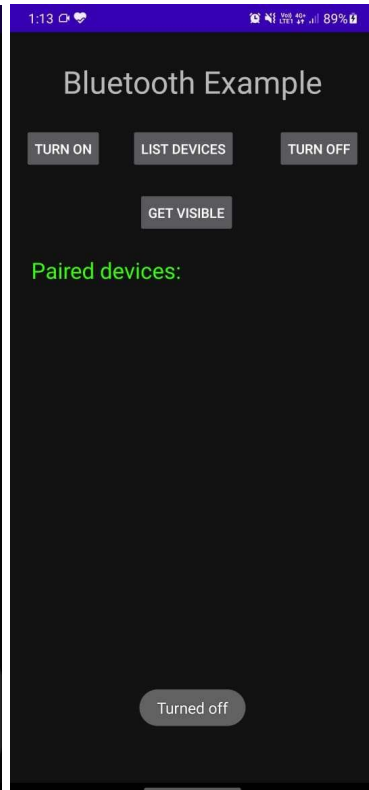
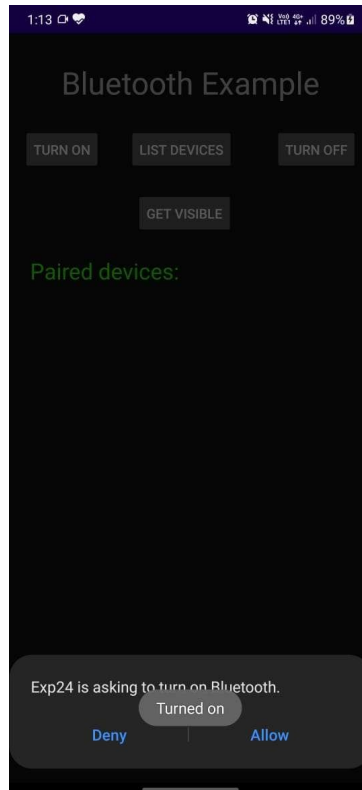
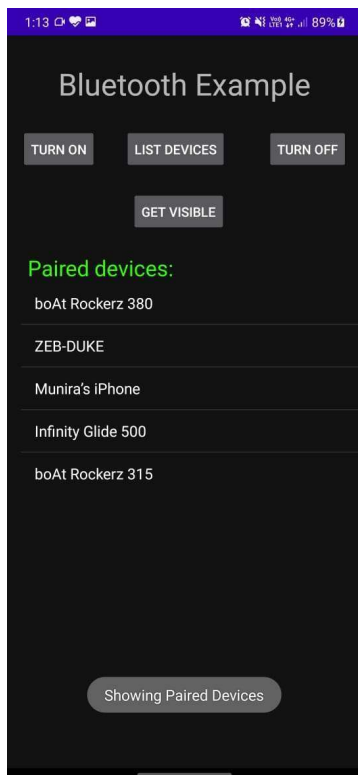
 package="com.example.exp24">

```

```

<uses-permission android:name="android.permission.BLUETOOTH"/>
<uses-permission android:name="android.permission.BLUETOOTH_ADMIN"/>
<application
 android:allowBackup="true"
 android:icon="@mipmap/ic_launcher"
 android:label="@string/app_name"
 android:roundIcon="@mipmap/ic_launcher_round"
 android:supportsRtl="true"
 android:theme="@style/Theme.Exp24">
 <activity android:name=".MainActivity">
 <intent-filter>
 <action android:name="android.intent.action.MAIN" />
 <category android:name="android.intent.category.LAUNCHER" />
 </intent-filter>
 </activity>
</application>
</manifest>

```



**XI. References / Suggestions for further Reading**

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. [https://www.tutorialspoint.com/android/android\\_advanced\\_tutorial.pdf](https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf)
4. <https://developer.android.com>

**XII. Assessment Scheme**

Performance indicators		Weightage
Process related (10 Marks)		30%
1 .	Logic Formation	10%
2 .	Debugging ability	15%
3 .	Follow ethical practices	5%
Product related (15 Marks)		70%
4 .	Interactive GUI	20%
5 .	Answer to Practical related questions	20%
6 .	Expected Output	20%
7 .	Timely Submission	10%
Total (25 Marks)		100%

***List of student Team Members***

1 [Heet Zatakia](#)

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

## Practical No. 26: Perform Async task using SQLite.

### I. Practical Significance

Android AsyncTask is an abstract class provided by Android which gives us the liberty to perform heavy tasks in the background and keep the UI thread light thus making the application more responsive.

### II. Relevant Program Outcomes (POs)

PO2- Discipline knowledge

PO3- Experiments and practice

### III. Competency and Practical Skills

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to develop the application using the Async task.
2. Able to keep the GUI light for heavy database applications.

### IV. Relevant Course Outcome(s)

1. Configure Android environment and development tools.
2. Develop rich user Interfaces by using layouts and controls.
3. Create Android application using database

### V. Practical Outcome (PrOs)

Demonstrate Async tasks using SQLite.

### VI. Relevant Affective Domain related Outcomes

1. Work collaboratively in team
2. Follow ethical Practices.

### VII. Minimum Theoretical Background

Android application runs on a single thread when launched. Due to this single thread model, tasks that take longer time to fetch the response can make the application non-responsive. To avoid this, we use android AsyncTask to perform the heavy tasks in background on a dedicated thread and pass the results back to the UI thread. Hence use of AsyncTask in android application keeps the UI thread responsive at all times.

### VIII. Resources required (Additional)

S r · N o.	Instrument /Object	Specificatio n	Quantit y	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

## IX. Practical related Questions

*Note: Below given are a few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.*

1. List the basic methods used in an android AsyncTask class.
2. Differentiate between AsyncTask and Services.
3. Name the method used, if a process takes a long time to do its work?

(Space for answers)

### **Ans 1) Basic methods used in an android AsyncTask class are as follows:**

1. `doInBackground()` : This method contains the code which needs to be executed in background. In this method we can send results multiple times to the UI thread by the `publishProgress()` method. To notify that the background processing has been completed we just need to use the return statements
2. `onPreExecute()` : This method contains the code which is executed before the background processing starts
3. `onPostExecute()` : This method is called after the `doInBackground` method completes processing. Result from `doInBackground` is passed to this method
4. `onProgressUpdate()` : This method receives progress updates from `doInBackground` method, which is published via `publishProgress` method, and this method can use this progress update to update the UI thread

**Ans 2)** In some cases it is possible to accomplish the same task with either an AsyncTask or a Service however usually one is better suited to a task than the other. AsyncTasks are designed for once-off time-consuming tasks that cannot be run on the UI thread. A common example is fetching/processing data when a button is pressed. Services are designed to be continually running in the background. In the example above of fetching data when a button is pressed, you could start a service, let it fetch the data, and then stop it, but this is inefficient. It is far faster to use an AsyncTask that will run once, return the data, and be done.

If you need to be continually doing something in the background, though, a Service is your best bet. Examples of this include playing music, continually checking for new data, etc.

**Ans 3)** AsyncTasks should ideally be used for short operations (a few seconds at the most.) If you need to keep threads running for long periods of time, it is highly recommended you use the various APIs provided by the `java.util.concurrent` package such as `Executor`, `ThreadPoolExecutor` and `FutureTask`.

## X. Exercise

**Note:** Faculty must ensure that every group of students use different input values.

(Use blank space for answers or attach more pages if needed)

1. Write a program to insert data in SQLite database using AsyncTask

(Space for answers)

Ans1) 1. activity\_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
 xmlns:android="http://schemas.android.com/apk/res/android"
 xmlns:app="http://schemas.android.com/apk/res-auto"
 xmlns:tools="http://schemas.android.com/tools"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 tools:context=".MainActivity">
 <EditText
 android:id="@+id/editTextRecordNum"
 android:layout_width="318dp"
 android:layout_height="50dp"
 android:layout_alignParentLeft="true"
 android:layout_alignParentTop="true"
 android:layout_marginStart="44dp"
 android:layout_marginLeft="44dp"
 android:layout_marginTop="124dp"
 android:ems="10"
 android:fontFamily="@font/poppins_medium"
 android:inputType="number"
 android:singleLine="true"
 android:textColor="#FFFFFF"
 app:layout_constraintStart_toStartOf="parent"
 app:layout_constraintTop_toTopOf="parent"
 tools:ignore="MissingConstraints" />
 <Button
```

```
 android:id="@+id/buttonNormalInsert"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_below="@+id/editTextRecordNum"
 android:layout_alignLeft="@+id/editTextRecordNum"
 android:layout_marginStart="44dp"
 android:layout_marginLeft="44dp"
 android:layout_marginTop="44dp"
 android:fontFamily="@font/poppins_medium"
 android:text="Normal Insert"
 app:layout_constraintStart_toStartOf="parent"
 app:layout_constraintTop_toBottomOf="@+id/editTextRecordNum" />
 <Button
 android:id="@+id/buttonFastInsert"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_alignBaseline="@+id/buttonNormalInsert"
 android:layout_alignBottom="@+id/buttonNormalInsert"
 android:layout_marginTop="44dp"
 android:layout_marginEnd="48dp"
 android:layout_marginRight="48dp"
 android:layout_toRightOf="@+id/buttonN
```

```

ormalInsert"

android:fontFamily="@font/poppins_medium"
 android:text="Fast Insert"

app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintTop_toBottomOf="@+id/editTextRecordNum" />
<TextView
 android:id="@+id/textViewStatus"

android:layout_width="wrap_content"

android:layout_height="wrap_content"

android:layout_below="@+id/buttonNormalInsert"

```

```

android:layout_alignLeft="@+id/buttonNormalInsert"
 android:layout_marginStart="16dp"
 android:layout_marginLeft="16dp"
 android:layout_marginTop="36dp"

android:fontFamily="@font/poppins_medium"
 android:padding="10dp"
 android:text="Status"
 android:textColor="#FFFFFF"
 android:textSize="18sp"

app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/buttonNormalInsert" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

## 2. MainActivity.java

```

package com.example.coexp26;

import android.os.AsyncTask;
import android.os.Bundle;
import android.app.Activity;
import android.view.View;
import android.widget.EditText;
import android.widget.TextView;
public class MainActivity extends Activity
{
 final String TAG = "MainActivity.java";
 EditText editTextRecordNum;
 TextView tvStatus;

 @Override
 protected void onCreate(Bundle savedInstanceState) {
 super.onCreate(savedInstanceState);

 setContentView(R.layout.activity_main);
 View.OnClickListener handler = new
 View.OnClickListener() {
 public void onClick(View v) {
 switch (v.getId()) {
 case R.id.buttonNormalInsert:
 new
 AsyncInsertData("normal").execute();
 break;

```

```

 case R.id.buttonFastInsert:
 new
 AsyncInsertData("fast").execute();
 break;
 }
 }
 };
 // EditText for entering desired
 number of records to be inserted
 editTextRecordNum = (EditText)
 findViewById(R.id.editTextRecordNum);
 // Button for normal and fast insert

 findViewById(R.id.buttonNormalInsert).set
 OnClickListener(handler);

 findViewById(R.id.buttonFastInsert).setO
 nClickListener(handler);
 // status TextView
 tvStatus = (TextView)
 findViewById(R.id.textViewStatus);
 }
 // we used AsyncTask so it won't block
 the UI thread during inserts.
 class AsyncInsertData extends
 AsyncTask<String, String, String> {
 DatabaseHandler databaseHandler;
 String type;

```

```

 long timeElapsed;
 protected AsyncInsertData(String
type){
 this.type = type;
 this.databaseHandler = new
DatabaseHandler(MainActivity.this);
 }
 // @type - can be 'normal' or 'fast'
 @Override
 protected void onPreExecute() {
 super.onPreExecute();
 tvStatus.setText("Inserting " +
editTextRecordNum.getText() + "
records...");
 }
 @Override
 protected String
doInBackground(String... aurl) {
 try {
 // get number of records to be
inserted
 int insertCount =
Integer.parseInt(editTextRecordNum.getT
ext().toString());

 // empty the table

 databaseHandler.deleteRecords();
 // keep track of execution time

```

```

 long lStartTime =
System.nanoTime();
 if (type.equals("normal")) {

 databaseHandler.insertNormal(insertCount
);
 } else {

 databaseHandler.insertFast(insertCount);
 }
 // execution finished
 long lEndTime =
System.nanoTime();
 // display execution time
 timeElapsed = lEndTime -
lStartTime;
 } catch (Exception e) {
 e.printStackTrace();
 }
 return null;
}
protected void onPostExecute(String
unused) {
 tvStatus.setText("Done inserting "
+ databaseHandler.countRecords() + "
records. Time elapsed: " + timeElapsed /
1000000 + " ms.");
} } }

```

### 3. DatabaseHandler.java

```

package com.example.coexp26;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import
android.database.sqlite.SQLiteDatabase;
import
android.database.sqlite.SQLiteOpenHelper
;
import
android.database.sqlite.SQLiteStatement;
import android.util.Log;
public class DatabaseHandler extends
SQLiteOpenHelper {
 // for our logs

 public static final String TAG =

```

```

"DatabaseHandler.java";
 // database version
 private static final int
DATABASE_VERSION = 7;
 // database name
 protected static final String
DATABASE_NAME = "NinjaDatabase2";
 // table details
 public String tableName = "locations";
 public String fieldObjectId = "id";
 public String fieldObjectName =
"name";
 public String fieldObjectDescription =
"description";
 // constructor

 public DatabaseHandler(Context

```



```

context) {
 super(context, DATABASE_NAME,
 null, DATABASE_VERSION);
}
// creating table
@Override
public void onCreate(SQLiteDatabase
db) {
 String sql = "";
 sql += "CREATE TABLE " +
tableName;
 sql += " (";
 sql += fieldObjectId + " INTEGER
PRIMARY KEY AUTOINCREMENT, ";
 sql += fieldObjectName + " TEXT, ";
 sql += fieldObjectDescription + "
TEXT ";
 sql += ") ";
 db.execSQL(sql);
 String INDEX = "CREATE UNIQUE
INDEX locations_index ON "
 + tableName + " (name,
description)";
 db.execSQL(INDEX);
}
// When upgrading the database, it will
drop the current table and recreate.
@Override
public void onUpgrade(SQLiteDatabase
db, int oldVersion, int newVersion) {
 String sql = "DROP TABLE IF
EXISTS " + tableName;
 db.execSQL(sql);
 onCreate(db);
}
// insert data using transaction and
prepared statement
public void insertFast(int insertCount) {
 // you can use INSERT only
 String sql = "INSERT OR REPLACE
INTO " + tableName + " (name,
description) VALUES (?, ?)";
 SQLiteDatabase db =
this.getWritableDatabase();
 db.beginTransactionNonExclusive();
 // db.beginTransaction();
 SQLiteStatement stmt =
db.compileStatement(sql);
 for(int x=1; x<=insertCount; x++){
 stmt.bindString(1, "Name # " + x);
 stmt.bindString(2, "Description # "
+ x);
 stmt.execute();
 stmt.clearBindings();
 }
}

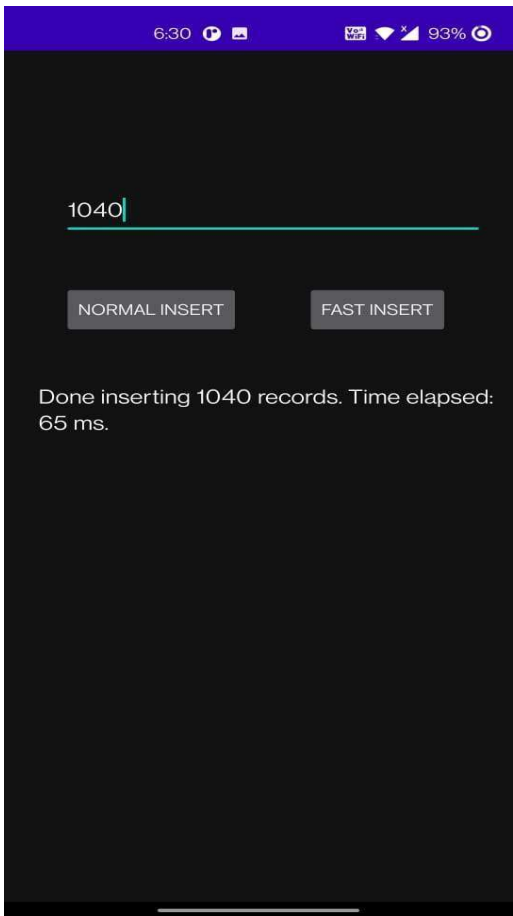
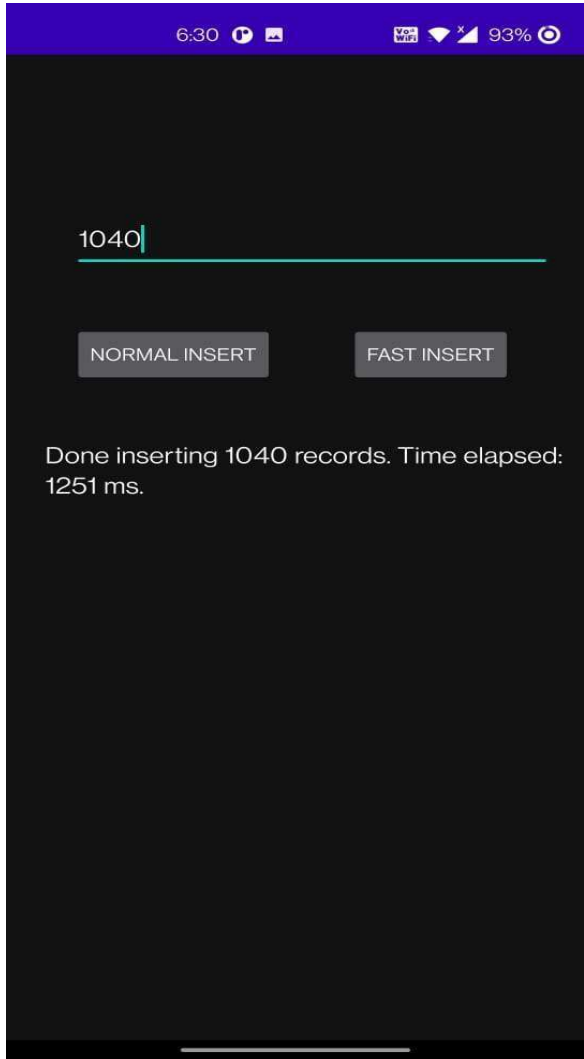
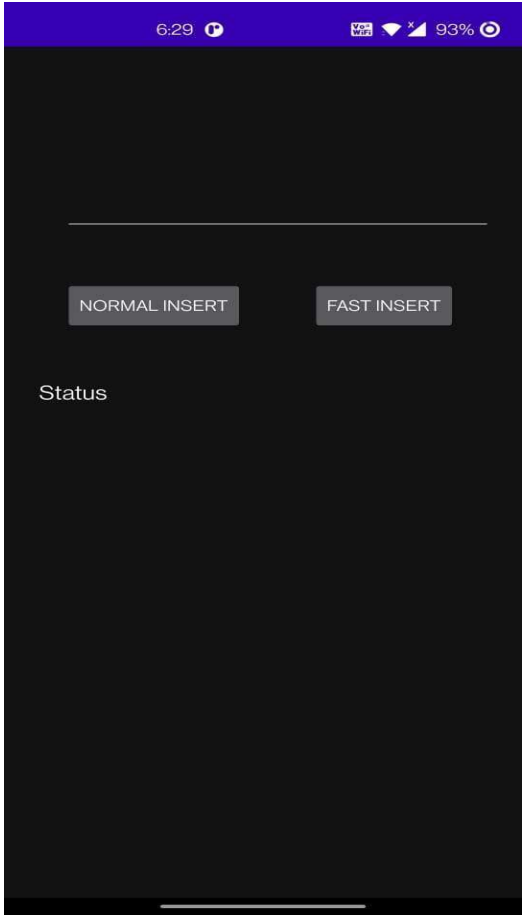
```

```

}
db.setTransactionSuccessful();
db.endTransaction();
db.close();
}
// inserts the record without using
transaction and prepare statement

public void insertNormal(int
insertCount){
 try{
 SQLiteDatabase db =
this.getWritableDatabase();
 for(int x=1; x<=insertCount; x++){
 ContentValues values = new
ContentValues();
 values.put(fieldObjectName,
"Name # " + x);
 values.put(fieldObjectDescription,
"Description # " + x);
 db.insert(tableName, null,
values);
 }
 db.close();
 } catch (Exception e) {
 e.printStackTrace();
 }
}
// deletes all records
public void deleteRecords() {
 SQLiteDatabase db =
this.getWritableDatabase();
 db.execSQL("delete from " +
tableName);
 db.close();
}
// count records
public int countRecords() {
 SQLiteDatabase db =
this.getWritableDatabase();
 Cursor cursor =
db.rawQuery("SELECT count(*) from " +
tableName, null);
 cursor.moveToFirst();
 int recCount = cursor.getInt(0);
 cursor.close();
 db.close();
 return recCount;
}
}

```



**XI. References / Suggestions for further Reading**

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. [https://www.tutorialspoint.com/android/android\\_advanced\\_tutorial.pdf](https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf)
4. <https://developer.android.com>

**XII. Assessment Scheme**

Performance indicators		Weightage
Process related (10 Marks)		30%
1 .	Logic Formation	10%
2 .	Debugging ability	15%
3 .	Follow ethical practices	5%
Product related (15 Marks)		70%
4 .	Interactive GUI	20%
5 .	Answer to Practical related questions	20%
6 .	Expected Output	20%
7 .	Timely Submission	10%
Total (25 Marks)		100%

***List of student Team Members***

1 [Heet Zatakia](#)

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

**Practical No. 28: Create login application where you will have to validate username and password till the username and password is not validated, login button should remain disabled.**

**I. Practical Significance**

Today Login and Registration form in Android are part of every application out there. So, when we are programming, we work with many registration forms. Forms can be very different from a simple login or registration to a complex ordering form for any application.

**II. Relevant Program Outcomes (POs)**

PO1- Basic knowledge

PO2- Discipline  
knowledge

PO3- Experiments and  
practice PO7- Ethics

**III. Competency and Practical Skills**

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to develop various registration forms.
2. Able to develop the application with various kinds of validations.

**IV. Relevant Course Outcome(s)**

1. Configure Android environment and development tools.
2. Develop rich user Interfaces by using layouts and controls.
3. Use User Interface components for android application development.

**V. Practical Outcome (PrOs)**

Create a sample application with a login module. (Check username and password) On successful login, Change Text View “Login Successful”. And on login fail, alert the user using Toast “Login fail”.

**VI. Relevant Affective Domain Related Outcome(s)**

1. Work collaboratively in team
2. Follow ethical Practices.

**VII. Minimum Theoretical Background**

With registration, how you can check data that the user has entered with simple validation. Validation can check many conditions. We can verify if an email address is a valid email and if a user entered all the required data, for instance, we check if Edit Text is empty for the first and last name. We can prepare a way to notify the user that the data is not valid. On login activity, we should check for password length. There are a few things login and registration form need:

- Clean user interface.
- Validation (check if the email is an email and if the user entered all the data).
- Notifications for the user that the data is incorrect.
- Instructions for the user (e.g. how many characters are required for password).

#### VIII. Resources required (Additional)

S r. N o.	Instru ment /Object	Specificati on	Quanti ty	Remarks
1	Android enable d smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

#### IX. Practical related Questions

*Note: Below given are a few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.*

1. Explain validation of user input?
2. List and explain various GUI components used to design the login form with validation.
3. Differentiate between Text View and Edit Text View.

**(Space for answers)**

**Ans 1)** Input Validation eliminates the errors that can be done by the user while giving inputs to our app. For example if we want to get the user's email we can check if the entered email is a valid email or not before storing it inside the database.

**Ans 2)** Various GUI components used to design the login form with validation are as follows:

1. Add the first EditText for the first name.
  - a. Name it "firstName" (use the id attribute to name the component).
  - b. Set text to "".
  - c. Set the hint to "First name".
  - d. This one should have 3 constraints; one links to the top of the screen, second to left border and third to the right border.
2. Add EditText for the last name.
  - a. Name it "lastName".
  - b. Set text to "".
  - c. Set the hint to "First name".
  - d. This one should have 3 constraints; one linked to the bottom of "First name", second to left border and third to the right border.

3. Add EditText for postal address.
  - a. Name it "address".
  - b. Set text to "".
  - c. Set a hint to "Postal address".
  - d. This one should have 3 constraints; one linked to the bottom of "Last name", second to left border and third to the right border.
4. Add EditText for email.
  - a. Name it "email".
  - b. Set text to "".
  - c. Set a hint to "Email".
  - d. This one should have 3 constraints; one linked to the bottom of "Postal address", second to left border and third to the right border.
5. Add a Button for registration.
  - a. Name it "registration".
  - b. Set text to "Registration".

**Ans 3)** EditText is used for user input. TextView is used to display text and is not editable by the user. TextView is the widget used when you want the user to View the Text (such as a label, etc) and EditText used when you want the user to be able to edit the text.

#### X. Exercise

**Note:** Faculty must ensure that every group of students use different input values.

(Use blank space for answers or attach more pages if needed)

1. Write a program to create the login form with necessary validations like length of username and password, empty text fields, count of unsuccessful login attempts. Display the login successful/Unsuccessful toastmessage.

**(Space for answers)**

Ans1) 1. activity\_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout

xmlns:android="http://schemas.android.com/apk/res/android"

xmlns:app="http://schemas.android.com/apk/res-auto"
```

```
xmlns:tools="http://schemas.android.com/tools"

 android:layout_width="match_parent"
 android:layout_height="match_parent"
 tools:context=".MainActivity">
 <EditText

 android:id="@+id/User_password"

 android:layout_width="wrap_content"
```

```

 android:layout_height="wrap_content"

 android:layout_below="@+id/User_email"

 android:layout_alignParentStart="true"

 android:layout_alignParentLeft="true"

 android:layout_alignParentEnd="true"

 android:layout_alignParentRight="true"
 android:layout_marginStart="49dp"
 android:layout_marginLeft="49dp"
 android:layout_marginTop="25dp"
 android:layout_marginEnd="53dp"
 android:layout_marginRight="53dp"
 android:ems="10"
 android:hint="Password:"
 android:inputType="textPassword"
 android:padding="10dp" />
 <TextView
 android:id="@+id/User_Text"

 android:layout_width="wrap_content"

 android:layout_height="wrap_content"

 android:layout_alignParentTop="true"

 android:layout_alignParentEnd="true"

 android:layout_alignParentRight="true"
 android:layout_marginTop="113dp"
 android:layout_marginEnd="106dp"
 android:layout_marginRight="106dp"

 android:fontFamily="@font/poppins_semi
 bold"
 android:text="User Login"

```

```

 android:textColor="#FFFFFF"
 android:textSize="40dp" />
 <Button
 android:id="@+id/User_login"
 android:layout_width="150dp"

 android:layout_height="wrap_content"

 android:layout_below="@+id/User_passw
 ord"

 android:layout_centerHorizontal="true"
 android:layout_marginTop="25dp"
 android:text="Login" />
 <EditText
 android:id="@+id/User_email"
 android:layout_width="310dp"

 android:layout_height="wrap_content"

 android:layout_below="@+id/User_Text"

 android:layout_alignParentStart="true"

 android:layout_alignParentLeft="true"

 android:layout_alignParentEnd="true"

 android:layout_alignParentRight="true"
 android:layout_marginStart="49dp"
 android:layout_marginLeft="49dp"
 android:layout_marginTop="80dp"
 android:layout_marginEnd="52dp"
 android:layout_marginRight="52dp"
 android:hint="Email:"

 android:inputType="textEmailAddress"
 android:padding="10dp" />
</RelativeLayout>

```

## 2. MainActivity.java

```
package com.example.coexp28;
import
androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import
android.icu.text.UnicodeSetSpanner;
import android.os.Bundle;
import android.util.Patterns;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
public class MainActivity extends
AppCompatActivity {
 Button login;
 EditText Uemail,Upass;
 public String Useremail,Upassword;
 @Override
 protected void onCreate(Bundle
savedInstanceState) {
 super.onCreate(savedInstanceState);

 setContentView(R.layout.activity_main);
 login =
 findViewById(R.id.User_login);
 login.setOnClickListener(new
View.OnClickListener() {
 @Override
 public void onClick(View v) {
 check();
 }
 });

 Uemail =
 findViewById(R.id.User_email);
 Upass =
 findViewById(R.id.User_password);
 }
 public void check(){
 Useremail =
```

```
Uemail.getText().toString().trim();
 Upassword =
 Upass.getText().toString().trim();
 if(Useremail.isEmpty()) {
 Uemail.setError("Enter an e-mail,it
is blank");
 Uemail.requestFocus();
 return;
 }

 if(!Patterns.EMAIL_ADDRESS.matcher(U
useremail).matches()){
 Uemail.setError("Enter an e-
mail");
 Uemail.requestFocus();
 return;
 }
 if(Upassword.isEmpty()) {
 Upass.setError("Enter a password ,
it is blank");
 Upass.requestFocus();
 return;
 }
 if(Upassword.length()<6){
 Upass.setError("Minimum length is
6 characters");
 Upass.requestFocus();
 return;
 }

 if(Useremail.equals("heetzatakia@vesp.c
om") && Upassword.equals("heetzatakia
@10")){
 Toast.makeText(MainActivity.this,"Welco
me User",Toast.LENGTH_LONG).show();
 Intent intent = new
Intent(MainActivity.this,MainActivity2.cla
ss);
 startActivity(intent);
 }
 }
}
```



```

 else{
 Toast.makeText(this,"Wrong email
or
password",Toast.LENGTH_LONG).show()
 }
 }
}

```

### 3. activity\_main2.xml

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
 xmlns:android="http://schemas.android.com/apk/res/android"
 xmlns:app="http://schemas.android.com/apk/res-auto"
 xmlns:tools="http://schemas.android.com/tools"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 tools:context=".MainActivity2">
</androidx.constraintlayout.widget.ConstraintLayout>

```

### 5. MainActivity2.java

```

package com.example.coexp28;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class MainActivity2 extends AppCompatActivity {
 @Override
 protected void onCreate(Bundle savedInstanceState) {
 super.onCreate(savedInstanceState);
 setContentView(R.layout.activity_main2);
 }
}

```

### 5. AndroidManifest.xml

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
 package="com.example.coexp28">

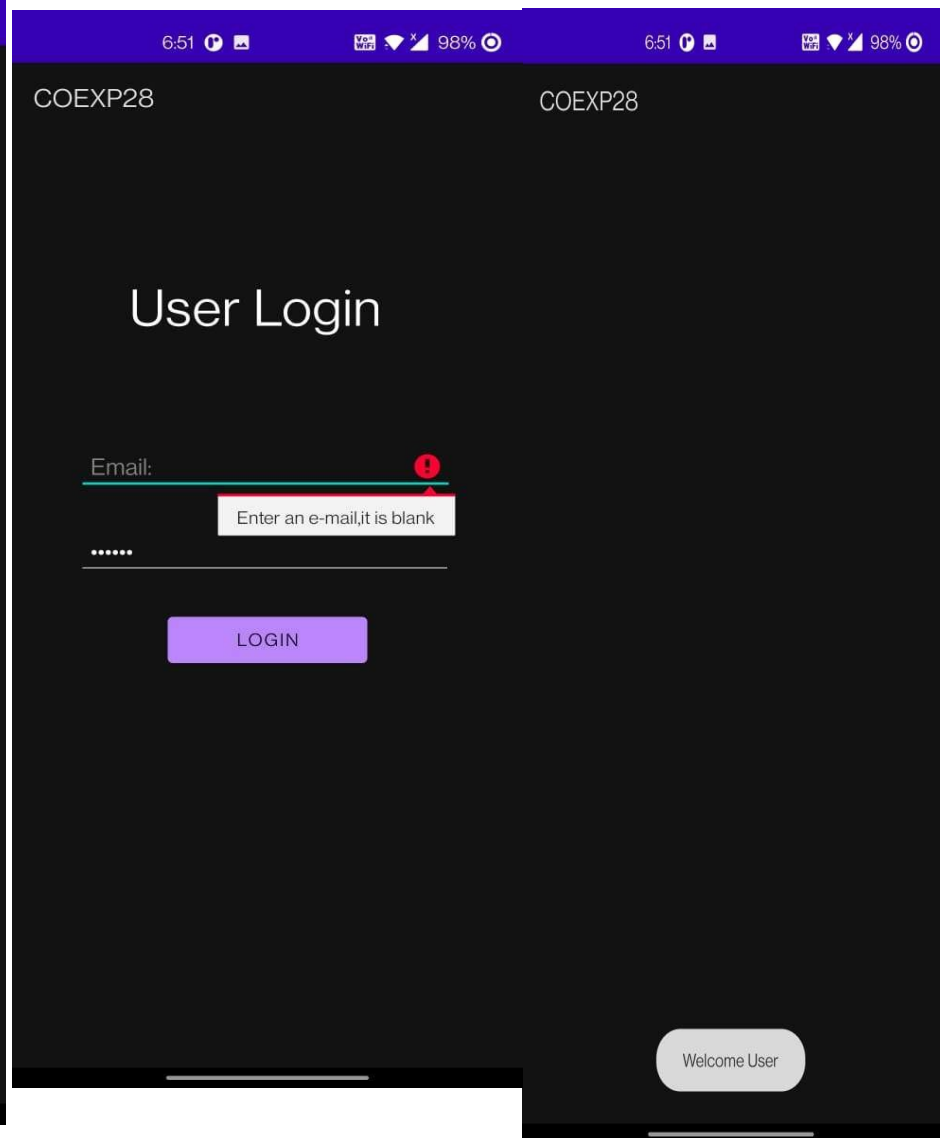
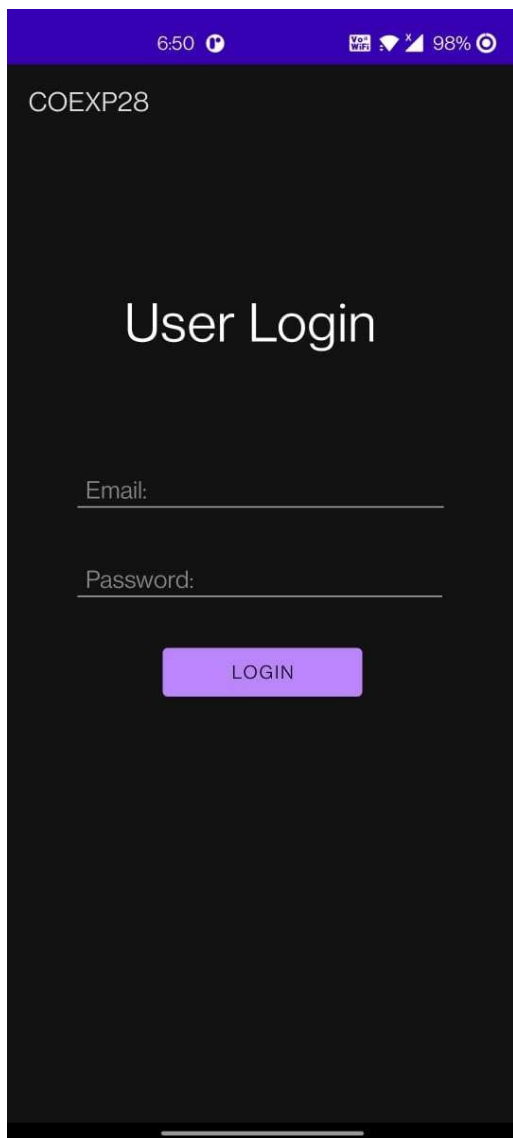
 <application
 android:allowBackup="true"
 android:icon="@mipmap/ic_launcher"
 android:label="@string/app_name"
 android:roundIcon="@mipmap/ic_launcher_round"
 android:supportsRtl="true"
 android:theme="@style/Theme.COEXP28">
 <activity android:name=".MainActivity2"></activity>
 <activity android:name=".MainActivity">

```

```
<intent-filter>
 <action android:name="android.intent.action.MAIN" />

 <category android:name="android.intent.category.LAUNCHER" />
</intent-filter>
</activity>
</application>

</manifest>
```



**XI. References / Suggestions for further Reading**

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. [https://www.tutorialspoint.com/android/android\\_advanced\\_tutorial.pdf](https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf)
4. <https://developer.android.com>

**XII. Assessment Scheme**

Performance indicators		Weightage
Process related (10 Marks)		30%
1 .	Logic Formation	10%
2 .	Debugging ability	15%
3 .	Follow ethical practices	5%
Product related (15 Marks)		70%
4 .	Interactive GUI	20%
5 .	Answer to Practical related questions	20%
6 .	Expected Output	20%
7 .	Timely Submission	10%
Total (25 Marks)		100%

***List of student Team Members***

1 [Heet Zatakia](#)

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

## **Practical No. 29: Develop a program to: a) Send SMS b) Receive SMS**

### **I. Practical Significance**

Android devices can send and receive messages to or from any other phone that supports Short Message Service (SMS). Android offers the Messenger application that can send and receive SMS messages.

### **II. Relevant Program Outcomes (POs)**

PO2- Discipline knowledge

PO3- Experiments and  
practice PO4-Engineering  
tools

PO7- Ethics

### **III. Competency and Practical Skills**

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Develop the application to send SMS.
2. Develop the application to receive SMS with customized GUI.

### **IV. Relevant Course Outcome(s)**

1. Configure Android environment and development tools.
2. Develop rich user Interfaces by using layouts and controls.
3. Interpret features of Android operating system

### **V. Practical Outcome (PrOs)**

Develop a program to: a) Send SMS b) Receive SMS.

### **VI. Relevant Affective Domain related Outcomes**

1. Work collaboratively in team
2. Follow ethical Practices.

### **VII. Minimum Theoretical Background**

A host of third-party apps for sending and receiving SMS messages are also available in Google Play. The SMS protocol was primarily designed for user-to-user communication and is not well-suited for apps that want to transfer data. You should not use SMS to send data messages from a web server to your app on a user device. SMS is neither encrypted nor strongly authenticated on either the network or the device. Access to the SMS features of an Android device is protected by user permissions. Just as your app needs the user's permission to use phone features, so also does an app need the user's permission to directly use SMS features.

You have two choices for sending SMS messages:

- Use an implicit Intent to launch a messaging app such as Messenger, with the ACTION\_SENDTO action.
- Send the SMS message using the sendTextMessage() method or other methods of the SmsManager class.

To receive SMS messages, the best practice is to use the `onReceive()` method of the Broadcast Receiver class. The Android framework sends out system broadcasts of events such as receiving an SMS message, containing intents that are meant to be received using a Broadcast Receiver. Your app receives SMS messages by listening for the `SMS_RECEIVED_ACTION` broadcast.

**Methods :**

- 1 `ArrayList<String> divideMessage(String text) :-` This method divides a message text into several fragments, none bigger than the maximum SMS message size.
- 2 `static SmsManager getDefault() :-` This method is used to get the default instance of the Sms Manager
- 3 `void sendDataMessage(String destination Address, String scAddress, short destinationPort, byte[] data, PendingIntent sentIntent, PendingIntent deliveryIntent):-` This method is used to send a data based SMS to a specific application port.
- 4 `void sendTextMessage(String destinationAddress, String scAddress, String text, PendingIntent sentIntent, PendingIntent deliveryIntent) :-` Send a text based SMS.

**VIII. Resources required (Additional)**

S r. N o.	Instrume nt /Object	Specificati on	Quantit y	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

**IX. Practical related Questions**

*Note: Below given are a few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.*

1. Explain the use of `SmsManagerClass`.
2. List changes that need to be done in `AndroidManifest.XML` file to send and receive messages.

**(Space for answers)**

**Ans 1)** `SmsManager` that supports both GSM and CDMA. Manages SMS operations such as sending data, text, and pdu SMS messages. Get this object by calling the static method `SmsManager`.

**Ans 2)** Changes that need to be made in `AndroidManifest.xml` file to send and receive messages are as follows:

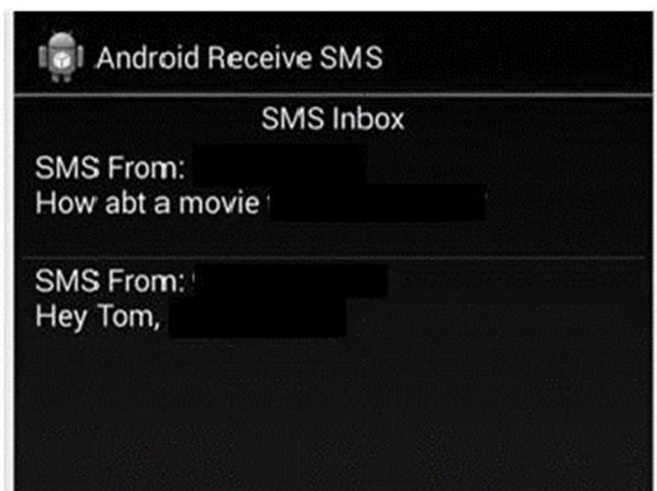
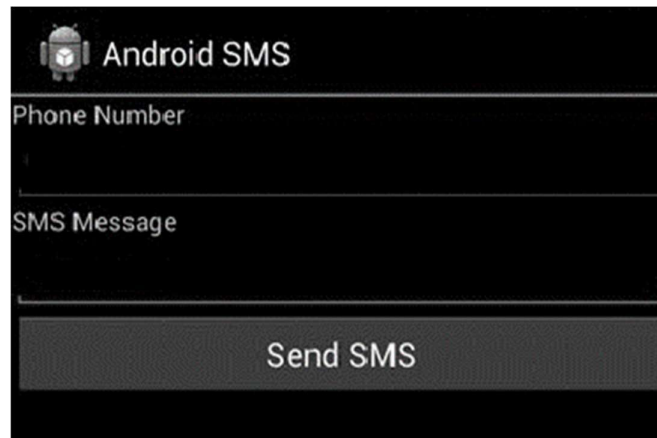
1. Pass data (the phone number) and the message with the implicit intent.
2. Create an app that sends SMS messages using the `SmsManager` class.
3. Check for the SMS permission, which can change at any time.
4. Request permission from the user, if necessary, to send SMS messages.

## X. Exercise

**Note:** Faculty must ensure that every group of students use different input value.

(Use blank space for answers or attach more pages if needed)

1. Write a program to send and receive SMS, make use of the following GUI.



(Space for answers)

Ans1) 1. activity\_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout
```

```
xmlns:android="http://schemas.android.co
m/apk/res/android"
```

```
xmlns:app="http://schemas.android.com/a
pk/res-auto"
```

```
xmlns:tools="http://schemas.android.com/t
```

```
ools"
```

```
android:layout_width="match_parent"
```

```
android:layout_height="match_parent"
```

```
android:orientation="vertical"
```

```
android:padding="16dp"
```

```
android:gravity="center_horizontal"
```

```
tools:context=".MainActivity">
```

```
<EditText
```

```
android:id="@+id/et_phone"
```

```

 android:layout_width="match_parent"

 android:layout_height="wrap_content"
 android:hint="Enter Phone Number"
 android:padding="12dp"
 android:maxLength="10"
 android:inputType="phone"

 android:background="@color/cardview_s
 hadow_start_color"/>

 <EditText
 android:id="@+id/et_message"

 android:layout_width="match_parent"

 android:layout_height="wrap_content"
 android:hint="Enter Message"

 android:padding="12dp"
 android:minLines="6"
 android:inputType="textMultiLine"
 android:gravity="top"
 android:layout_marginTop="8dp"

 android:background="@color/cardview_s
 hadow_start_color"/>

 <Button
 android:id="@+id/bt_send"

 android:layout_width="wrap_content"

 android:layout_height="wrap_content"
 android:text="Send SMS"
 android:layout_marginTop="32dp"/>

 </LinearLayout>

```

## 2. MainActivity.java

```

package com.example.coexp29;
import androidx.annotation.NonNull;
import
 androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import
 androidx.core.content.ContextCompat;

import android.Manifest;
import android.content.pm.ActivityInfo;
import
 android.content.pm.PackageManager;
import android.os.Bundle;
import android.telephony.SmsManager;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends
 AppCompatActivity {

 EditText etPhone, etMessage;
 Button btSend;

 @Override
 protected void onCreate(Bundle
 savedInstanceState) {
 super.onCreate(savedInstanceState);

 setContentView(R.layout.activity_main);

 etPhone =
 findViewById(R.id.et_phone);
 etMessage =
 findViewById(R.id.et_message);
 btSend =
 findViewById(R.id.bt_send);

 btSend.setOnClickListener(new
 View.OnClickListener() {
 @Override
 public void onClick(View view) {

```

```

 if
(ContextCompat.checkSelfPermission(Main
nActivity.this

 ,

Manifest.permission.SEND_SMS)

 ==

PackageManager.PERMISSION_GRANTED)
{

 sendMessage();

 }else {

ActivityCompat.requestPermissions(Main
Activity.this, new
String[] {Manifest.permission.SEND_SMS
}, 100);

 }

 }

 });

}

private void sendMessage() {

 String sPhone =
etPhone.getText().toString().trim();

 String sMessage =
etMessage.getText().toString().trim();

 if (!sPhone.equals("") &&
!sMessage.equals("")) {

 SmsManager smsManager =
SmsManager.getDefault();

 smsManager.sendTextMessage(sPhone,
null, sMessage, null, null);

 Toast.makeText(getApplicationContext(),
"SMS sent successfully!",

```

```

Toast.LENGTH_LONG).show();

 }else {

 Toast.makeText(getApplicationContext(),
"Enter value first.",
Toast.LENGTH_SHORT).show();

 }

}

@Override

public void

onRequestPermissionsResult(int

requestCode, @NonNull

@org.jetbrains.annotations.NotNull

String[] permissions, @NonNull

@org.jetbrains.annotations.NotNull int[]

grantResults) {

 super.onRequestPermissionsResult(request

Code, permissions, grantResults);

 if (requestCode == 100 &&

grantResults.length > 0 &&

grantResults[0]

 ==

PackageManager.PERMISSION_GRANTED)
{

 sendMessage();

 }else {

 Toast.makeText(getApplicationContext(),
"Permission Denied!",
Toast.LENGTH_SHORT).show();

 }

 }

}

```



### 3. AndroidManifest.xml

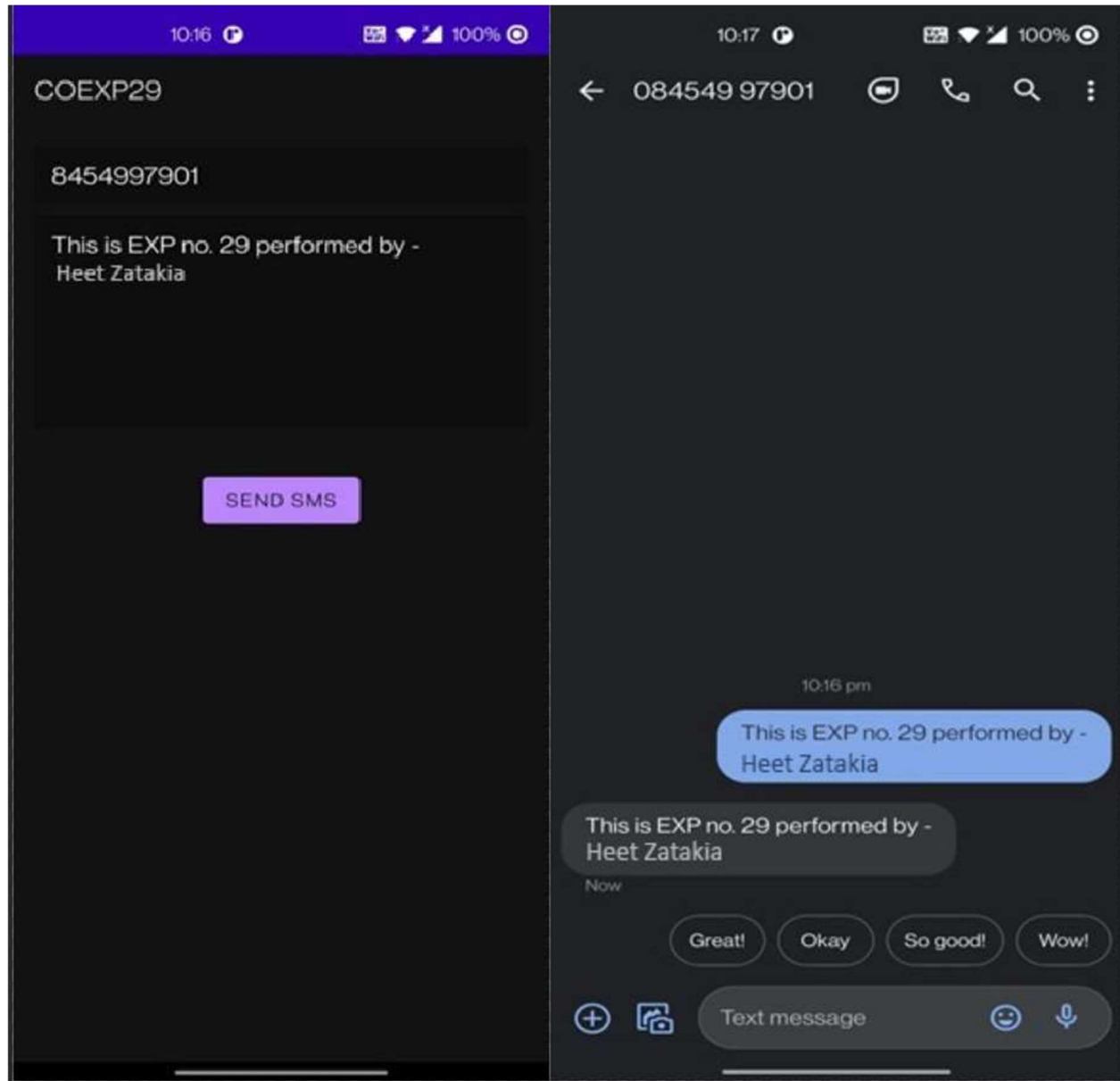
```
<?xml version="1.0" encoding="utf-8"?>
<manifest
xmlns:android="http://schemas.android.com/ap
k/res/android"
 package="com.example.coexp29">
 <uses-permission
android:name="android.permission.SEND_SM
S"/>
 <application
 android:allowBackup="true"
 android:icon="@mipmap/ic_launcher"
 android:label="@string/app_name"

android:roundIcon="@mipmap/ic_launcher_ro
und"
 android:supportsRtl="true"

 android:theme="@style/Theme.COEXP29">
 <activity android:name=".MainActivity">
 <intent-filter>
 <action
android:name="android.intent.action.MAIN" />

 <category
android:name="android.intent.category.LAUN
CHER" />
 </intent-filter>
 </activity>
 </application>

</manifest>
```



#### **XI. References / Suggestions for further Reading**

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. [https://www.tutorialspoint.com/android/android\\_advanced\\_tutorial.pdf](https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf)
4. <https://developer.android.com>

## XII. Assessment Scheme

Performance indicators		Weightage
Process related (10 Marks)		30%
1 .	Logic Formation	10%
2 .	Debugging ability	15%
3 .	Follow ethical practices	5%
Product related (15 Marks)		70%
4 .	Interactive GUI	20%
5 .	Answer to Practical related questions	20%
6 .	Expected Output	20%
7 .	Timely Submission	10%
Total (25 Marks)		100%

### *List of student Team Members*

1 [Heet Zatakia](#)

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

## **Practical No. 30: Develop a program to send and receive e-mail.**

### **I. Practical Significance**

To send email from your Android application, you have to write an Activity that needs to launch an email client and send an email using your Android device. This practice focuses on integrating the existing email clients in the new applications.

### **II. Relevant Program Outcomes (POs)**

PO1- Basic knowledge

PO2- Discipline  
knowledge

PO3- Experiments and  
practice PO7- Ethics

### **III. Competency and Practical Skills**

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to use the existing email clients in any android applications.
2. Able to develop the application for sending the emails.

### **IV. Relevant Course Outcome(s)**

1. Configure Android environment and development tools.
2. Develop rich user Interfaces by using layouts and controls.
3. Use User Interface components for android application development.

### **V. Practical Outcome (PrOs)**

Develop a program to send and receive e-mail.

### **VI. Relevant Affective Domain Related Outcome(s)**

1. Work collaboratively in team
2. Follow ethical Practices.

### **VII. Minimum Theoretical Background**

#### **Intent Object - Action to send Email:**

You will use ACTION\_SEND action to launch an email client installed on your Android device. Following is simple syntax to create an intent with ACTION\_SEND action.

```
Intent emailIntent = new Intent(Intent.ACTION_SEND);
```

#### **Intent Object – Data Type to send Email**

To send an email you need to specify mailto: as URI using setData() method and data type will be to text/plain using setType() method as follows:

```
emailIntent.setData(Uri.parse("mailto:"));
emailIntent.setType("text/plain");
```

### VIII. Resources required(Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

### IX. Practical related Questions

*Note: Below given are a few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.*

1. Why it becomes necessary to have an inbuilt email module in mobile applications.
2. List the extra fields that can be used in an application to send emails.

**(Space for answers)**

**Ans 1)** Before starting Email Activity, You must know Email functionality with intent, Intent is carrying data from one component to another component with-in the application or outside the application.

To send an email from your application, you don't have to implement an email client from the beginning, but you can use an existing one like the default Email app provided from Android, Gmail, Outlook, K-9 Mail etc. For this purpose, we need to write an Activity that launches an email client, using an implicit Intent with the right action and data. In this example, we are going to send an email from our app by using an Intent object that launches existing email clients.

**Ans 2)** Extra fields that can be used in an application to send emails are as follows:

EXTRA\_EMAIL - It's an array of email addresses

EXTRA\_SUBJECT - The subject of the email that we want to send

EXTRA\_TEXT - The body of the email

## X. Exercise

*Note: Faculty must ensure that every group of students use different input value.*

(Use blank space for answers or attach more pages if needed)

1. Write a program to send email.

**(Space for answers)**

Ans1) 1. Activity\_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
 xmlns:android="http://schemas.android.com/apk/res/android"
 xmlns:tools="http://schemas.android.com/tools"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 android:orientation="vertical"
 android:padding="16dp"
 tools:context=".MainActivity">

 <TextView
 android:layout_width="match_parent"
 android:layout_height="wrap_content"
 android:text="To:"
 android:textAppearance="@style/TextAppearance.AppCompat.Large" />

 <EditText
 android:id="@+id/edit_text_to"
 android:layout_width="match_parent"
 android:layout_height="wrap_content"
 android:inputType="textEmailAddress" />

 <TextView
 android:layout_width="match_parent"
 android:layout_height="wrap_content"
 android:text="Subject:"
 android:textAppearance="@style/TextAppearance.AppCompat.Large" />

 <EditText
 android:id="@+id/edit_text_subject"
 android:layout_width="match_parent"
 android:layout_height="wrap_content"
 android:inputType="textEmailSubject" />
```

```

<TextView
 android:layout_width="match_parent"
 android:layout_height="wrap_content"
 android:text="Message:"
 android:textAppearance="@style/TextAppearance.AppCompat.Large" />

<EditText
 android:id="@+id/edit_text_message"
 android:layout_width="match_parent"
 android:layout_height="wrap_content"
 android:gravity="start|top"
 android:lines="10" />

<Button
 android:id="@+id/button_send"
 android:layout_width="match_parent"
 android:layout_height="wrap_content"
 android:text="send" />

</LinearLayout>

```

## 2. MainActivity.java

```

package com.example.coexp30;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

public class MainActivity extends AppCompatActivity {
 private EditText mEditTextTo,mEditTextSubject,mEditTextMessage;
 @Override
 protected void onCreate(Bundle savedInstanceState) {
 super.onCreate(savedInstanceState);
 setContentView(R.layout.activity_main);

 mEditTextTo = findViewById(R.id.edit_text_to);
 mEditTextSubject = findViewById(R.id.edit_text_subject);
 mEditTextMessage = findViewById(R.id.edit_text_message);

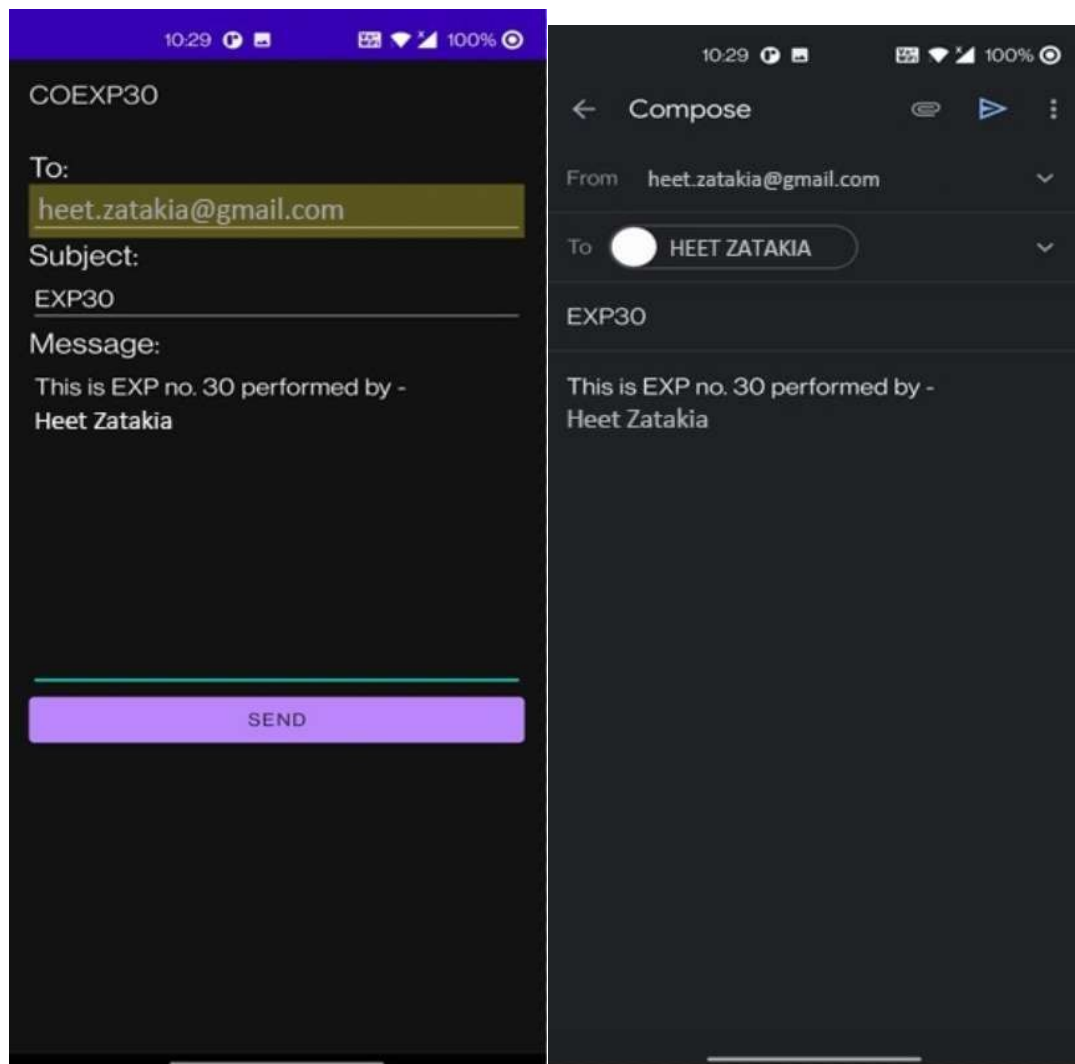
 Button buttonSend = findViewById(R.id.button_send);
 buttonSend.setOnClickListener(new View.OnClickListener() {

```

```

@Override
public void onClick(View v) {
 sendMail();
}
});
}
private void sendMail() {
 String recipientList = mEditTextTo.getText().toString();
 String[] recipients = recipientList.split(",");
 String subject = mEditTextSubject.getText().toString();
 String message = mEditTextMessage.getText().toString();
 Intent intent = new Intent(Intent.ACTION_SEND);
 intent.putExtra(Intent.EXTRA_EMAIL, recipients);
 intent.putExtra(Intent.EXTRA_SUBJECT, subject);
 intent.putExtra(Intent.EXTRA_TEXT, message);
 intent.setType("message/rfc822");
 startActivity(Intent.createChooser(intent, "Choose an email client"));
}
}

```





**XI. References / Suggestions for further Reading**

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. [https://www.tutorialspoint.com/android/android\\_advanced\\_tutorial.pdf](https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf)
4. <https://developer.android.com>

**XII. Assessment Scheme**

Performance indicators		Weightage
<b>Process related (10 Marks)</b>		<b>30%</b>
1 .	Logic Formation	10%
2 .	Debugging ability	15%
3 .	Follow ethical practices	5%
<b>Product related (15 Marks)</b>		<b>70%</b>
4 .	Interactive GUI	20%
5 .	Answer to Practical related questions	20%
6 .	Expected Output	20%
7 .	Timely Submission	10%
	<b>Total (25 Marks)</b>	<b>100%</b>

***List of student Team Members***

1 [Heet Zatakia](#)

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

## **Practical No. 31: Deploy map-based application. Part I**

### **I. Practical Significance**

Android allows us to integrate google maps in our application. You can show any location on the map, or can show different routes on the map. You can also customize the map according to your choices.

### **II. Relevant Program Outcomes (POs)**

PO1- Basic knowledge

PO2- Discipline  
knowledge

PO3- Experiments and  
practice PO4- Engineering  
tools

### **III. Competency and Practical Skills**

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to integrate Maps in any application.
2. Demonstrate the benefits of Maps and its uses.

### **IV. Relevant Course Outcome(s)**

1. Configure Android environment and development tools.
2. Develop rich user Interfaces by using layouts and controls.
3. Create an Android application using a database.
4. Publish Android applications.

### **V. Practical Outcome (PrOs)**

Deploy map-based application. Part I

### **VI. Relevant Affective Domain related Outcomes**

1. Work collaboratively in team
2. Follow ethical Practices.

### **VII. Minimum Theoretical Background**

Add the map fragment into the xml layout file. Its syntax is given below –

```
<fragment
android:id="@+id/map"
"

android:name="com.google.android.gms.maps.MapFragment"
android:layout_width="match_parent"
android:layout_height="match_parent"/>
```

Add some permissions along with the Google Map API key in the AndroidManifest.XML file. Its syntax is given below:

```
<!--Permissions-->
<uses-permission
android:name="android.permission.ACCESS_NETWORK_STATE"
<uses-permission android:name="android.permission.INTERNET"
```

```

/>
<uses-permission
android:name="com.google.android.providers.gsf.permission.
READ_GSERVICES" />
<uses-permission
android:name="android.permission.WRITE_EXTERNAL_STORAGE" />

```

### VIII. Resources required (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

### IX. Practical related Questions

*Note: Below given are a few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.*

1. List the names of map types and write the syntax to change it.
2. Name the methods used to enable and disable zoom features.

(Space for answers)

**Ans 1)** The following map types are supported in Google Maps API:

1. ROADMAP (normal, default 2D map)
2. SATELLITE (photographic map)
3. HYBRID (photographic map + roads and city names)
4. TERRAIN (map with mountains, rivers, etc.)

The map type is specified either within the Map properties object, with the **mapTypeId** property:

```

var mapOptions = {
 center:new google.maps.LatLng(51.508742,-0.120850),
 zoom:7,
 mapTypeId: google.maps.MapTypeId.HYBRID
};

```

Or by calling the map's **setMapTypeId()** method:

```
map.setMapTypeId(google.maps.MapTypeId.HYBRID);
```

**Ans 2)** To enable or disable all interactive features, we use the **interactivity()** method. Set true or false to the method depending if you'd like to enable the map interactivity or disable it.

## X. Exercise

*Note: Faculty must ensure that every group of students use different input values.*

(Use blank space for answers or attach more pages if needed)

1. Write a program to locate the user's current location.

**(Space for answers)**

Ans1) 1. activity\_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
 xmlns:app="http://schemas.android.com/apk/res-auto"
 xmlns:tools="http://schemas.android.com/tools"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 tools:context=".MainActivity">

 <fragment
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 android:id="@+id/google_map"
 android:name="com.google.android.gms.maps.SupportMapFragment"
 />
</RelativeLayout>
```

2. build.gradle (:app)

```
plugins {
 id 'com.android.application'
}

android {
 compileSdkVersion 30
 buildToolsVersion "30.0.3"

 defaultConfig {
 applicationId "com.example.coexp31"
 minSdkVersion 17
 targetSdkVersion 30
 versionCode 1
 versionName "1.0"

 testInstrumentationRunner
 "androidx.test.runner.AndroidJUnitRunner"
 }

 buildTypes {
 release {
 minifyEnabled false
 proguardFiles
 getDefaultProguardFile('proguard-android-
 optimize.txt'), 'proguard-rules.pro'
 }
 }

 compileOptions {
 sourceCompatibility
 JavaVersion.VERSION_1_8
 targetCompatibility
 JavaVersion.VERSION_1_8
 }
}

dependencies {
 implementation
```

```
'androidx.appcompat:appcompat:1.3.0'
 implementation
'com.google.android.material:material:1.3.0'
 implementation
'androidx.constraintlayout:constraintlayout:2.0.4'
 implementation
'com.google.android.gms:play-services-maps:17.0.1'
```

```
 implementation
'com.google.android.gms:play-services-location:18.0.0'
 testImplementation 'junit:junit:4.+
 androidTestImplementation
'androidx.test.ext:junit:1.1.2'
 androidTestImplementation
'androidx.test.espresso:espresso-core:3.3.0'
}
```

### 3. strings.xml

```
<resources>
 <string name="app_name">COEXP31</string>
 <string name="map_key"
 translatable="false">AIzaSyB_qsP8AOP_P0MdIPz-48TDaJYjTP3vbjo</string>
</resources>
```

### 4. MainActivity.java

```
package com.example.coexp31;

import
androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import androidx.annotation.NonNull;
import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import
androidx.fragment.app.FragmentActivity;

import android.Manifest;
import android.content.pm.PackageManager;
import android.location.Location;
import android.os.Bundle;
import android.widget.Toast;

import
com.google.android.gms.location.FusedLocationProviderClient;
import
com.google.android.gms.location.LocationServi
```

```
ces;
import
com.google.android.gms.maps.CameraUpdateFactory;
import
com.google.android.gms.maps.GoogleMap;
import
com.google.android.gms.maps.OnMapReadyCallback;
import
com.google.android.gms.maps.SupportMapFragment;
import
com.google.android.gms.maps.model.LatLng;
import
com.google.android.gms.maps.model.MarkerOptions;
import
com.google.android.gms.tasks.OnSuccessListener;
import com.google.android.gms.tasks.Task;

public class MainActivity extends
FragmentActivity implements
OnMapReadyCallback {
```

```

 Location currentLocation;
 FusedLocationProviderClient
 fusedLocationProviderClient;

 private static final int REQUEST_CODE =
 101;

 @Override
 protected void onCreate(Bundle
 savedInstanceState) {
 super.onCreate(savedInstanceState);
 setContentView(R.layout.activity_main);

 fusedLocationProviderClient =
 LocationServices.getFusedLocationProviderCli
 ent(this);//CLIENT REGISTER
 fetchLastLocation();
 }

 private void fetchLastLocation() {
 if
 (ActivityCompat.checkSelfPermission(this,
 Manifest.permission.ACCESS_FINE_LOCATI
 ON) !=
 PackageManager.PERMISSION_GRANTED &&
 ActivityCompat.checkSelfPermission(this,
 Manifest.permission.ACCESS_COARSE_LOCA
 TION) !=
 PackageManager.PERMISSION_GRANTED) {

 ActivityCompat.requestPermissions(this, new
 String[]{Manifest.permission.ACCESS_FINE_
 LOCATION}, REQUEST_CODE);
 return;

 }

 Task<Location> task =
 fusedLocationProviderClient.getLastLocation();
 task.addOnSuccessListener(new
 OnSuccessListener<Location>() {
 @Override
 public void onSuccess(Location
 location) {
 if (location != null){
 currentLocation = location;
 Toast.makeText(getApplicationContext(),curren

```

```

 tLocation.getLatitude()+" "+currentLocation.get
 Longitude(),Toast.LENGTH_SHORT).show();
 SupportMapFragment
 supportMapFragment=(SupportMapFragment)
 getSupportFragmentManager().findFragmentBy
 Id(R.id.google_map);
 // SupportMapFragment
 supportMapFragment = (SupportMapFragment)
 getSupportFragmentManager().findFragmentBy
 Id(R.id.google_map);

 supportMapFragment.getMapAsync(MainActiv
 ity.this);
 }
 }
});
}

@Override
public void onMapReady(GoogleMap
googleMap) {
 LatLng latLng = new
 LatLng(currentLocation.getLatitude(),currentLo
 cation.getLongitude());
 MarkerOptions markerOptions = new
 MarkerOptions().position(latLng).title("I am
 here");

 googleMap.animateCamera(CameraUpdateFact
 ory.newLatLng(latLng));

 googleMap.animateCamera(CameraUpdateFact
 ory.newLatLngZoom(latLng,5));
 googleMap.addMarker(markerOptions);
}

@Override
public void onRequestPermissionsResult(int
requestCode, @NonNull String[] permissions,
@NonNull int[] grantResults) {
 switch (requestCode){
 case REQUEST_CODE:
 if(grantResults.length>0 &&
 grantResults[0] ==
 PackageManager.PERMISSION_GRANTED){
 fetchLastLocation();
 }
 }
}

```

```
 break;
 }
}
```

## 5. AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
 package="com.example.coexp31">

 <uses-permission android:name="android.permission.INTERNET"/>
 <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>
 <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/>

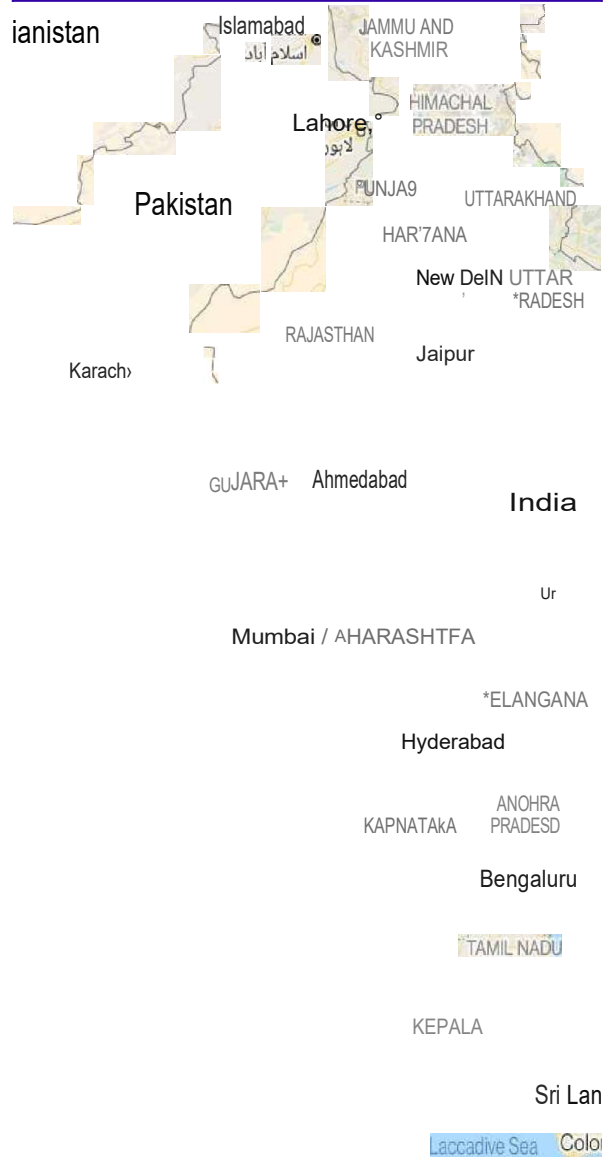
 <application
 android:allowBackup="true"
 android:icon="@mipmap/ic_launcher"
 android:label="@string/app_name"
 android:roundIcon="@mipmap/ic_launcher_round"
 android:supportsRtl="true"
 android:theme="@style/Theme.COEXP31">

 <meta-data
 android:name="com.google.android.geo.API_KEY"
 android:value="@string/map_key"/>

 <activity android:name=".MainActivity">
 <intent-filter>
 <action android:name="android.intent.action.MAIN" />

 <category android:name="android.intent.category.LAUNCHER" />
 </intent-filter>
 </activity>
 </application>

</manifest>
```



19.043085172.8572711

Google



Google





**XI. References / Suggestions for further Reading**

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. [https://www.tutorialspoint.com/android/android\\_advanced\\_tutorial.pdf](https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf)
4. <https://developer.android.com>

**XII. Assessment Scheme**

Performance indicators		Weightage
Process related (10 Marks)		30%
1 .	Logic Formation	10%
2 .	Debugging ability	15%
3 .	Follow ethical practices	5%
Product related (15 Marks)		70%
4 .	Interactive GUI	20%
5 .	Answer to Practical related questions	20%
6 .	Expected Output	20%
7 .	Timely Submission	10%
Total (25 Marks)		100%

***List of student Team Members***

1 [Heet Zatakia](#)

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	

## **Practical No. 32: Deploy map-based application. Part II**

### **I. Practical Significance**

Android allows us to integrate google maps in our application. You can show any location on the map, or can show different routes on the map. You can also customize the map according to your choices.

### **II. Relevant Program Outcomes (POs)**

PO1- Basic knowledge

PO2- Discipline  
knowledge

PO3- Experiments and

practice PO4- Engineering  
tools

### **III. Competency and Practical Skills**

“Create simple Android applications.”

This practical is expected to develop the following skills

1. Able to integrate Maps in any application.
2. Demonstrate the benefits of Maps and its uses.

### **IV. Relevant Course Outcome(s)**

1. Configure Android environment and development tools.
2. Develop rich user Interfaces by using layouts and controls.
3. Create an Android application using a database.
4. Publish Android applications.

### **V. Practical Outcome (PrOs)**

Deploy map-based application. Part II

### **VI. Relevant Affective Domain Related Outcome(s)**

1. Work collaboratively in team
2. Follow ethical Practices.

### **VII. Minimum Theoretical Background**

Methods available in the Google Map class are given below.

- 1.addCircle(CircleOptions options) : This method add a circle to the map
- 2.addPolygon(PolygonOptions options) : This method add a polygon to the map
- 3.addTileOverlay(TileOverlayOptions options) : This method add tile overlay to the map
4. animateCamera(CameraUpdate update) : This method Moves the map according to the update with an animation
5. clear() : This method removes everything from the map
6. getLocation() : This method returns the currently displayed user location
- 7.moveCamera(CameraUpdate update) : This method repositions the camera according to the instructions defined in the update
- 8.setTrafficEnabled(boolean enabled) : This method Toggles the traffic layer on or off
- 9.snapshot(GoogleMap.SnapshotReadyCallback callback) : This method Takes a snapshot of the map

10. stopAnimation() : This method stops the camera animation if there is one in progress

### VIII. Resources required (Additional)

Sr. No.	Instrument /Object	Specification	Quantity	Remarks
1	Android enabled smartphone / Android version supporting emulator	2 GB RAM	1	Data cable is mandatory for emulators

### IX. Practical related Questions

*Note: Below given are a few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.*

1. Explain the ways to add Markers on the Google Map.
2. Write the syntax for the method which is used to add a compass in Google Map.

(Space for answers)

**Ans 1)** For adding a custom marker to Google Maps navigate to the app > res > drawable > Right-Click on it > New > Vector Assets and select the icon which we have to show on your Map. You can change the color according to our requirements. After creating this icon now we will move towards adding this marker to our Map.

**Ans 2)** The Maps API provides a compass graphic which appears in the top left corner of the map under certain circumstances. The compass will only ever appear when the camera is oriented such that it has a non-zero bearing or non-zero tilt. When the user clicks on the compass, the camera animates back to a position with bearing and tilt of zero (the default orientation) and the compass fades away shortly afterwards. You can disable the compass appearing altogether by calling **UiSettings.setCompassEnabled(boolean)**. However, you cannot force the compass to always be shown.

### X. Exercise

*Note: Faculty must ensure that every group of students use different input value.*

(Use blank space for answers or attach more pages if needed)

1. Write a program to draw a route between two locations.

(Space for answers)

Ans1) 1. activity\_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
 xmlns:android="http://schemas.android.com/apk/res/android"
 xmlns:app="http://schemas.android.com/apk/res-auto"
 xmlns:tools="http://schemas.android.com/tools"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 android:orientation="vertical"
 android:padding="16dp"
 android:gravity="center_horizontal"
 tools:context=".MainActivity">
 <EditText
 android:id="@+id/et_source"
 android:layout_width="317dp"
 android:layout_height="56dp"
 android:layout_marginTop="220dp"
 android:background="@android:drawable/editbox_background"
 android:fontFamily="@font/poppins_medium"
 android:hint="Enter Source Location"
 android:padding="12dp"
 app:layout_constraintEnd_toEndOf="parent"
 app:layout_constraintStart_toStartOf="parent"
 app:layout_constraintTop_toTopOf="parent" />
 <EditText
 android:id="@+id/et_destination"
 android:layout_width="317dp"
 android:layout_height="52dp"
 android:layout_marginTop="36dp"
 android:background="@android:drawable/editbox_background"
 android:fontFamily="@font/poppins_medium"
 android:hint="Enter Destination Location"
 android:padding="12dp"
 app:layout_constraintEnd_toEndOf="parent"
 app:layout_constraintStart_toStartOf="parent"
 app:layout_constraintTop_toBottomOf="@+id/et_source" />
 <Button
 android:id="@+id/bt_track"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_marginTop="56dp"
 android:fontFamily="@font/poppins_medium"
 android:text="Display Location"
 app:backgroundTint="#3F51B5"
 app:layout_constraintEnd_toEndOf="parent"
 app:layout_constraintHorizontal_bias="0.497"
```

```

 app:layout_constraintStart_toStartOf="parent"
 app:layout_constraintTop_toBottomOf="@+id/et_destination" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

## 2. MainActivity.java

```

package com.example.coexp32;
import androidx.appcompat.app.AppCompatActivity;

import android.content.ActivityNotFoundException;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.Display;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
 EditText etSource,etDestination;
 Button btTrack;
 @Override
 protected void onCreate(Bundle savedInstanceState) {
 super.onCreate(savedInstanceState);
 setContentView(R.layout.activity_main);
 etSource = findViewById(R.id.et_source);
 etDestination = findViewById(R.id.et_destination);
 btTrack = findViewById(R.id.bt_track);
 btTrack.setOnClickListener(new View.OnClickListener() {
 @Override
 public void onClick(View v) {
 String sSource = etSource.getText().toString().trim();
 String sDestination = etDestination.getText().toString().trim();
 if(sSource.equals("") && sDestination.equals("")){
 Toast.makeText(MainActivity.this, "Enter both location",
Toast.LENGTH_SHORT).show();
 }else{
 DisplayTrack(sSource,sDestination); } } } } }
 private void DisplayTrack(String sSource, String sDestination) {
 try{

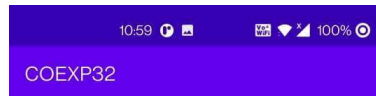
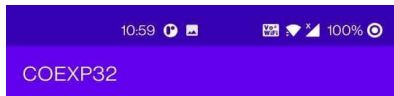
 Uri uri = Uri.parse("https://www.google.co.in/maps/dir/" + sSource + "/"
 + sDestination);
 Intent intent= new Intent(Intent.ACTION_VIEW,uri);
 intent.setPackage("com.google.android.apps.maps");

```

```

 intent.setFlags(Intent.FLAG_ACTIVITY_NEW_TASK);
 startActivity(intent);
 } catch (ActivityNotFoundException e){
 Uri uri =
Uri.parse("https://play.google.com/store/apps/details?id=com.google.android.apps.maps");
 Intent intent= new Intent(Intent.ACTION_VIEW,uri);
 intent.setFlags(Intent.FLAG_ACTIVITY_NEW_TASK);
 startActivity(intent);
 }
}
}

```

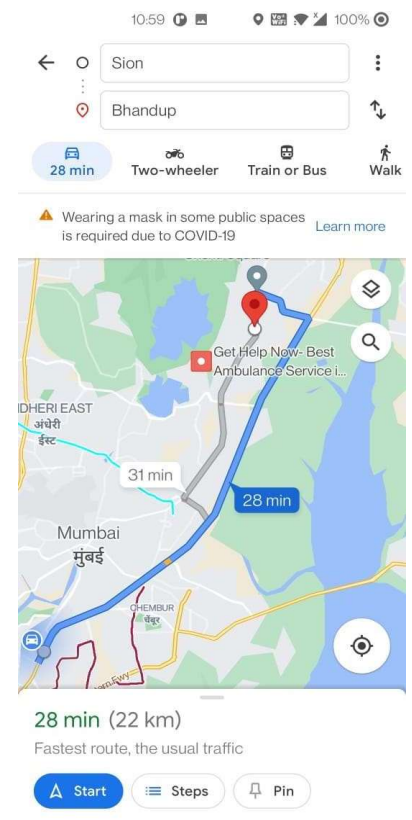




DISPLAY LOCATION



DISPLAY LOCATION



**XI. References / Suggestions for further Reading**

1. <https://www.tutorialspoint.com/android>
2. <https://stuff.mit.edu>
3. [https://www.tutorialspoint.com/android/android\\_advanced\\_tutorial.pdf](https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf)
4. <https://developer.android.com>

**XII. Assessment Scheme**

Performance indicators		Weightage
Process related (10 Marks)		30%
1 .	Logic Formation	10%
2 .	Debugging ability	15%
3 .	Follow ethical practices	5%
Product related (15 Marks)		70%
4 .	Interactive GUI	20%
5 .	Answer to Practical related questions	20%
6 .	Expected Output	20%
7 .	Timely Submission	10%
Total (25 Marks)		100%

***List of student Team Members***

1 [Heet Zatakia](#)

Marks Obtained			Dated signature of Teacher
Process Related(10)	Product Related(15)	Total (25)	